

riemann-channel: lab book

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1 Frontmatter, Status Vocabulary, and Scope

1.1 Aim

This lab book records an attempt to formulate the Riemann hypothesis in the language of tensor networks and operational quantum mechanics: channels, Kraus operators, transfer matrices, matrix-product states. The aim is a formulation that captures the mechanisms of the proofs that exist (Weil for curves, Deligne for varieties over finite fields, Ihara–Bass and the Ramanujan property for regular graphs) and connects more directly to quantum mechanics than the Hilbert–Pólya programme does. Nothing here proves anything about the Riemann hypothesis. The Markdown notes under `notes/` are the exploratory record; this book is their distillation, and every statement here is tagged.

1.2 The organising picture

Every zeta function in sight has two sides. Side A is a nonnegative count of closed orbits (prime cycles of a graph, points of a curve, prime powers with weight $\log p$) and its Euler product; it is a product state, the primes do not talk to each other. Side B is a small non-Hermitian operator whose power sums reproduce the counts; its eigenvalues are the zeros. The Riemann hypothesis, in every instance, is the statement that all nontrivial eigenvalues of side B share one modulus. Section 2 fixes the notation and Section 3 the definitions used for both sides throughout.

1.3 Status vocabulary

Every theorem-like environment in this book carries exactly one status tag, printed at its right margin, equal to the status derived by `scripts/labbook_check.py` from `db/claims.tsv`:

- **proved**: a written proof exists under `notes/` and a reviewer other than the author has returned VERDICT `<id>`: VALID in a file under `notes/reviews/`. Both author and reviewers to date are Claude models (single family); no second model family has been used.
- **sketched**: written by the author, not yet attacked.
- **numerical**: established by a script whose output is captured under `outputs/`; never more than numerical evidence.
- **cited**: a statement of the literature, backed by a quote in `db/provenance.tsv` that the gate byte-checks against the arXiv TeX source under `refs/src/`.
- **conjectured**, **open**, **refuted**, **assumed**: as the words say; a claim that depends on any of the first three prints **unsupported**, and one that depends on an assumption prints with the suffix **-conditional**.

Definitions carry no tag; they are **stipulated** (ours) or **quoted** (a source’s, with provenance) in `db/definitions.tsv`. The derived-status table generated from the claims database is reproduced here.

id	kind	derived status	assumptions
prop:bc-residue-state	proposition	sketched	-
prop:bc-no-finite-phase-representation	proposition	sketched	-
prop:bc-symmetry-state-extension	proposition	sketched	-
thm:bc-covariant-gks-cone	theorem	sketched	-

id	kind	derived status	assumptions
thm:bc-critical-cone-dimensions	theorem	sketched	-
prop:bc-weyl-covariance-collapse	proposition	sketched	-
prop:bc-symplectic-odd-counterexample	proposition	sketched	-
prop:bc-compatible-prime-coupling	proposition	sketched	-
num:bc-symmetry-generators	numerical	numerical	-
obs:one-identity	observation	sketched	-
obs:rh-finite-permutation	observation	sketched	-
obs:rh-small-matrix	observation	sketched	-
obs:rh-graph-ramanujan	observation	sketched	-
obs:rh-mps-correlation-length	observation	sketched	-
obs:rh-flow-pairing	observation	sketched	-
obs:hilbert-polya-hermitian	observation	sketched	-
prop:scattering-inner	proposition	sketched	-
num:scattering-symbol	numerical	numerical	-
prop:functional-model-modes	proposition	sketched	-
prop:phase-identity	proposition	sketched	-
prop:ringnorm-trace	proposition	proved-conditional	asm:explicit-formula-trace
num:ringnorm-3000	numerical	numerical	-
prop:bc-mpo	proposition	sketched	-
num:bc-mpo	numerical	numerical	-
conj:stinespring-jump-form	conjecture	open	-
num:weil-lps-ramanujan	numerical	numerical	-
num:weil-lps-rh	numerical	numerical	-
prop:weil-lps-exactly-ramanujan	proposition	sketched	-
conj:weil-lps-hecke	conjecture	open	-
prop:as-transfer-matrix	proposition	proved-conditional	asm:normal-basis
prop:as-unitarity	proposition	proved	-
num:artin-schreier-mps	numerical	numerical	-
obs:as-nonquadratic	observation	sketched	-
obs:deligne-ingredients	observation	sketched	-
obs:abelian-obstruction	observation	sketched	-
thm:qihara-general	theorem	proved	-
cor:qihara-kraus	corollary	proved	-
cor:qihara-unitary	corollary	proved	-
cor:qihara-poles	corollary	proved	-
obs:rh-iff-ramanujan-channel	observation	sketched	-
num:qihara-unitary	numerical	numerical	-
num:qihara-general	numerical	numerical	-
cit:mo-extended-ihara	citedfact	cited	-
cit:mo-adjoint-weights	citedfact	cited	-
cit:wf-arbitrary-weights	citedfact	cited	-
cit:bc-nonbacktracking	citedfact	cited	-
cit:harrow-transfer	citedfact	cited	-
cit:iyer-exact-ramanujan	citedfact	cited	-
cit:sunada-attribution	citedfact	cited	-
obs:prior-art-verdict	observation	sketched	-
conj:kraus-ramanujan	conjecture	open	-
conj:assemble-over-p	conjecture	open	-
obs:prime-by-prime-blind	observation	sketched	-
asm:fuchsian-geometry	assumption	assumed	-
asm:laplace-spectrum-compact	assumption	assumed	-
asm:selberg-zeta-entire	assumption	assumed	-

id	kind	derived status	assumptions
asm:heat-semigroup	assumption	assumed	-
asm:zeta-standard	assumption	assumed	-
asm:gamma-standard	assumption	assumed	-
cit:dz-guillemin	citedfact	cited	-
cit:dfg-band	citedfact	cited	-
cit:dz-ruelle-selberg	citedfact	cited	-
cit:marklof-trace	citedfact	cited	-
cit:fgkp-casimir-laplacian	citedfact	cited	-
cit:fjs-scattering-det	citedfact	cited	-
cit:mv-continuous-term	citedfact	cited	-
prop:circle-comb	proposition	proved	-
prop:prime-circles-orbital	proposition	proved	-
prop:finite-euler-spectrum	proposition	proved-conditional	asm:gamma-standard, asm:ze
prop:lindblad-sum-of-squares	proposition	proved	-
prop:casimir-laplacian	proposition	proved	-
prop:quantum-lindblad-tensor	proposition	proved	-
prop:poincare-jacobian	proposition	proved	-
prop:flat-tower	proposition	proved-conditional	asm:fuchsian-geometry
prop:tower-divisor-band	proposition	proved-conditional	asm:fuchsian-geometry, asm
prop:graph-tower-ruelle	proposition	proved-conditional	asm:fuchsian-geometry, asm
prop:cuspidal-prime-comb	proposition	proved-conditional	asm:gamma-standard, asm:ze
prop:damping-identity	proposition	proved-conditional	asm:gamma-standard, asm:ze
num:selberg-lindblad	numerical	numerical	-
obs:continuous-dictionary	observation	sketched-conditional	asm:fuchsian-geometry, asm
conj:quantum-lindblad-gap	conjecture	open	-
obs:spt-protection-is-sign-data	observation	sketched	-
thm:weil-positivity-finite	theorem	proved	-
thm:weil-duality-pairing	theorem	proved	-
prop:weil-orbit-negative	proposition	proved	-
prop:weil-reciprocal-only	proposition	proved	-
prop:kraus-conjugation-symmetry	proposition	proved	-
thm:kraus-inverse-pairing-duality	theorem	proved	-
prop:kraus-both-pairings-unitary	proposition	proved	-
thm:kraus-weil-criterion	theorem	proved	-
prop:kraus-no-duality-example	proposition	proved	-
thm:weil-positivity-continuous	theorem	proved	-
thm:weil-line-duality	theorem	proved	-
thm:cmps-ring-norms	theorem	proved	-
prop:hp-inner-product-discrete	proposition	proved	-
prop:hp-inner-product-continuous	proposition	proved	-
prop:weil-blind-jordan	proposition	proved	-
obs:weil-zeta-dictionary	observation	sketched	-
obs:huang-boundedness	observation	proved	-
obs:kraus-dichotomy	observation	proved	-
num:weil-positivity	numerical	numerical	-
cit:weil-criterion	citedfact	cited	-
cit:li-criterion	citedfact	cited	-
cit:huang-graph-criterion	citedfact	cited	-
cit:hastings-bound	citedfact	cited	-
cit:herglotz-bochner	citedfact	cited	-
cit:suzuki-kernel	citedfact	cited	-
cit:pt-real-spectrum	citedfact	cited	-

id	kind	derived status	assumptions
obs:ringnorm-sign-correction	observation	sketched-conditional	asm:explicit-formula-trace
asm:explicit-formula-trace	assumption	assumed	-
asm:zero-count-location	assumption	assumed	-
asm:positive-distributions	assumption	assumed	-
asm:landau-laplace	assumption	assumed	-
asm:lax-phillips-modes	assumption	assumed	-
asm:weil-curves	assumption	assumed	-
cit:lidskii-trace	citedfact	cited	-
cit:landau-dirichlet	citedfact	cited	-
cit:deninger-graded-lefschetz	citedfact	cited	-
cit:connes-absorption	citedfact	cited	-
lem:supertrace-distribution	lemma	proved	-
thm:supertrace-rigidity	theorem	proved	-
cor:forced-net-spectrum	corollary	proved-conditional	asm:explicit-formula-trace
thm:parity-forced-finite	theorem	proved-conditional	asm:explicit-formula-trace
prop:parity-infinite-dominated	proposition	proved-conditional	asm:explicit-formula-trace
obs:parity-infinite-open	observation	sketched	-
thm:no-ungraded-trace	theorem	proved	-
prop:trace-supertrace-criterion	proposition	proved	-
cor:curve-counts-graded	corollary	proved-conditional	asm:weil-curves
prop:riemann-graded-generator	proposition	proved-conditional	asm:explicit-formula-trace
prop:ladder-parity-free	proposition	proved-conditional	asm:explicit-formula-trace
obs:phantasm-scope	observation	sketched	-
cit:fmps-supertrace	citedfact	cited	-
cit:fcmps-graded-bond	citedfact	cited	-
cit:cmps-schmidt-fixed-point	citedfact	cited	-
cit:bc-kms-unique	citedfact	cited	-
thm:vacuum-decay-finite	theorem	proved	-
cor:vacuum-decay-traces	corollary	proved	-
prop:absorbing-vacuum-channel	proposition	proved-conditional	asm:lax-phillips-modes
obs:zero-coherences-populations	observation	sketched	-
prop:vacuum-renewal-trivial-entanglement	proposition	proved	-
prop:normal-kms-gibbs	proposition	proved	-
lem:mm1-spectrum	lemma	proved	-
prop:prime-chain-lindbladian	proposition	proved	-
prop:prime-chain-offdiagonal	proposition	proved	-
cor:gibbs-delocalisation	corollary	proved	-
num:riemann-graded-trace	numerical	numerical	-
num:bc-entropy	numerical	numerical	-
obs:gibbs-entanglement-hagedorn	observation	sketched	-
obs:character-sector-grading	observation	sketched-conditional	asm:as-lpolynomial, asm:fr
obs:complementary-halves	observation	sketched	-
obs:phantasm-established-scope	observation	sketched	-
conj:phantasm-both-halves	conjecture	open	-
conj:galois-graded-bond	conjecture	open	-
asm:normal-basis	assumption	assumed	-
asm:as-lpolynomial	assumption	assumed	-
asm:weil-implementer-exact	assumption	assumed	-
asm:frobenius-semisimple	assumption	assumed	-
cit:weil-character-formula	citedfact	cited	-
cit:weil-egorov-uniqueness	citedfact	cited	-
lem:gauss-quadratic-form	lemma	proved	-

id	kind	derived status	assumptions
thm:as-sign-law	theorem	proved-conditional	asm:normal-basis
cor:as-periodic-sign	corollary	proved-conditional	asm:as-lpolynomial, asm:no
prop:as-sign-counterexamples	proposition	proved-conditional	asm:normal-basis
thm:as-frobenius-block	theorem	proved-conditional	asm:as-lpolynomial, asm:no
thm:as-super-transfer	theorem	proved-conditional	asm:as-lpolynomial, asm:fr
thm:as-symplectic-map	theorem	proved-conditional	asm:weil-implementer-exact
cor:as-weil-bound-fixed-space	corollary	proved-conditional	asm:normal-basis, asm:weil
num:artin-schreier-super	numerical	numerical	-
cit:kl-identity	citedfact	cited	-
cit:kl-rh-four-way	citedfact	cited	-
cit:kangyu-alternating	citedfact	cited	-
cit:llp-geodesic-flow	citedfact	cited	-
cit:llp-one-sided-rh	citedfact	cited	-
cit:kamber-lp	citedfact	cited	-
cit:storm-reduction	citedfact	cited	-
cit:bcds-triangulation	citedfact	cited	-
cit:no-general-complex-zeta	citedfact	cited	-
cit:deitmar-alternating	citedfact	cited	-
cit:dz-order-chi	citedfact	cited	-
cit:hashimoto-hprime	citedfact	cited	-
cit:knill-sdet	citedfact	cited	-
cit:mo-dirac	citedfact	cited	-
asm:h-point	assumption	assumed	-
asm:h-llp	assumption	assumed	-
asm:h-kl	assumption	assumed	-
asm:h-ky	assumption	assumed	-
asm:h-ky-local	assumption	assumed	-
asm:h-kl-table	assumption	assumed	-
asm:h-lsv	assumption	assumed	-
asm:h-strong	assumption	assumed	-
asm:h-kamber	assumption	assumed	-
thm:ordered-vs-building	theorem	proved-conditional	asm:h-kl, asm:h-llp, asm:h
thm:total-zeta-vertex-l	theorem	proved-conditional	asm:h-kl
thm:cochain-cancellation	theorem	proved	-
thm:pointed-euler-exponent	theorem	proved-conditional	asm:h-ky
thm:universal-bass-schur	theorem	proved	-
cor:bass-every-graph	corollary	proved	-
prop:tetrahedron-obstruction	proposition	proved	-
thm:graded-superdeterminant	theorem	proved	-
prop:vertex-l-no-zeros	proposition	proved-conditional	asm:h-kl
prop:parity-only	proposition	proved	-
prop:three-circles	proposition	proved-conditional	asm:h-kl, asm:h-kl-table
thm:incidence-two-schur	theorem	proved	-
thm:berezinian-total	theorem	proved	-
thm:twisted-bass-schur	theorem	proved	-
thm:kraus-ring-positivity	theorem	proved	-
prop:positivity-no-go-scope	proposition	proved	-
obs:graded-determinant-natural	observation	proved	-
num:small-complex-tests	numerical	numerical	-
prop:pure-gauge-full-complex	proposition	proved	-
thm:artin-voltage-decomposition	theorem	proved	-
thm:euler-block-bookkeeping	theorem	proved-conditional	asm:h-ky, asm:h-ky-local

id	kind	derived status	assumptions
thm:quantum-vertex-transfer	theorem	proved-conditional	asm:h-kl, asm:h-ky, asm:h-
prop:coverage-converse	proposition	proved	-
prop:one-sided-rh-variable	proposition	proved-conditional	asm:h-kamber, asm:h-llp
prop:rank-three-dualities	proposition	proved-conditional	asm:h-kl-table, asm:h-lsv
prop:weil-positivity-blocks	proposition	proved	-
num:a2-complex	numerical	numerical	-
num:a2-vertex-spectrum	numerical	numerical	-
num:a2-identity	numerical	numerical	-
num:a2-flow-operators	numerical	numerical	-
num:a2-block-spectra	numerical	numerical	-
num:a2-quantum-twist	numerical	numerical	-
num:a2-voltage-euler	numerical	numerical	-
conj:vertex-collapse-characterisation	conjecture	conjectured	-

1.4 Lockstep rules

The gate (`make check`; `make ci` adds a fresh build) fails unless: every note under `notes/` is named by a shard header and every shard header names an existing note; every evidence script is named by a shard and has a captured output; every theorem-like environment has one label, one claims row, and one status tag equal to the derived status; every definition has one row and appears once; the notation macros are exactly the generated image of `db/notation.tsv` with no duplicate symbols; every provenance quote is found whole and contiguous at its stated locus; and the build has no undefined references or citations.

2 Conventions and Notation

Every symbol below is a macro generated from `db/notation.tsv`; shards may not define macros of their own, and the gate warns whenever a shard writes a reserved control sequence (such as the bare Greek letters) bare instead of through its macro. The table is regenerated by `make regen` and the gate fails if it is stale.

2.1 Standing conventions

1. **Channel normalisation.** A channel Φ is normalised, $\Phi(X) = \frac{1}{D} \sum_i U_i X U_i^\dagger$, so that its leading eigenvalue is 1. The unnormalised sum is $\Sigma = \sum_i \mathcal{E}_i = D\Phi$. Every Ihara–Bass statement is written with Σ or $D\Phi$; the notes that wrote Φ without the $1/D$ are read with this convention.
2. **The size parameter q .** One letter, three roles that never co-occur: the field size in Section 14, the LPS prime in Section 13, and $D - 1$ for graphs and channels. In every role the critical modulus is $\sqrt{q}$. For a channel with D Kraus operators, $q := D - 1$ is a standing identification.
3. **Ramanujan units.** Spectral bounds are reported in adjacency units: a channel eigenvalue is multiplied by $D = q + 1$ and compared with $2\sqrt{q}$. Hastings’ bound $\lambda_H = 2\sqrt{D - 1}/D$ is the same statement in channel units.
4. **The zeta variable is u ,** in $\zeta_X(u)$, $\zeta_T(u)$, $Z_C(u)$ and $\det(1 - uT)$. The ring-length variable of the length flow is L , never $u = t/2$.
5. **Zeros.** A nontrivial zero is $\rho = \sigma + i\gamma$ with σ its real part; β is reserved for the inverse temperature of the Bost–Connes system. The Riemann hypothesis reads $\sigma = \frac{1}{2}$.
6. **Indices.** n is the matrix size in M_n and, in the ring pictures, the number of sites, which for Artin–Schreier curves is the extension degree of $\mathbb{F}_{q^n}$. Walk, word and prime-cycle lengths are ℓ . Counts are N_ℓ .
7. **Column stacking.** $\text{Ad}(A) = \bar{A} \otimes A$ as an $n^2 \times n^2$ matrix, so $\text{Tr Ad}(A) = |\text{Tr } A|^2$.
8. **Non-backtracking operator.** $T(v \otimes |i\rangle) = \sum_{j \neq \bar{i}} \mathcal{E}_i v \otimes |j\rangle$: the label i names the superoperator applied, and the next label may not be its reversal. Word products are composed right to left, $\mathcal{E}_{i_\ell} \cdots \mathcal{E}_{i_1}$, with the cyclic condition $i_{k+1} \neq \bar{i}_k$ for $k \in \mathbb{Z}/\ell$.
9. **Artin–Schreier sign.** $\alpha_i = -\lambda_i(E)$, so that $\mathcal{S}_n = -\sum_i \alpha_i^n$ for all n and $L = \prod_i (1 - \alpha_i u)$, the standard normalisation of the L -function of an exponential sum. The curve count is $N_n = q^n + 1 - \sum_i \alpha_i^n$.
10. **Canonical form.** An MPS is in (left) canonical form when $\sum_k A_k^\dagger A_k = 1$; then the transfer matrix has leading eigenvalue 1. The spectral normalisation used in the translation note (leading eigenvalue 1, rest inside the unit disc) is a consequence, not the definition.
11. **Status vocabulary.** The book uses only the tags of Section 3 (Definition 4.17). The two-value “established / not established” of the notes and the prior-art verdicts (**SAME**, **RELATED**, **NOT**) are read as **sketched/numerical** versus **open**, and as a provenance vocabulary confined to Section 23, respectively.
12. **Trace.** Tr is the operator trace; the field trace is $\text{tr}_{q^n/q}$.

2.2 Clashes found and resolved

The extraction of 2026-09-12 (`notes/extract/notation-and-definitions.md`) found 146 symbols across the seven notes, with these collisions. Each resolution below is what the table enforces.

- ψ carried five meanings: Chebyshev function, additive character, Lévy exponent, digamma, MPS ring state. Now: ψ is Chebyshev; e_q the additive character; κ_σ the Lévy exponent; (Γ'/Γ) the digamma; Ψ_n the ring state.
- T carried five: the non-backtracking operator, the length-flow generator, the curve-zeta variable, the Heisenberg–Weyl operator, the Hecke operator. Now: T is the non-backtracking operator; $\mathcal{G}$ the generator; u the variable; $T(u, v)$ the Heisenberg–Weyl operator (only with two arguments); A_q the Hecke adjacency operator.
- ρ carried three: zero, density matrix, twist. Now: ρ is a zero; X a channel input; ϱ a twisting representation.
- β carried three: real part of a zero, inverse temperature, Deligne weight. Now: σ , β , w .
- ξ was both the completed zeta and the correlation length, within two pages. Now: ξ is the completed zeta; ℓ_k^c the correlation length; ϑ_k the phase.
- S was the scattering symbol, the exponential sum and the start map. Now: S , $\mathcal{S}_n$, $\mathbf{S}$.
- J was the Artin–Schreier range, the reversal operator and a splitting matrix. Now: J (italic), J (sans-serif), $\mathbf{J}$ (bold).
- D was the Kraus count, the dilation unitary, a diagonal degree matrix and the degree operator. Now: D , $\mathbf{D}$, and $\mathcal{D}(u)(u)$; the diagonal degree matrix of a general graph is never needed on a bouquet.
- E was the MPS transfer matrix and the Artin–Schreier transfer matrix (the same concept, kept as E) and the superoperators of a Kraus family (now $\mathcal{E}_i$, a different glyph).
- W, V : the edge and vertex spaces of the non-backtracking construction are $\mathcal{W}, \mathcal{V}$; $W(g)$ and $W_\pm$ are the Weil representation and its blocks; $|V|$ is a vertex count.
- A : the adjacency matrix A ; Kraus operators A_i ; the LPS Hecke adjacency A_q .
- F : the Fourier transform $\mathcal{F}$; Frobenius Fr ; the numerical window sum F ; Weil’s zero-side function F_W .
- M : a generic matrix M ; the chirp $\mathbf{M}$.
- σ : the real part of s ; the reversal involution is now ι . Σ is only ever $\Sigma = D\Phi$.
- α : Frobenius eigenvalues carry a subscript, α_i ; a quaternion is bare α .
- m : the number of Kraus pairs; walk length is ℓ . N : $\dim \mathcal{V}$; counts are N_ℓ .
- μ : an edge eigenvalue; the Bost–Connes isometries are $\mathfrak{m}_n$. G : a group; the window width is G_w . θ : the normal-basis generator θ ; phases are ϑ_k . s : the complex variable; a square root of $q \bmod p$ is s_q .

2.3 Notation table

symbol	meaning	domain	where
ϱ_b	prescribed stationary density in a finite BC bond model	finite BC states	RC-02E
ϱ^{res}	diagonal matrix extension of a finite BC phase marginal	finite BC states	RC-02E
P	local parity operator $x \rightarrow -x$ on $l\{2(F_p)$	finite Weil model	RC-02E
$\mathfrak{C}$	cone of stationary covariant finite GKLS generators	finite channels	RC-02E
C	unnormalised Choi matrix of a linear map	finite channels	RC-02E
$\mathcal{D}$	completely depolarizing channel $D_d(X) = \text{Tr}(X) I/d$	finite channels	RC-02E
Tr	trace	general	RC-01
Ad	adjoint action, $\text{Ad}(A): X \rightarrow A X A^\dagger$	channels	RC-01
Re	real part	general	RC-01
Im	imaginary part	general	RC-01
M_n	$n \times n$ complex matrices with the Hilbert-Schmidt inner product	channels	RC-01
$\mathcal{V}$	vertex space of the non-backtracking construction ($V = M_n$ in the channel case), $N = \dim V$	quantum Ihara	RC-08
$\mathcal{W}$	edge space $V(x) C\{D$	quantum Ihara	RC-08
T	non-backtracking (Hashimoto) edge operator, classical on directed edges or quantum on the edge space	graphs, quantum Ihara	RC-02
$\mathcal{G}$	generator of the length flow whose trace is the prime measure, eigenvalues 1, ρ , -2, -4, ...	Riemann channel	RC-03
$\mathcal{E}_i$	i -th superoperator of a family ($E_i = \text{Ad}(A_i)$ in the Kraus case)	quantum Ihara	RC-08
A_i	i -th Kraus operator	channels	RC-01
U_i	i -th unitary Kraus operator	channels	RC-01
Σ	unnormalised transfer matrix $\sum_i E_i (= D \Phi$ for a unital channel)	channels, MPS	RC-01
Φ	unital channel $(1/D) \sum_i U_i X U_i^\dagger$, or a general channel	channels	RC-01
D	number of Kraus operators (= degree of the bouquet); $D = 2m$	channels	RC-01
m	number of Kraus pairs, $D = 2m$	quantum Ihara	RC-08
ι	fixed-point-free reversal involution on the index set $[D]$, $\bar{i} = \iota(i)$	quantum Ihara	RC-08
$\bar{\cdot}$	reversed index $\iota(i)$	quantum Ihara	RC-08
J	backtracking operator $J(v(x) i\rangle) = E_i v(x) \bar{i}\rangle$	quantum Ihara	RC-08
R	propagate-along-edge map $R: W \rightarrow V$	quantum Ihara	RC-08
S	start-any-edge map $S: V \rightarrow W$	quantum Ihara	RC-08
$\mathcal{A}(u)$	deformed adjacency $u \sum_i E_i (1 - u\{2 E_{\{\bar{i}\}} E_i)\{-1\}$	quantum Ihara	RC-08
$\mathcal{D}(u)$	deformed degree $u\{2 \sum_i E_{\{\bar{i}\}} E_i (1 - u\{2 E_{\{\bar{i}\}} E_i)\{-1\}$	quantum Ihara	RC-08
ζ_T	zeta function $\det(1 - uT)\{-1\}$ of a non-backtracking operator	quantum Ihara	RC-08

symbol	meaning	domain	where
ζ_Φ	quantum Ihara zeta of a unital channel with unitary Kraus operators	quantum Ihara	RC-02
ζ_X	Ihara zeta of a graph X	graphs	RC-02
A	adjacency matrix of a graph	graphs	RC-02
$A.$	adjacency operator of the LPS Cayley graph for norm-q quaternions (Hecke operator)	Weil-LPS	RC-05
q	size parameter: field size (Artin-Schreier), LPS prime (Weil-LPS), and $D - 1 = \text{degree} - 1$ for graphs and channels	all	RC-01
p	a rational prime; the modulus of the Weil representation in RC-05	all	RC-01
λ_2	second largest eigenvalue (or singular value) of a channel or graph, normalised to leading eigenvalue 1	expanders	RC-02
λ_H	Hastings bound $2 \sqrt{(D-1)/D}$	expanders	RC-02
μ	eigenvalue of the edge operator T; pairs $\mu \mu' = q$ with an adjacency eigenvalue λ	graphs	RC-02
ℓ	length of a walk, word or prime cycle	graphs, quantum Ihara	RC-02
$N.$	closed non-backtracking walk count of length ℓ , or point count over F_{q^ℓ}	graphs, curves	RC-02
P	prime cycle of a graph (primitive closed non-backtracking walk up to rotation)	graphs	RC-02
Fr	Frobenius	curves	RC-06
ϱ	representation twisting a zeta (holonomy, Artin-Ihara twist, adjoint representation)	graphs, Deligne	RC-07
X	density matrix or general operator input of a channel	channels	RC-01
ρ	nontrivial zero of zeta, $\rho = \sigma + i \gamma$	number theory	RC-01
σ	real part of s (or of a zero)	number theory	RC-01
$\gamma.$	imaginary part of the n -th zero	number theory	RC-01
ξ	completed zeta $\xi(s) = (1/2) s (s-1) \pi^{\hat{s}/2} \Gamma(s/2) \zeta(s)$	number theory	RC-04
S	Lax-Phillips scattering symbol $S(\tau) = \xi(1 - 2i\tau)/\xi(1 + 2i\tau)$	Riemann channel	RC-04
τ	Mellin variable on the dilation group; dilation by $e^{\hat{t}}$ acts as $e^{\hat{t}}$	Riemann channel	RC-04
K_S	model space $H^{\hat{2}}_2 \text{ minus } S H^{\hat{2}}_2$	Riemann channel	RC-04
$Z(t)$	compressed dilation semigroup $P_{\{K_S\}} e^{\hat{t}}$ restricted to K_S	Riemann channel	RC-04
Λ	von Mangoldt function	number theory	RC-01
ψ	Chebyshev function $\psi(x) = \sum_{n \leq x} \Lambda(n)$	number theory	RC-03
e_q	additive character of F_q composed with the trace	curves	RC-06
$\kappa.$	Bost-Connes Levy exponent $\kappa_\sigma(\tau) = \log \zeta(\sigma - i\tau) - \log \zeta(\sigma)$	Bost-Connes	RC-04
$\nu.$	von Mangoldt jump measure $\nu_\sigma = \sum p^{\hat{k}} \{-k\sigma\} / k \delta_{\{k \log p\}}$	Bost-Connes	RC-04
β	inverse temperature of the Bost-Connes system	Bost-Connes	RC-04
L	ring length variable of the length flow (prime powers sit at $L = \log n$)	Riemann channel	RC-03

symbol	meaning	domain	where
F	windowed zero sum $F(u)$ of the ring-norm test	Riemann channel	RC-04
E	transfer matrix of a matrix product state (the Artin-Schreier transfer matrix is the instance of RC-06)	MPS, curves	RC-02
Ψ	periodic matrix product state on n sites	MPS	RC-02
$\langle \Psi \Psi \rangle$	ring norm of the periodic n -site state, $= \text{Tr } E^{\hat{\cdot}}_n$	MPS	RC-02
ℓ^c	k -th correlation length of a transfer matrix	MPS	RC-03
h	entropy per site (log of the leading eigenvalue)	MPS	RC-03
$\mathbb{F}_q$	finite field with q elements	curves	RC-01
$\mathbb{F}_p$	finite field with p elements	curves, LPS	Weil-RC-01
$\mathbb{F}_{q^n}$	extension of degree n	curves	RC-06
$\mathcal{S}$	exponential sum $S_n(g) = \sum_{x \in \mathbb{F}_{q^n}} e_q(\text{Tr } g(x))$	curves	RC-06
J	range of the quadratic Artin-Schreier polynomial, $\dim E = q^{\hat{\cdot}} J$	curves	RC-06
α	Frobenius eigenvalues (roots of the L-polynomial), $ \alpha = \sqrt{q}$	curves	RC-06
SL_2	special linear group	Weil-LPS	RC-05
PSL_2	projective special linear group	Weil-LPS	RC-05
W	Weil representation $W(g)$ of $SL_2(\mathbb{F}_p)$ on $\ell^{\hat{\cdot}}_2(\mathbb{F}_p)$	Weil-LPS	RC-05
$W_{\pm}$	even and odd blocks of the Weil representation, dimensions $(p \pm 1)/2$	Weil-LPS	RC-05
Φ	Weil-LPS channel for the LPS prime q (with superscript pm for the block)	Weil-LPS	RC-05
α	integer quaternion of norm q in the LPS normalisation	Weil-LPS	RC-05
g_{α}	image of the quaternion α in $SL_2(\mathbb{F}_p)$	Weil-LPS	RC-05
Z	clock operator on $\ell^{\hat{\cdot}}_2(\mathbb{F}_p)$	Weil-LPS	RC-05
X	shift operator on $\ell^{\hat{\cdot}}_2(\mathbb{F}_p)$	Weil-LPS	RC-05
$\mathcal{F}$	Fourier transform on $\ell^{\hat{\cdot}}_2(\mathbb{F}_p)$	Weil-LPS	RC-05
M	chirp operator $\text{diag } e_p(x^{\hat{\cdot}}/2)$	Weil-LPS	RC-05
D	dilation unitary on $\ell^{\hat{\cdot}}_2(\mathbb{F}_p)$	Weil-LPS	RC-05
I	splitting matrix $I = ((x, y), (y, -x))$ with $x^{\hat{\cdot}}_2 + y^{\hat{\cdot}}_2 + 1 = 0 \pmod{p}$	Weil-LPS	RC-05
J	splitting matrix $J = ((0, 1), (-1, 0))$	Weil-LPS	RC-05
$T(u, v)$	Heisenberg-Weyl operator $X^{\hat{\cdot}}_u Z^{\hat{\cdot}}_v$	Weil-LPS	RC-05
c	pointwise weight scalar of a twisted coefficient system ($ \text{eigenvalue} = c^{\hat{\cdot}}_{\ell}$)	Deligne	RC-07
L	L-function (of a curve with coefficients, or Artin-Ihara)	graphs, curves	RC-02
(Γ'/Γ)	digamma function in the archimedean term of the explicit formula	number theory	RC-04
w	weight of a pure coefficient system ($ \text{Frobenius eigenvalue} = q^{\hat{\cdot}}_{w/2}$)	Deligne	RC-07
$\text{tr}_{q^n/q}$	field trace from $\mathbb{F}_{q^n}$ to $\mathbb{F}_q$	curves	RC-06
Z_C	zeta function of a curve C over $\mathbb{F}_q$, $\exp \sum_n N_n u^{\hat{\cdot}}_n / n$	curves	RC-06

symbol	meaning	domain	where
G_w	Gaussian window width of the ring-norm test	Riemann channel	RC-04
F_W	zero-side function $\sum_{\rho} e^{\frac{1}{2}(\rho - 1/2)u}$ whose positive-definiteness is Weil's criterion	number theory	RC-03
$\vartheta.$	k-th oscillation frequency of a transfer matrix, $\lambda_k = \exp(-1/\ell \{c_k + i \vartheta_k\})$	MPS	RC-03
θ	generator of a normal basis of $F_{\{q\}^n}$	curves	RC-06
s_q	a square root of q modulo p	Weil-LPS	RC-05
$m.$	Bost-Connes isometry μ_n	Bost-Connes	RC-04
$U.$	Shor's unitary $x \rightarrow ax \bmod N$, the level-N Galois symmetry of Bost-Connes	Bost-Connes	RC-04
$\hat{g}$	test window in the explicit formula (Fourier transform g)	number theory	RC-04
t	semigroup time in $Z(t)$; dilation by $e^{\hat{t}}$	Riemann channel	RC-04
u	the zeta variable, in $\zeta_X(u)$, $\zeta_T(u)$, $Z_C(u)$, $\det(1 - uT)$	all	RC-01
$\ell^2(\cdot)$	Hilbert space of square-summable functions on a set	general	RC-01
$\mathcal{O}_{Dw}$	Dwork's p-adic transfer operator, multiplication by the splitting function then decimation	curves	RC-06
dec_q	decimation $x^{\hat{m}} \rightarrow x^{\{m/q\}}$ on p-adic power series	curves	RC-06
ℓ -adic	ell-adic (cohomology); this ell is not a walk length	Deligne	RC-06
$\mathcal{X}$	generator of the geodesic flow on $\Gamma \backslash G$, $X = H/2$ through left-invariant fields; $(e^{\hat{t}}\{-tX\}f)(m) = f(\phi_{\{-t\}} m)$	Selberg	RC-09B
H	left-invariant field of $H = \text{diag}(1, -1)$	Selberg	RC-09B
E	left-invariant field of $E = ((0,1), (1,0))$	Selberg	RC-09B
W	left-invariant field of $W = ((0,1), (-1,0))$, the compact direction	Selberg	RC-09B
Ω	Casimir $(1/4)(H^{\hat{2}} + E^{\hat{2}} - W^{\hat{2}})$	Selberg	RC-09B
Δ	positive hyperbolic Laplacian $-y^{\hat{2}}(d_x^{\hat{2}} + d_y^{\hat{2}})$	Selberg	RC-09B
$\lambda.$	j-th Laplace eigenvalue, $\lambda_j = 1/4 + r_j^{\hat{2}}$	Selberg	RC-09B
$r.$	spectral parameter of λ_j ; $r_0 = i/2$ for the constant mode, $r_j = i \nu_j$ for exceptional eigenvalues	Selberg	RC-09B
Z_{Sel}	Selberg zeta $\prod_{\gamma} \prod_{k \geq 0} (1 - e^{\hat{t}}\{-(s+k) \ell_{\gamma}\})$	Selberg	RC-09B
ζ_R	Ruelle zeta $\prod_{\gamma} (1 - e^{\hat{t}}\{-s \ell_{\gamma}\}) = Z_{\text{Sel}}(s)/Z_{\text{Sel}}(s+1)$	Selberg	RC-09B
P_{γ}	linearised Poincare map of the closed geodesic γ	Selberg	RC-09B
Tr^b	Guillemin flat trace	Selberg	RC-09B
ς	Laplace variable of the flat trace (the resonance variable), never the Selberg variable s	Selberg	RC-09B
Λ_{fl}	Laplace transform of the flat trace, $\int_0^{\infty} e^{\hat{t}}\{-\text{varsigma } t\} \text{Tr}\{\text{flat } e^{\hat{t}}\}\{-tX\} dt$	Selberg	RC-09B
D_{tow}	the tower $\prod_{j \geq 1} Z_{\text{Sel}}(\text{varsigma} + j)$, the flat dynamical determinant	Selberg	RC-09B

symbol	meaning	domain	where
$\mathcal{B}_0$	nonconstant first band $\{-1/2 + i r_j : j \geq 1\}$	Selberg	RC-09B
φ	scattering determinant of $\text{PSL}(2, \mathbb{Z})$, $\sqrt{\pi} \Gamma(s-1/2) \zeta(2s-1) / (\Gamma(s) \zeta(2s))$	modular	RC-09B
m_{sc}	scattering multiplier $-(1/2)(\text{varphi}'/\text{varphi})(1/2 + i r)$	modular	RC-09B
C_{cusp}	inverse Fourier transform of the scattering multiplier	modular	RC-09B
P_+	positive-time prime measure $\sum_n \Lambda(n)/n \delta(t - 2 \log n)$	modular	RC-09B
P_{ch}	prime measure $\sum_n \Lambda(n) n^{\hat{\cdot}} \{-1/2\} \delta(t - 2 \log n)$ of the Riemann-channel note	modular	RC-09B
A_{arch}	archimedean-plus-boundary distribution $1/2 - 1/(4 \sinh(t /2))$ away from zero	modular	RC-09B
S_{triv}	trivial sub-multiset of the spectrum, removed before rescaling	Weil positivity	RC-08B
r	critical radius; $\sqrt{D-1}$ in the unitary Kraus case	Weil positivity	RC-08B
$\mathcal{A}_{\text{ret}}$	retained multiset $\text{spec}(X)$ minus S_{triv} (algebraic multiplicities)	Weil positivity	RC-08B
$n.$	rescaled trace sequence $n_l = r^{\hat{\cdot}} \{-1\} (\text{Tr } X^{\hat{\cdot}} l - \sum \{S_{\text{triv}}\} \mu^{\hat{\cdot}} l)$, $n_{-1} = \text{conj}(n_l)$	Weil positivity	RC-08B
W	Weil (Toeplitz) form $\sum_{l,m} c_l \text{conj}(c_m) \nu_{\{-l-m\}}$	Weil positivity	RC-08B
M	mode-pairing form $\sum_{\mu} f_c(\mu/r) \text{conj}(f_c(J(\mu)/r))$	Weil positivity	RC-08B
J	reflection in the critical circle, $J(\mu) = r^{\hat{\cdot}} \{2/\text{conj}(\mu)\}$; fixed points $ \mu = r$	Weil positivity	RC-08B
$\mathcal{L}$	Lindblad-type generator $K x + x K^{\hat{\cdot}} \dagger + \sum_j R_j x R_j^{\hat{\cdot}} \dagger$ on M_n	Weil positivity	RC-08B
F	rescaled continuous trace $F(t) = e^{\hat{\cdot}} \{-at\} (\text{Tr } e^{\hat{\cdot}} \{tL\} - \text{trivial part})$, $F(-t) = \text{conj}(F(t))$	Weil positivity	RC-08B
a	critical real part, the line $\text{Re } \rho = a$; $a = -1/4$ for the Riemann channel	Weil positivity	RC-08B
$f.$	polynomial $f_c(z) = \sum_l c_l z^{\hat{\cdot}} l$ of a finitely supported coefficient vector c	Weil positivity	RC-08B
$R.$	reflection in the line $\text{Re } \rho = a$, $R_a(\rho) = 2a - \text{conj}(\rho)$	Weil positivity	RC-08B
ϖ	generic eigenvalue (mode) of a continuous transfer generator L	Weil positivity	RC-08C
$\varkappa$	eigenvalue of the unnormalised transfer matrix Σ (the prover's α); pairs with the two roots of $\mu^{\hat{\cdot}} \{2 - \alpha \mu + q\}$	Weil positivity	RC-08B
str	supertrace: trace on the even part minus trace on the odd part of a $\mathbb{Z}_2$ -graded space	graded	RC-04B
comb	prime comb $\sum_{n \geq 2} \Lambda(n) \delta(u - \log n)$ on $(0, \infty)$	number theory	RC-04B
Tr_{dist}	distributional trace of the compressed semigroup, $\sum_{\rho} e^{\hat{\cdot}} \{-\text{conj}(\rho) t/2\}$, a definition by fiat	Riemann channel	RC-04B
ν_{net}	net signed multiplicity $m(\lambda, +) - m(\lambda, -)$ of a graded spectral datum	graded	RC-04B
$\mathfrak{B}$	Riemann graded bond $C_+ (+) (K_S (+) l^{\hat{\cdot}} \{2(N_{\geq 1})\})_-$	Riemann channel	RC-04B

symbol	meaning	domain	where
G	graded generator 0 (+) B (+) $\text{diag}(-(k+1/2))$ on the Riemann graded bond	Riemann channel	RC-04B
B	generator of the compressed semigroup, $Z(t) = e^{\hat{\{tB\}}}$	Riemann channel	RC-04B
$\mathcal{L}_{\text{vac}}$	vacuum-decay Lindbladian on $B(C (+) H)$: no-event part 0 (+) B, jump $ \text{vac}\rangle\langle j $	channels	RC-04C
vac	vacuum vector spanning the even line of the vacuum-decay model	channels	RC-04C
Ω_{vac}	vacuum density $ \text{vac}\rangle\langle\text{vac} $, the stationary state of the vacuum-decay Lindbladian	channels	RC-04C
$\mathcal{L}^{\text{pc}}$	prime-chain detailed-balance Lindbladian at inverse temperature β , jumps μ_p and $\mu_p\hat{\{}}\dagger$	Bost-Connes	RC-04C
ϱ^G	Bost-Connes Gibbs density $N\hat{\{}}\{-\beta\}/\zeta(\beta)$ on $l\hat{\{}}2(N)$	Bost-Connes	RC-04C
$\mathcal{S}_{\text{ent}}$	von Neumann entropy of a bond density (entanglement entropy of the half-line in the cMPS reading)	Bost-Connes	RC-04C
TFD.	thermofield-double purification of the Gibbs density	Bost-Connes	RC-04C
$\mathbb{E}$	super-transfer matrix of an Artin-Schreier curve: even block $E_0 (+) (1)$, odd block the Frobenius blocks	curves	RC-06B
F	corrected Frobenius block of the character sum $S_n(g)$, $L(g, T) = \det(1 - T F_g)$	curves	RC-06B
Q°	ring quadratic form $\sum_j a_j \sum_i x_i x_{\{i+j\}}$ on $F_q\hat{\{}}n$ (indices mod n), matrix $P(S)$	curves	RC-06B
Q^{tr}	trace quadratic form $\text{tr}(g(x))$ in normal-basis coordinates, matrix $C P(S)$	curves	RC-06B
P	symmetrised coefficient Laurent polynomial $P_g(z) = \sum_j (a_j/2)(z^{\hat{\{}}j} + z^{\hat{\{}}\{-j\}})$	curves	RC-06B
S	cyclic shift $(S x)_i = x_{\{i+1\}}$ on $F_q\hat{\{}}n$, the Frobenius in normal-basis coordinates	curves	RC-06B
$\mathcal{R}$	radical $\ker P_g(S)$ of the ring form on $F_q\hat{\{}}n$	curves	RC-06B
δ	radical sign $\det(S \text{ restricted to the radical } R_n)$ in $\{+1, -1\}$	curves	RC-06B
h	period parameter: the smallest power of q strictly greater than the order of vanishing of $z\hat{\{}}J P_g(z)$ at -1	curves	RC-06B
d	rank deficiency of the ring form, $= \dim \ker(M_g\hat{\{}}n - 1)$, $= \log_q(S_n \hat{\{}}2/q\hat{\{}}n)$	curves	RC-06B
M	symplectic matrix in $\text{Sp}(2J, F_q)$ implemented by the transfer unitary $E_g/\sqrt{q}$	curves	RC-06B
X	centred Weyl operator $X_{\{u,v\}} s\rangle = e_q(v.(s + u/2)) s + u\rangle$ on $l\hat{\{}}2(F_q\hat{\{}}J)$	curves	RC-06B
$\mathcal{U}$	transfer unitary $E_g/\sqrt{q}$	curves	RC-06B
Sp	symplectic group	curves	RC-06B
D	transfer determinant $D_g(T) = \det(1 - T E_g)$	curves	RC-06B
$\mathcal{Z}$	graded geodesic determinant (total ζ) of a complex with specified geodesic data	complexes	RC-02D
sdet	superdeterminant (Berezinian) of an even operator on a Z_2 -graded space	graded, complexes	RC-02D

3 Definitions

Each definition appears exactly once in the book. The column `merged_from` of `db/definitions.tsv` lists the notes and sections whose wordings were merged; where the wordings differed in substance, the difference is stated in the definition.

3.1 The two sides

Definition 3.1 (Side A and side B). For a zeta function written as $\exp \sum_{\ell \geq 1} N_\ell u^\ell / \ell$ with nonnegative counts N_ℓ , *side A* is the gas of rings: the counts, their Euler product over primitive closed orbits, and the product state they form. *Side B* is a transfer operator whose power sums reproduce the counts, $N_\ell = \text{Tr } T^\ell$ (or a semigroup $\text{Tr } e^{\text{L}\mathcal{G}}$ in continuous length); its eigenvalues are the poles and zeros.

Definition 3.2 (The one identity). For any matrix M with eigenvalues $\lambda_1, \dots, \lambda_N$, $1/\det(1-uM) = \prod_i (1 - \lambda_i u)^{-1} = \exp \left(\sum_{\ell \geq 1} \text{Tr}(M^\ell) u^\ell / \ell \right)$. The middle term is a bosonic partition function with one mode per eigenvector; the right term a sum over closed walks with the $1/\ell$ of a ring's rotation symmetry.

Definition 3.3 (Small matrix). For a shift or permutation with fixed-point counts $\text{fix}(\ell)$ of its ℓ -th power, a *small matrix* is any matrix M_{small} with $\text{Tr } M_{\text{small}}^\ell = \text{fix}(\ell)$ for all ℓ . It exists exactly when the zeta function is rational, and its eigenvalues are not those of the permutation matrix.

Definition 3.4 (RH analogue, Ramanujan property). A side-B operator with leading eigenvalue $\lambda_{\max}$ (or leading growth q per step) satisfies the *RH analogue* when every other nonzero eigenvalue has modulus $\sqrt{\lambda_{\max}}$ (respectively $\sqrt{q}$). In counting terms the fluctuation of the counts is the square root of the leading term. For graphs this is the Ramanujan property; for the Riemann zeta it is the Riemann hypothesis; for an MPS it is one correlation length equal to $2/h$.

3.2 Graphs

Definition 3.5 (Non-backtracking walk, prime cycle). A walk in a graph is non-backtracking if it never immediately retraces an edge; closed in the strong sense means the last step is not the reverse of the first. A *prime cycle* P is a closed non-backtracking walk that is not a repetition of a shorter one, taken up to choice of starting point; $\ell(P)$ is its length. On a bouquet with $D/2$ loops the directed edges are $[D]$ and a closed non-backtracking walk is a cyclically reduced word.

Definition 3.6 (Ihara zeta). For a finite graph X with N_ℓ closed non-backtracking walks of length ℓ , $\zeta_X(u) = \exp \left(\sum_\ell N_\ell u^\ell / \ell \right) = \prod_P (1 - u^{\ell(P)})^{-1}$.

Definition 3.7 (Hashimoto operator). The *Hashimoto* (non-backtracking edge) operator of a graph is the matrix T on directed edges with $T_{ef} = 1$ when f can follow e without backtracking; then $N_\ell = \text{Tr } T^\ell$ and $\zeta_X(u) = \det(1 - uT)^{-1}$.

Definition 3.8 (Ihara–Bass formula). For a $(q+1)$ -regular graph with adjacency matrix A on $|V|$ vertices and $|E|$ undirected edges, $\det(1 - uT) = (1 - u^2)^{|E| - |V|} \det(1 - uA + qu^2)$. Each adjacency eigenvalue λ produces two eigenvalues μ, μ' of T with $\mu\mu' = q$, $\mu + \mu' = \lambda$; the eigenvalue $q+1$ gives $\mu \in \{q, 1\}$.

Definition 3.9 (Ramanujan graph). A $(q+1)$ -regular graph is *Ramanujan* when every adjacency eigenvalue other than $\pm(q+1)$ satisfies $|\lambda| \leq 2\sqrt{q}$, the spectral radius of the universal covering tree. Equivalently all nontrivial μ satisfy $|\mu| = \sqrt{q}$, which is the RH analogue for ζ_X .

Definition 3.10 (Artin–Ihara L -function). For a graph with a representation ϱ of its fundamental group assigned to closed walks (a holonomy), $L(u, \varrho) = \prod_P \det(1 - \varrho(P)u^{\ell(P)})^{-1}$, where the product is over prime cycles. For a bouquet the fundamental group is free and ϱ is a choice of one operator per loop.

3.3 Channels and matrix product states

Definition 3.11 (Channel, Kraus operators). A *channel* on M_n is a completely positive trace-preserving map $\Phi(X) = \sum_i A_i X A_i^\dagger$ with $\sum_i A_i^\dagger A_i = 1$; the A_i are its Kraus operators. It is *unital* when $\Phi(1) = 1$ and *mixed-unitary* when $A_i = D^{-1/2} U_i$ with unitary U_i , in which case $\Phi(X) = \frac{1}{D} \sum_i U_i X U_i^\dagger$ and the family is taken closed under adjoint.

Definition 3.12 (Transfer matrix of an MPS). For a matrix product state with local tensors $A_1, \dots, A_d \in M_n$ (physical dimension d , bond dimension n), the *transfer matrix* is $E = \sum_i \text{Ad}(A_i) = \Sigma$, acting on M_n .

Definition 3.13 (Ring norm). The *ring norm* of the periodic n -site state Ψ_n built from the tensors is $\langle \Psi_n | \Psi_n \rangle = \text{Tr } E^n$; the zeta function of the state is the generating function $\exp \sum_n \langle \Psi_n | \Psi_n \rangle u^n / n = \det(1 - uE)^{-1}$.

Definition 3.14 (Canonical form). An MPS is in (left) canonical form when $\sum_k A_k^\dagger A_k = 1$, i.e. when E is the transfer matrix of a channel; then E has leading eigenvalue 1 and the remaining eigenvalues lie in the closed unit disc.

Definition 3.15 (Correlation length). In canonical form write the subleading eigenvalues as $\lambda_k = e^{-1/\ell_k^c + i\vartheta_k}$; the ℓ_k^c are the correlation lengths and the ϑ_k the phases. The entropy per site is $h = \log \lambda_{\max}$ of the unnormalised transfer matrix.

Definition 3.16 (Quantum expander). A channel with D Kraus operators is a (n, D, λ) *quantum expander* when 1 is its unique fixed point (up to scale) and every other eigenvalue has modulus at most $\lambda < 1$.

Definition 3.17 (Ramanujan quantum expander). A quantum expander is *Ramanujan* when $\lambda_2 \leq \lambda_H$, where, in Hastings' words, $\lambda_H = \frac{2\sqrt{D-1}}{D}$. (provenance `prov:hastings07-bound`); in adjacency units, $D\lambda_2 \leq 2\sqrt{D-1}$.

Definition 3.18 (Quantum Hashimoto operator, quantum Ihara zeta). For superoperators $\mathcal{E}_1, \dots, \mathcal{E}_D$ on $\mathcal{V}$ and a fixed-point-free reversal ι on $[D]$, the *quantum Hashimoto operator* is T on $\mathcal{W} = \mathcal{V} \otimes \mathbb{C}^D$, $T(v \otimes |i\rangle) = \sum_{j \neq \iota i} \mathcal{E}_i v \otimes |j\rangle$, and $\zeta_T(u) = \det(1 - uT)^{-1}$. For a mixed-unitary channel, $\mathcal{E}_i = \text{Ad}(U_i)$ with $U_{\bar{i}} = U_i^\dagger$, and the zeta is written ζ_Φ : the *quantum Ihara zeta* of the channel. It is the Artin–Ihara L -function of the bouquet twisted by $\varrho = \text{Ad} \circ U$.

4 Definitions, Continued

Continuation of Section 3; split for length only.

4.1 The Riemann channel

Definition 4.1 (Scattering symbol). With $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and τ the Mellin variable on the dilation group, the *scattering symbol* of the modular surface is $S(\tau) = \xi(1-2i\tau)/\xi(1+2i\tau)$.

Definition 4.2 (Inner function, model space, compressed semigroup). A function analytic and bounded by 1 in the lower half plane with unimodular boundary values is *inner*. For inner S the *model space* is $K_S = H^2 \ominus SH^2$ and the *compressed semigroup* is $Z(t) = P_{K_S} e^{it\tau}|_{K_S}$, $t > 0$.

Definition 4.3 (Riemann channel). The *Riemann channel* is the completely positive semigroup $X \mapsto Z(t)^* X Z(t)$ on $B(K_S)$, whose single Kraus operator is the compressed semigroup of the scattering symbol. Its modes are the zeros of ζ and its distributional trace is the prime measure; the name is also used loosely for the whole side-A/side-B package of the Riemann zeta.

Definition 4.4 (Weil explicit formula). For a test window $\hat{g}$ with transform g , $\sum_{\gamma} \hat{g}(\gamma) = \hat{g}(\frac{i}{2}) + \hat{g}(-\frac{i}{2}) - g(0) \log \pi + \frac{1}{2\pi} \int \hat{g}(r) \operatorname{Re}(\Gamma'/\Gamma)(\frac{1}{4} + \frac{ir}{2}) dr - 2 \sum_n \frac{\Lambda(n)}{\sqrt{n}} g(\log n)$, the sum on the left over the ordinates of the nontrivial zeros.

Definition 4.5 (Bost–Connes system, Lévy exponent). The *Bost–Connes system* is the C^* -dynamical system generated by the phase operators $e(a/b)$ and the isometries $\mathbf{m}_n$ on $\ell^2(\mathbb{N})$, with time evolution scaling $\mathbf{m}_n$ by n^{it} ; its $\operatorname{KMS}_{\beta}$ states for $\beta > 1$ have partition function $\zeta(\beta)$ and $\beta = 1$ is its phase transition. Its *Lévy exponent* at real part σ is $\kappa_{\sigma}(\tau) = \log \zeta(\sigma - i\tau) - \log \zeta(\sigma)$, the cumulant generating function of the zeta distribution, with jump measure ν_{σ} .

Definition 4.6 (Matrix product operator over the prime chain). Writing $\ell^2(\mathbb{N}) = \bigotimes_p$ (oscillator at p), an operator is a *matrix product operator over the prime chain* when it is a contraction of one local tensor per prime with a bond space carried between consecutive primes; for $e(a/b)$ the bond space is $\mathbb{Z}/b$ and the local transfer map is $r \mapsto p^k r$.

Definition 4.7 (Hilbert–Pólya operator). On the subleading sector of the length flow write $\mathcal{G} = -\frac{1}{2} + iH$; H is the *Hilbert–Pólya operator*, with eigenvalues $\gamma_n - i(\sigma_n - \frac{1}{2})$. It is not the entanglement Hamiltonian of any state.

Definition 4.8 (Weil positivity). *Weil’s criterion*: the Riemann hypothesis holds iff $F_W(u) = \sum_{\rho} e^{(\rho-1/2)u}$ is positive-definite on the dilation group, i.e. iff its spectral measure $\sum_{\rho} \delta_{t_{\rho}}$ is a positive measure on the real line.

4.2 Weil–LPS channels

Definition 4.9 (LPS generators). For a prime $q \equiv 1 \pmod{4}$, the *LPS generators* are the $q+1$ integer quaternions $\alpha = a_0 + a_1 i + a_2 j + a_3 k$ of norm q with $a_0 > 0$ odd and a_1, a_2, a_3 even; they generate a $(q+1)$ -regular tree, and their images g_{α} in $\operatorname{PSL}_2(\mathbb{F}_p)$ (for q a square mod p , after division by s_q) are the generators of the LPS Cayley graph.

Definition 4.10 (Weil representation). The *Weil representation* W of $\operatorname{SL}_2(\mathbb{F}_p)$ on $\ell^2(\mathbb{F}_p)$ is the projective unitary representation characterised by $W(g) T(u, v) W(g)^{\dagger} \propto T(g(u, v))$ for the Heisenberg–Weyl operators $T(u, v) = X^u Z^v$; it is generated by $\mathcal{F}$, $\mathbf{M}$ and $\mathbf{D}$ and splits into the blocks $W_{\pm}$ of dimensions $(p \pm 1)/2$ on which -1 acts as a scalar.

Definition 4.11 (Weil–LPS channel). For admissible $(p; q)$ the *Weil–LPS channel* is $\Phi_q^\pm(X) = \frac{1}{q+1} \sum_\alpha W_\pm(g_\alpha) X W_\pm(g_\alpha)^\dagger$ on $M_{(p\pm 1)/2}$: Harrow’s construction with the Weil representation and LPS generators.

Definition 4.12 (Hecke relation). The adjacency operators A_q of the LPS Cayley graph satisfy $A_q^2 - q1 = A_{q^2}$ and $A_{q_1}A_{q_2} = A_{q_1q_2}$, the tree relations “two steps without backtracking” and multiplicativity; the Weil–LPS channels for different q commute.

4.3 Curves

Definition 4.13 (Artin–Schreier curve, exponential sum). For $g \in \mathbb{F}_q[x]$ the *Artin–Schreier curve* is $y^q - y = g(x)$; its affine point count over $\mathbb{F}_{q^n}$ is $N_n = q^n + \sum_{a \in \mathbb{F}_q^\times} \mathcal{S}_n(ag)$ with the *exponential sum* $\mathcal{S}_n(g) = \sum_{x \in \mathbb{F}_{q^n}} e_q(\text{tr}_{q^n/q} g(x))$. The quadratic family is $g(x) = \sum_{j=0}^J a_j x^{1+q^j}$.

Definition 4.14 (Self-dual normal basis). A basis $\{\theta, \theta^q, \dots, \theta^{q^{n-1}}\}$ of $\mathbb{F}_{q^n}$ is *normal*; Frobenius is then the cyclic shift of coordinates. It is *self-dual* when $\text{tr}_{q^n/q}(ab) = \sum_i a_i b_i$; such bases exist for n odd.

Definition 4.15 (Dwork transfer operator). *Dwork’s operator* is $\mathcal{O}_{\text{Dw}} = \text{dec}_q \circ M_F$ on p -adic power series, multiplication by the splitting function M_F followed by the decimation $\text{dec}_q : x^m \mapsto x^{m/q}$; $\mathcal{S}_n^\times = (q^n - 1) \text{Tr } \mathcal{O}_{\text{Dw}}^n$ and L is its Fredholm determinant.

Definition 4.16 (Pure of weight, big monodromy, slicing, the sign). A coefficient system on a curve over $\mathbb{F}_q$ is *pure of weight* w when every eigenvalue of every holonomy of a prime cycle of length ℓ has modulus $q^{w\ell/2}$; it has *big monodromy* when the only invariants of its even tensor powers are the pairwise contractions by the given pairing. *Slicing* writes the middle block of a variety as the interesting block of a base curve with coefficients from hyperplane sections. *The sign* is the minus sign with which the interesting block enters the point count, so that its eigenvalues are zeros, not poles, of Z_C .

Definition 4.17 (Status vocabulary). The tags **proved**, **sketched**, **numerical**, **cited**, **conjectured**, **open**, **refuted**, **assumed**, with the derived forms **unsupported** and **-conditional**, as fixed in Section 2 of the frontmatter; definitions are **stipulated** or **quoted**.

4.4 The continuous dictionary

Definition 4.18 (Geodesic flow on the unit tangent bundle). For $\Gamma < \text{PSL}_2(\mathbb{R})$ discrete, torsion-free and cocompact, the unit tangent bundle of $\Gamma \backslash \mathbb{H}$ is $\Gamma \backslash G$ with $G = \text{PSL}_2(\mathbb{R})$, and the *geodesic flow* is $\phi_t(\Gamma g) = \Gamma g a_t$, $a_t = \text{diag}(e^{t/2}, e^{-t/2})$. Its generator is $\mathcal{X} = H/2$ through the left-invariant fields H, E, W of the basis H, E, W of $\mathfrak{sl}_2$, and the Koopman operator is $(e^{-t\mathcal{X}} f)(m) = f(\phi_{-t} m)$. Primitive closed geodesics γ have lengths ℓ_γ and opposite orientations are not identified.

Definition 4.19 (Selberg zeta, Ruelle zeta). $Z_{\text{Sel}}(s) = \prod_\gamma \prod_{k \geq 0} (1 - e^{-(s+k)\ell_\gamma})$, in Marklof’s words $Z(s) = \prod_{\{\gamma \in H_*\}} \prod_{m=0}^\infty (1 - e^{-(s+m)\ell_\gamma})$, (provenance **prov:marklof04-selbergzeta**), and the *Ruelle zeta* in the direct-product convention is $\zeta_R(s) = \prod_\gamma (1 - e^{-s\ell_\gamma})$.

Definition 4.20 (Guillemin flat trace). For an Anosov flow with closed orbits γ , primitive periods ℓ_γ and linearised Poincaré maps P_γ , the *flat trace* of the Koopman operator is the distribution on $t > 0$ $\text{Tr}^b e^{-t\mathcal{X}} = \sum_\gamma \sum_{k \geq 1} \ell_\gamma \delta(t - k\ell_\gamma) / |\det(1 - P_\gamma^k)|$ (Dyatlov–Zworski, provenance **prov:dz13-guillemin**). It is not a trace-class Hilbert-space trace.

Definition 4.21 (Flat dynamical determinant, first band). $\Lambda_{\text{fl}}(\varsigma) = \int_0^\infty e^{-\varsigma t} \text{Tr}^\flat e^{-t\mathcal{X}} dt$ for $\text{Re } \varsigma > 0$, and the *tower* $D_{\text{tow}}(\varsigma) = \prod_{j \geq 1} Z_{\text{Sel}}(\varsigma + j)$. *Trace resonances* are the poles of the continued Λ_{fl} . The *nonconstant first band* is $\mathcal{B}_0 = \{-\frac{1}{2} \pm ir_j : j \geq 1\}$, with $\lambda_j = \frac{1}{4} + r_j^2$ the Laplace eigenvalues and $r_0 = i/2$ for the constant mode excluded.

Definition 4.22 (Scattering determinant of the modular group). $\varphi(s) = \sqrt{\pi} \Gamma(s - \frac{1}{2}) \zeta(2s - 1) / (\Gamma(s) \zeta(2s))$, in the source's words In the case when Γ is the modular group $\text{PSL}(2, \mathbb{Z})$, the scattering determinant is given by $\phi(s) = \sqrt{\pi} \frac{\Gamma(s - 1/2)}{\Gamma(s)} \frac{\zeta(2s - 1)}{\zeta(2s)}$, (provenance prov:fjs16-scatt) the *scattering multiplier* is $m_{\text{sc}}(r) = -\frac{1}{2}(\varphi'/\varphi)(\frac{1}{2} + ir)$ and $C_{\text{cusp}} = \mathcal{F}^{-1} m_{\text{sc}}$ with $\mathcal{F}^{-1} b(t) = \frac{1}{2\pi} \int e^{irt} b(r) dr$.

Definition 4.23 (Prime comb measures). $P_+(t) = \sum_{n \geq 2} \frac{\Lambda(n)}{n} \delta(t - 2 \log n)$ and $P_{\text{ch}}(t) = \sum_{n \geq 2} \Lambda(n) n^{-1/2} \delta(t - 2 \log n)$ on $(0, \infty)$, locally finite measures; A_{arch} is the archimedean-plus-boundary distribution of Proposition 25.5, equal to $\frac{1}{2} - 1/(4 \sinh(|t|/2))$ away from $t = 0$.

Definition 4.24 (Trivial set, critical radius, rescaled trace sequence). For a linear operator X on a finite-dimensional complex space, $\text{spec}(X)$ is its spectrum as a multiset with algebraic multiplicities, so $\text{Tr } X^\ell = \sum_{\mu \in \text{spec}(X)} \mu^\ell$. A *trivial set* S_{triv} is a sub-multiset of $\text{spec}(X)$; the *retained multiset* is $\mathcal{A}_{\text{ret}} = \text{spec}(X) \setminus S_{\text{triv}}$ (multiplicities subtracted). For a *critical radius* $r > 0$ the *rescaled trace sequence* is $\mathbf{n}_\ell = r^{-\ell} (\text{Tr } X^\ell - \sum_{\mu \in S_{\text{triv}}} \mu^\ell) = \sum_{\mu \in \mathcal{A}_{\text{ret}}} (\mu/r)^\ell$ for $\ell \geq 0$ (with $0^0 = 1$), extended by $\mathbf{n}_{-\ell} = \overline{\mathbf{n}_\ell}$. A sequence with this symmetry is *positive definite* if $\sum_{\ell, k=0}^L c_\ell \bar{c}_k \mathbf{n}_{\ell-k} \geq 0$ for every L and every $c \in \mathbb{C}^{L+1}$.

Definition 4.25 (Weil form, reflection, mode-pairing form). With the data of Definition 4.24, the *Weil form* of X is the Toeplitz form $W(c) = \sum_{\ell, k} c_\ell \bar{c}_k \mathbf{n}_{\ell-k}$ on finitely supported coefficient vectors c , and $f_c(z) = \sum_\ell c_\ell z^\ell$. The *reflection in the critical circle* is $J(\mu) = r^2/\bar{\mu}$ on $\mathbb{C}^\times$; its fixed points are the circle $|\mu| = r$. When $0 \notin \mathcal{A}_{\text{ret}}$ the *mode-pairing form* is $M(c) = \sum_{\mu \in \mathcal{A}_{\text{ret}}} f_c(\mu/r) \overline{f_c(J(\mu)/r)}$: each mode paired with its reflected partner.

Definition 4.26 (Adjoint pairing and inverse pairing). A family $\mathcal{E}_i = \text{Ad}(B_i)$, $i \in [D]$, with reversal ι is *adjoint-paired* if $B_{\bar{i}} = B_i^\dagger$ and *inverse-paired* if every B_i is invertible and $B_{\bar{i}} = B_i^{-1}$ (so $\mathcal{E}_{\bar{i}} = \mathcal{E}_i^{-1}$, the hypothesis of Corollary 17.3). The Kraus families of Definition 3.18 are the adjoint-paired ones.

Definition 4.27 (Rescaled continuous trace). For a linear operator $\mathcal{L}$ on a finite-dimensional complex space, a trivial set $S_{\text{triv}} \subseteq \text{spec}(\mathcal{L})$ and a real number a , the *rescaled continuous trace* is $F(t) = e^{-at} (\text{Tr } e^{t\mathcal{L}} - \sum_{\varpi \in S_{\text{triv}}} e^{t\varpi}) = \sum_{\varpi \in \mathcal{A}_{\text{ret}}} e^{t(\varpi-a)}$ for $t \geq 0$, extended by $F(-t) = \overline{F(t)}$. A function with this symmetry is *positive definite* if $\sum_{j, k} c_j \bar{c}_k F(t_j - t_k) \geq 0$ for all finite families of real times and complex coefficients. The *reflection in the line* $\text{Re } \varpi = a$ is $R_a(\varpi) = 2a - \bar{\varpi}$.

5 Definitions, Continued: the Phantasm Campaign

Continuation of Section 4; split for length only.

Definition 5.1 (Graded spectral datum, supertrace distribution, net multiplicity). A *graded spectral datum* is a countable multiset of pairs (λ, ϵ) , $\lambda \in \mathbb{C}$, $\epsilon \in \{+1, -1\}$ the *parity*, with multiplicities $m(\lambda, \epsilon) \in \mathbb{N}$, such that $\operatorname{Re} \lambda \leq c$ for some real c and $\sum m(\lambda, \epsilon)(1 + |\lambda|)^{-N} < \infty$ for some N . Its *supertrace distribution* is $\operatorname{str} e^{tG} := \sum \epsilon m(\lambda, \epsilon) e^{\lambda t}$ on $(0, \infty)$, and its *net multiplicity* is $\nu_{\text{net}}(\lambda) = m(\lambda, +1) - m(\lambda, -1)$. Two data with the same ν_{net} are *equivalent up to cancelling pairs*. For a $\mathbb{Z}_2$ -graded operator the supertrace is the trace on the even part minus the trace on the odd part.

Definition 5.2 (Distributional trace of the compressed semigroup, prime comb). $\operatorname{Tr}_{\text{dist}} Z(t) := \sum_{\rho} e^{-\bar{\rho}t/2}$, the sum over all nontrivial zeros with multiplicity, is a distribution on $(0, \infty)$ (Lemma 10.3); this is a definition by fiat, not an application of a trace theorem to $Z(t)$. The *prime comb* is $\operatorname{comb}(u) = \sum_{n \geq 2} \Lambda(n) \delta(u - \log n)$, so that $e^{-t/2} \operatorname{comb}(t/2) = 2P_+(t)$.

Definition 5.3 (Riemann graded bond and generator). $\mathfrak{B} = \mathbb{C}_+ \oplus (K_S \oplus \ell^2(\mathbb{N}_{\geq 1}))_-$, the even line spanned by the fixed vector and the odd part the model space together with an *archimedean ladder*; $G = 0 \oplus B \oplus \operatorname{diag}(-(k + \frac{1}{2}))_{k \geq 1}$, where $Z(t) = e^{tB}$ is the compressed semigroup.

Definition 5.4 (Vacuum-decay Lindbladian, absorbing-vacuum channel). For a Hilbert space H , B on H and an exit vector j with $-(B + B^\dagger) = |j\rangle\langle j|$, the *vacuum-decay Lindbladian* on $B(\mathbb{C}\text{vac} \oplus H)$ is $\mathcal{L}_{\text{vac}}(x) = \bar{B}x + x\bar{B}^\dagger + JxJ^\dagger$ with $\bar{B} = 0 \oplus B$ and $J = |\text{vac}\rangle\langle j|$; $\Omega_{\text{vac}} = |\text{vac}\rangle\langle \text{vac}|$. The grading is vac even, H odd. The *absorbing-vacuum channel* of a strongly stable contraction semigroup Z_t is $x \mapsto (1 \oplus Z_t)x(1 \oplus Z_t)^\dagger + \operatorname{Tr}[(1 - Z_t^\dagger Z_t)X] \Omega_{\text{vac}}$, X the H -block of x .

Definition 5.5 (Prime-chain detailed-balance Lindbladian). With $\mathfrak{m}_p|n\rangle = |pn\rangle$ the isometries of the Bost–Connes representation and rates $r_p > 0$,

$$\mathcal{L}_\beta^{\text{pc}}(x) = \sum_p r_p (\mathfrak{m}_p x \mathfrak{m}_p^\dagger - \tfrac{1}{2} \{\mathfrak{m}_p^\dagger \mathfrak{m}_p, x\}) + \sum_p r_p p^\beta (\mathfrak{m}_p^\dagger x \mathfrak{m}_p - \tfrac{1}{2} \{\mathfrak{m}_p \mathfrak{m}_p^\dagger, x\}),$$

in the Schrödinger convention; its diagonal part is the multiplicative birth–death process $n \rightarrow pn$ at rate r_p , $n \rightarrow n/p$ at rate $r_p p^\beta$. The Gibbs density is $\varrho_\beta^G = N^{-\beta}/\zeta(\beta)$.

Definition 5.6 (Entanglement Hamiltonian and entropy of a bond density). For a bond density X (the fixed point of a cMPS transfer generator, which by cited fact 11.3 carries the squared Schmidt coefficients of a bipartition) the *entanglement Hamiltonian* is $-\log X$, its spectrum the *entanglement spectrum*, and the *entanglement entropy* is $\mathcal{S}_{\text{ent}}(X) = -\operatorname{Tr} X \log X$; TFD_β denotes the thermofield-double purification of ϱ_β^G .

Definition 5.7 (Ring form, trace form, radical, radical sign, period parameter). For $g(x) = \sum_{j=0}^J a_j x^{1+q^j}$ over $\mathbb{F}_q$, q an odd prime, on $V = \mathbb{F}_q^n$ with cyclic indices let $\mathbb{S}$ be the shift $(\mathbb{S}x)_i = x_{i+1}$ and $P_g(z) = \sum_j \frac{a_j}{2} (z^j + z^{-j})$. The *ring form* is $Q_g^\circ(x) = \sum_j a_j \sum_i x_i x_{i+j} = x^t P_g(\mathbb{S})x$; the *trace form* in the coordinates of a normal basis θ is $Q_g^{\text{tr}}(x) = \operatorname{tr}_{q^n/q} g(x) = x^t C P_g(\mathbb{S})x$ with $C_{ik} = \operatorname{tr}_{q^n/q}(\theta^{q^i} \theta^{q^k})$ the (symmetric, invertible, circulant) Gram matrix. The *radical* is $\mathcal{R}_n = \ker P_g(\mathbb{S})$, the *radical sign* $\delta_n = \det(\mathbb{S}|_{\mathcal{R}_n}) \in \{\pm 1\}$ ($= 1$ if $\mathcal{R}_n = 0$), $d_n = \dim \mathcal{R}_n$, and the *period parameter* h is the smallest power of q strictly greater than the order of vanishing of $z^J P_g(z)$ at $z = -1$.

Definition 5.8 (Super-transfer matrix of an Artin–Schreier curve). $\mathbb{E} = (E_0 \oplus 1)_+ \oplus (\bigoplus_{a \in \mathbb{F}_q^\times} F_{ag})_-$, where E_0 is the transfer matrix of the zero polynomial (the de Bruijn adjacency matrix of memory

J , or the scalar q when $J = 0$), the summand 1 is the point at infinity, and F_g is the *Frobenius block* of the character sum: the diagonal $q^J \times q^J$ matrix with $L(g, u) = \det(1 - uF_g)$ (Theorem 15.7). The point count is its supertrace, $N_n = \text{str } \mathbb{E}^n$.

Definition 5.9 (Centred Weyl operators, symplectic transfer map, Weil implementer). On $\ell^2(\mathbb{F}_q^J)$ the *centred Weyl operators* are $X_{u,v}|s\rangle = e_q(v \cdot (s + u/2))|s + u\rangle$, $(u, v) \in \mathbb{F}_q^{2J}$. The *transfer unitary* is $\mathcal{U}_g = E_g/\sqrt{q}$; its *symplectic transfer map* $M_g \in \text{Sp}(2J, \mathbb{F}_q)$ is defined by $\mathcal{U}_g X_{u,v} \mathcal{U}_g^{-1} = X_{M_g(u,v)}$ (Theorem 15.13). A *Weil implementer* of $M \in \text{Sp}(2J, \mathbb{F}_q)$ is a unitary $W(M)$ with $W(M)X_{u,v}W(M)^{-1} = X_{M(u,v)}$; it is determined up to a scalar (cited fact 15.10), and $D_g(u) = \det(1 - uE_g)$.

6 Definitions, Continued: Flows on Complexes and their Twists

Continuation of Section 5; split for length only. Throughout, X is a finite abstract simplicial complex of dimension d with face numbers $f_0, \dots, f_d$ and Euler characteristic $\chi(X) = \sum_i (-1)^i f_i$, and $s_k = (-1)^{k+1}$.

Definition 6.1 (Ordered k -cell, ordered geodesic flow). An *ordered k -cell* of X is a tuple $[v_0, \dots, v_k]$ of distinct vertices whose underlying set is a k -simplex of X . Write $\mathcal{O}_k$ for the set of ordered k -cells, so $|\mathcal{O}_k| = (k+1)! f_k$, and $H_k = \mathbb{C}^{\mathcal{O}_k}$. The *ordered geodesic flow* is the operator T_k on H_k , $1 \leq k \leq d$,

$$T_k[v_0, \dots, v_k] = \sum_{\substack{w \notin \{v_0, \dots, v_k\} \\ \{v_1, \dots, v_k, w\} \in X, \{v_0, \dots, v_k, w\} \notin X}} [v_1, \dots, v_k, w] :$$

drop the oldest vertex, append a new one keeping the ordered overlap, and forbid the step that a $(k+1)$ -cell would short-cut. For $d = 1$ this is exactly the Hashimoto operator of Definition 3.7, and the excluded vertex is the immediately previous one.

Definition 6.2 (Pointed facet, opposition successor, algebraic length). Let X be a finite quotient of the Bruhat–Tits building $\mathcal{B}$ of PGL_n over a local field with residue order q , carrying the cyclic type structure of the affine diagram $\tilde{A}_{n-1}$. A *pointed k -facet* is a k -simplex with a distinguished vertex; ordering it cyclically in increasing type from that vertex gives its *ordered type*, a positive composition $(\lambda_0, \dots, \lambda_k)$ of n , and its *algebraic step length* is λ_0 . Write P_k for the set of pointed k -facets, so $|P_k| = (k+1)f_k$. A *successor* shifts the ordering and crosses to a vertex *opposite* in the spherical link of the common $(k-1)$ -facet; the *pointed opposition flow* is

$$\mathsf{T}_k(u)[\sigma] = u^{\lambda_0(\sigma)} \sum_{\tau \in \mathrm{Succ}(\sigma)} [\tau] \quad \text{on } \mathbb{C}^{P_k}.$$

Opposition is strictly stronger than the non-cell condition of Definition 6.1, and the exponent u^{λ_0} is a *length*, never a matrix power or a multiplicity.

Definition 6.3 (Graded geodesic determinant, total zeta of a complex). *Geodesic data* $\mathfrak{g}$ on X is, in each degree $1 \leq k \leq d$: a finite state set $\mathcal{S}_k$, a successor relation with incidence multiplicities, a positive integer algebraic length $\ell_k(\sigma)$, and a step transport $U_{\sigma\tau}$ on a finite-dimensional fibre V . Give the whole degree- k space $V \otimes \mathbb{C}^{\mathcal{S}_k}$ parity $k+1$, put $\mathsf{T}_k^U(u)(v \otimes [\sigma]) = \sum_{\sigma \rightsquigarrow \tau} u^{\ell_k(\sigma)} U_{\sigma\tau} v \otimes [\tau]$, and let $W = \bigoplus_{k=1}^d V \otimes \mathbb{C}^{\mathcal{S}_k}$. The *graded geodesic determinant* (the total zeta) is

$$\mathcal{Z}(X, \mathfrak{g}, U; u) = \mathrm{sdet}_W \left(I - \bigoplus_{k=1}^d s_k \mathsf{T}_k^U(u) \right)^{-1} = \prod_{k=1}^d \det(I - s_k \mathsf{T}_k^U(u))^{(-1)^k},$$

with the sign $s_k = (-1)^{k+1}$ inserted once per successor arrow. Odd k (edges) sits in the denominator, even k (chambers) in the numerator. The intrinsic default for an arbitrary complex is $\mathcal{S}_k = \mathcal{O}_k$, unit length, the rule of Definition 6.1 and $V = \mathbb{C}$, written $\mathcal{Z}_{\mathrm{ord}}(X, u)$; the building default is Definition 6.2, written $\mathcal{Z}_{\mathrm{pt}}(X, u)$.

Definition 6.4 (Bruhat–Tits building of PGL_n , colour-shift Hecke operators, Ramanujan complex). The vertices of $\mathcal{B}$ carry a colour (type) function with values in $\mathbb{Z}/n\mathbb{Z}$, and the *colour-shift Hecke operators* A_k , $1 \leq k \leq n-1$, sum a function over the neighbours of colour shift k : we may look at the $d-1$ colored adjacency operators A_k , $k = 1, \dots, d-1$ on $L^2(\mathcal{B} \setminus \{0\})$, called the Hecke

operators. (math/0406208:main.tex:541-549). They commute, are normal, and satisfy $A_k^* = A_{n-k}$. A finite quotient is a *Ramanujan complex* when its nontrivial joint spectrum lies in the tempered set $\text{Aspec}_n = \{(q^{k(n-k)/2} e_k(z))_k : |z_i| = 1, \prod_i z_i = 1\}$: A finite quotient $\mathbb{X}$ of $\mathbb{B}$ is called a Ramanujan complex if the eigenvalues of every non-trivial simultaneous eigenvector $v \in \bigwedge^2 \mathbb{X}$, $A_k v = \lambda_k v$, satisfy $(\lambda_1, \dots, \lambda_{d-1}) \in \text{Aspec}_d$. (math/0406208:main.tex:564-572). The condition is on the *joint* spectrum, which is strictly smaller than the product of the separate spectra.

Definition 6.5 (Cayley complex). For a finite group G with generating set S , the *Cayley complex* is the clique complex of the Cayley graph: The Cayley complex of G with respect to a set of generators S is the simplicial complex whose 1-skeleton is the Cayley graph $\text{Cayley}(G, S)$, where a subset of $i+1$ vertices is an i -cell iff every two vertices comprise an edge. (math/0406217: Lubotzky, Samuels and Vishne give explicit S , graded into colour classes $S_1, \dots, S_{n-1}$ with $|S_k| = \binom{n}{k}_q$ and $S_k^{-1} = S_{n-k}$, for which this complex is a Ramanujan complex covered by $\mathcal{B}$.

Definition 6.6 (Kraus connection on the 1-skeleton; flat, pure gauge, paired). A *connection* on the 1-skeleton of X with fibre V assigns to each directed edge (x, y) an arbitrary $E_{(x,y)} \in \text{End}(V)$; it is a *Kraus connection* when $V = M_n$ and each E_e is completely positive, for instance $E_e = \text{Ad}(B_e)$. Twist Definition 6.1 by anchoring the fibre at the *last* vertex, $T_k^E(v \otimes [\sigma]) = \sum E_{(v_k, w)} v \otimes [\tau]$. The connection is *flat* when $E_{yx} = E_{xy}^{-1}$ and $E_{yz} E_{xy} = E_{xz}$ on every 2-cell, and *pure gauge* when $E_{(x,y)} = F_y F_x^{-1}$ for some invertible F . It is *adjoint-paired* when $E_{\bar{e}} = E_e^\dagger$ and *inverse-paired* when $E_{\bar{e}} = E_e^{-1}$ (Definition 4.26). No positivity, invertibility, pairing or flatness is assumed unless said.

Definition 6.7 (Voltage quotient and Artin block). Let a finite group G act on X preserving the successor and length data and freely on the state set $\mathcal{S}_k$. Choose orbit representatives σ_a and encode each transition $\sigma_a \rightarrow h\sigma_b$ by its *voltage* $h \in G$; this is the *voltage quotient*, a digraph on the orbits with a group label on every arrow. For a unitary ϱ of G , the ϱ -block $\mathsf{T}_{k,\varrho}(u)$ replaces that arrow by $\varrho(h^{-1})$ on the fibre, keeping the length weight. For $R = \pi \otimes \bar{\pi} = \bigoplus_{\varrho} a_{\varrho} \varrho$ the *Artin determinant* is $D_{k,R}(u) = \prod_{\varrho} \det(I - s_k \mathsf{T}_{k,\varrho}(u))^{a_{\varrho}}$. This is not the same object as the connection of Definition 6.6 on the full cover.

Definition 6.8 (Ramanujan quantum expander of type $\tilde{A}_{n-1}$; exceptional space). Let $X = \text{Cay}(G, S)$ have the colour classes of Definition 6.5, let $\pi: G \rightarrow U(N)$ be unitary and $R = \text{Ad } \pi$ on M_n . Put $\Phi_k(X) = |S_k|^{-1} \sum_{s \in S_k} \pi(s) X \pi(s)^\dagger$ and $\hat{A}_k = |S_k| \Phi_k = \sum_{s \in S_k} R(s)$. Let $\mathcal{F}_R$ be the common fixed space and let $\mathcal{E}_R \supseteq \mathcal{F}_R$ be the sum, inside R , of the type-character subrepresentations excluded by the convention of Definition 6.4. The tuple $(\Phi_1, \dots, \Phi_{n-1})$ is *Ramanujan of type $\tilde{A}_{n-1}$ relative to $\mathcal{E}_R$* when its joint spectrum on $\mathcal{E}_R^\perp$ lies in Aspec_n . Only when $\mathcal{E}_R = \mathcal{F}_R$ is this the fixed-point-complement condition; on a partite complex $\mathcal{E}_R$ can be strictly larger.

7 Definitions, Continued: Finite BC States and Covariant Generators

Definition 7.1 (Finite phase marginal and matrix extension). For an odd prime p , the phase algebra generated by $e(k/p)$ is $C(\mathbb{F}_p)$, represented diagonally on $H_p = \ell^2(\mathbb{F}_p)$. Let w_β be the restriction of the unique BC state for $0 < \beta \leq 1$, or of the Galois average of extremal BC states for $\beta > 1$. Its *diagonal matrix extension* is $\varrho_{p,\beta}^{\text{res}} = \text{diag}(w_\beta(x))$ in $B(H_p)$. A matrix extension is a density with these diagonal entries; its off-diagonal entries are additional data. This does not specify a representation of the full BC algebra on H_p .

Definition 7.2 (Local arithmetic covariance and parity). The centred Weyl operators X_v , $v \in \mathbb{F}_p^2$, are those of Definition 5.9 with one register. Three requirements on a superoperator are distinguished: commuting with $\text{Ad}(U_a)$, $U_a|x\rangle = |ax\rangle$, $a \in \mathbb{F}_p^\times$ (*Galois torus*); commuting with $\text{Ad}(W(g))$, $g \in \text{Sp}(2, \mathbb{F}_p)$ (*linear symplectic covariance*); and commuting with all $\text{Ad}(X_v)$ (*Weyl-translation covariance*). The first action sends (x, y) to $(ax, a^{-1}y)$, and induces the inverse-labelled BC action on the phase generators. Weyl-translation covariance is an extra condition, not a consequence of carrying Weyl operators. Parity is $P|x\rangle = |-x\rangle$; operator parity is $X \mapsto PXP$.

Definition 7.3 (Stationary covariant GKLS cone). For a density ϱ_b on $\mathbb{C}^d$ and a unitary group action U , let $\mathfrak{C}_U(\varrho_b)$ be the cone of Schrödinger generators of CPTP semigroups with $\mathcal{L}(\varrho_b) = 0$ and $[\mathcal{L}, \text{Ad}(U_g)] = 0$ for every g . Its dimension means the real dimension of its linear span, including the overall time scale. The identity is fixed by the dual Heisenberg semigroup; fixing I also in the Schrödinger picture is an additional condition, equivalent to stationarity of the tracial density I/d .

Definition 7.4 (Conditional Choi matrix). For a complex-linear map L on M_d , set

$$\mathbf{C}_L = \sum_{i,j} |i\rangle\langle j| \otimes L(|i\rangle\langle j|), \quad b = d^{-1/2} \sum_i |i, i\rangle, \quad Q = I - |b\rangle\langle b|.$$

The *conditional Choi matrix* is $Q\mathbf{C}_L Q$. The completely depolarizing channel is $\mathcal{D}_d(X) = \text{Tr}(X)I/d$.

8 What the Riemann Hypothesis Has Become

Nothing here is new mathematics; it is a translation, written for TJO after the founding session from the ingredients built up there. Its point is one sentence, isolated in observation 8.3 and repeated in every later costume: the Riemann hypothesis says certain counts fluctuate no more than a random sequence would.

8.1 The one identity, and its two readings

Observation 8.1 (The one identity). [sketched] For any matrix M with eigenvalues $\lambda_1, \dots, \lambda_N$,

$$\frac{1}{\det(1 - uM)} = \prod_i \frac{1}{1 - \lambda_i u} = \exp \left(\sum_{\ell \geq 1} \frac{\text{Tr}(M^\ell)}{\ell} u^\ell \right).$$

The middle expression is a bosonic partition function, one mode per eigenvector with Boltzmann factor $\lambda_i u$; the right expression is a sum over closed walks of length ℓ , carrying the $1/\ell$ of a ring's rotation symmetry. One object, two descriptions: a gas of modes and a gas of rings (Definitions 3.1 and 3.2).

The whole subject is the tension between the two sides. For a permutation matrix the ring side is concrete: $\text{Tr}(M^\ell)$ counts the fixed points of the ℓ -th power, the closed walks are the cycles and their repeats, and the identity collapses to $\det(1 - uM)^{-1} = \prod_c (1 - u^{|c|})^{-1}$ over cycles c : a Bose gas with one mode per cycle, energy the cycle length. That product is the Euler product, and the primes will be the cycles.

8.2 A finite permutation: the analogue is free

Observation 8.2 (RH for a finite permutation). [sketched]

A permutation matrix permutes an orthonormal basis, hence is unitary, hence has every eigenvalue on the unit circle. The RH analogue at this level costs nothing.

8.3 An infinite permutation: a statement about a different matrix

For infinitely many cycles the permutation matrix is still unitary and still useless, its spectrum being roots of unity with infinite multiplicity. The count $\text{fix}(\ell) = \sum_{d|\ell} d c_d$ of fixed points of the ℓ -th power, with c_d the number of d -cycles, remains meaningful, and the interesting object is a *small matrix* (Definition 3.3) reproducing them.

Observation 8.3 (RH for the small matrix). [sketched] A small matrix M_{small} with $\text{Tr } M_{\text{small}}^\ell = \text{fix}(\ell)$ for all ℓ exists exactly when the zeta function is rational, and then its eigenvalues are new numbers, off the unit circle. Writing $\text{fix}(\ell) = \lambda_{\text{max}}^\ell + \sum_{i \geq 2} \lambda_i^\ell$, the RH analogue is that every other nonzero λ_i has $|\lambda_i| = \sqrt{\lambda_{\text{max}}}$ (Definition 3.4), so that the fluctuation has size $\sqrt{\lambda_{\text{max}}^\ell}$: the square root of the leading term, which is the fluctuation of a sum of $\lambda_{\text{max}}^\ell$ independent coin tosses. In counting language, RH says the cycle counts fluctuate no more than a random sequence would.

So the RH analogue is here a property some systems have and others do not. The note records two shift examples, both perfectly good infinite permutations.

shift	eigenvalues of the small matrix	RH analogue
4-symbol cyclic rule	2, $1 + i$, $1 - i$, 0	holds
golden-mean shift	ϕ , $-1/\phi$	fails

8.4 A graph: Ramanujan

Observation 8.4 (RH for a graph is the Ramanujan property). [sketched]

For a $(q+1)$ -regular graph the permutation is the shift on non-backtracking closed walks, the small matrix is the adjacency matrix A , and the Ihara–Bass formula (Definition 3.8) compresses one to the other: each eigenvalue λ of A produces a pair μ, μ' of eigenvalues of the edge operator T through

$$\mu^2 - \lambda\mu + q = 0, \quad \text{so} \quad \mu\mu' = q.$$

The leading $\lambda = q + 1$ gives $\mu = q, \mu' = 1$. For the rest, $\lambda^2 < 4q$ forces a complex conjugate pair with $|\mu| = |\mu'| = \sqrt{q}$, on the critical circle, while $\lambda^2 > 4q$ gives a real pair, one factor above $\sqrt{q}$ and one below. Hence RH for ζ_X holds if and only if $|\lambda| \leq 2\sqrt{q}$ for every nontrivial λ , i.e. iff the graph is Ramanujan (Definition 3.9), and the closed non-backtracking walk counts are $N_\ell = q^\ell + O(q^{\ell/2})$.

Three remarks carry over to the primes. First, **the pairing comes from a symmetry, the hypothesis from a bound**: $\mu\mu' = q$ is a functional equation pairing eigenvalues symmetrically about $\sqrt{q}$ on a logarithmic scale, and RH is the extra statement that each pair is degenerate in modulus, which happens exactly when it is a conjugate pair. Second, **self-adjointness is not enough**: A is symmetric, so λ is real, which guarantees the pairing but not the bound; the prism graph $C_{16} \times K_2$ is symmetric and fails it. Third, **the bound is geometric**: $[-2\sqrt{q}, 2\sqrt{q}]$ is the spectrum of the infinite tree covering the graph, so RH says the graph has no eigenvalue outside the spectrum of its universal cover; the word is tempered.

8.5 A matrix product state: one correlation length

Let $E = \Sigma$ be the transfer matrix of a matrix product state (Definition 3.12), so that $\text{Tr } E^n = \langle \Psi_n | \Psi_n \rangle$ is the ring norm of the periodic state on n sites (Definition 3.13) and the zeta function of observation 8.1 is the generating function of ring norms. In canonical form (Definition 3.14) the leading eigenvalue is 1 and the rest lie inside the unit disc, $\lambda_k = e^{-1/\ell_k^c + i\vartheta_k}$ (Definition 3.15), so $\langle \Psi_n | \Psi_n \rangle = 1 + \sum_k e^{-n/\ell_k^c} e^{in\vartheta_k}$.

Observation 8.5 (RH for an MPS). [sketched] Translating observation 8.4, where the leading growth is q per step and the subleading modulus $\sqrt{q}$: in normalised units the subleading modulus is $q^{-1/2}$, so $e^{-1/\ell_k^c} = q^{-1/2}$ and $\ell_k^c = 2/\log q = 2/h$ for every k . The RH analogue for an MPS reads: all correlation lengths are equal, and equal to twice the inverse entropy rate. The phases ϑ_k are unconstrained and there may be infinitely many of them; only the moduli are pinned.

The hierarchy this sits in has three levels, and each has a classical number-theoretic counterpart, anticipating observation 8.6 below.

statement about E (or $\mathcal{G}$)	name	counting meaning	number theory
a unique eigenvalue on the unit circle	primitivity (Perron–Frobenius)	ring norms follow the leading term	$\zeta(1+it) \neq 0$, the prime number theorem
all others inside a smaller circle	spectral gap	fluctuation exponentially smaller than the leading term	the zero-free region, $\psi(x) = x + O(xe^{-c\sqrt{\log x}})$
all others on one circle of radius e^{-1/ℓ_k^c} with $\ell_k^c = 2/h$	Ramanujan / RH	fluctuation is the square root of the leading term	all zeros on $\text{Re } s = \frac{1}{2}$

8.6 The primes: continuous length, and the sign

For the primes the cycle lengths are $\log p$, rationally independent, so the matrix powers become a semigroup $e^{\mathbf{L}\mathcal{G}}$ in a continuous length $\mathbf{L}$ and the ring norms become a measure in $\mathbf{L}$. Start from the Riemann–von Mangoldt formula for ψ , valid for x not a prime power, put $x = e^{\mathbf{L}}$ and differentiate in $\mathbf{L}$: the left side becomes a sum of delta functions, one per prime power, of weight $\log p$, and the right side a sum of exponentials.

Observation 8.6 (The explicit formula as a trace, and the pairing). [sketched] With $\psi(x) = x - \sum_{\rho} x^{\rho} / \rho - \log 2\pi - \frac{1}{2} \log(1 - x^{-2})$,

$$\sum_n \Lambda(n) \delta(\mathbf{L} - \log n) = e^{\mathbf{L}} - \sum_{\rho} e^{\rho \mathbf{L}} = (e^{-2\mathbf{L}} + e^{-4\mathbf{L}} + \dots),$$

which is $\text{Tr } e^{\mathbf{L}\mathcal{G}}$ for a generator $\mathcal{G}$ whose eigenvalues are 1 (the pole of ζ , the leading term, entropy 1), every zero ρ , and the trivial zeros $-2, -4, \dots$: the trace identity of observation 8.1 with M^{ℓ} replaced by $e^{\mathbf{L}\mathcal{G}}$, ring side the prime powers at lengths $\log n$, mode side the zeros. The functional equation $\xi(s) = \xi(1-s)$ makes ρ and $1-\rho$ both zeros, so after dividing by the leading $e^{\mathbf{L}}$ the normalised decay rates come in pairs $(1-\sigma, \sigma)$ summing to 1, exactly as $\mu\mu' = q$ paired the graph eigenvalues around $\sqrt{q}$. That pairing is a theorem; RH is the extra statement that every pair is degenerate, $\sigma = 1 - \sigma = \frac{1}{2}$.

Two features of this trace identity are visible immediately. The first is *the sign*: the zeros enter with a minus sign relative to the leading term, the same minus sign that marks the fermionic H^1 block in the elliptic-curve case (Section 16) and the absorption spectrum in Connes’s language. The total measure stays positive because $\Lambda(n) \geq 0$, but the subleading sector of the bond space contributes negatively, so whatever state this is, its bond space is graded, not plainly bosonic. The second is the pairing just stated. The numerical check against several thousand zeros is recorded in Section 9. Collecting the costumes:

language	the Riemann hypothesis says
counting	$\psi(x) = x + O(x^{1/2+\epsilon})$: the prime powers fluctuate like coin tosses
small matrix	every subleading mode has modulus the square root of the leading one
MPS	the state whose rings are the primes has <i>one</i> correlation length, $\ell_k^c = 2$ since $h = 1$, and an infinite tower of frequencies γ_n
channel	the transfer semigroup $e^{\mathbf{L}\mathcal{G}}$ has a single uniform decay rate $\frac{1}{2}$

8.7 The Hilbert space where this is literally true

The channel version is not only an analogy. It is a theorem on a concrete Hilbert space: the model space K_S of the scattering symbol S of the modular surface, on which the compressed dilation semigroup $Z(t)$ has one mode per zero, with decay rate and frequency read off from that zero. This is Lax–Phillips [52], and “RH iff every mode of $Z(t)$ decays at the same rate” is Faddeev–Pavlov [24]. What the founding session added is where S comes from: its phase is the imaginary part of the Bost–Connes Lévy exponent at the critical temperature $\beta = 1$ [9], so the ring gas and the transfer semigroup are one object seen from its two ends, bridged by the critical phase. Construction, normalisations (the factor $\frac{1}{2}$ from the doubled variable) and numerics are in Section 9; Definitions 4.1 to 4.3 fix the objects.

8.8 Why it is still hard, in this language

The graph case is the honest guide. There side B exists, the trace identity holds and the small matrix is symmetric for every graph, and most graphs fail the bound. Being Ramanujan is never a consequence of the structure; it is an extra fact, and in every known case it comes from arithmetic input of Deligne's type [17]. For Selberg's surfaces the same statement, no Laplace eigenvalue below $\frac{1}{4}$, is open.

What the translation settles: RH is a statement about the *sizes* of the subleading modes of a transfer semigroup, not about its existence. The semigroup exists, the zeros are its modes, the primes are its rings, the functional equation is the pairing of its rates. What it does not settle: a mechanism forcing the pairs to be degenerate. In positivity form (Definition 4.8), RH holds iff F_W is positive-definite on the dilation group, iff its spectral measure is positive on the real line, which by Bochner is the same as all its atoms being real.

Observation 8.7 (Hermiticity of the Hilbert–Pólya operator).

[sketched]

Write the length-flow generator on the subleading sector as $\mathcal{G} = -\frac{1}{2} + iH$ (Definition 4.7); then H has eigenvalues $\gamma_n - i(\sigma_n - \frac{1}{2})$, and RH says exactly that H is Hermitian, i.e. that the transfer semigroup is a uniform damping times an oscillation with real frequencies. H is the Hilbert–Pólya operator. The non-Hermitian generator carries the complex pairs and the Hermitian operator their symmetric combination, here $\rho(1 - \rho) = \sigma(1 - \sigma) + \gamma^2 + i\gamma(1 - 2\sigma)$, which is real iff $\sigma = \frac{1}{2}$; $\mathcal{G}$ is to H what the Hashimoto edge operator T is to the adjacency matrix A of a graph. For graphs the Hermitian object is given and its reality is a theorem; for the primes it is the conjecture.

Proving it would mean exhibiting that state, or a Stinespring form of $Z(t)$ with the prime dilations as Kraus operators, in a way that makes the positivity manifest; that is conjecture 9.10. The graph lesson says the Kraus structure alone will not do it, and that some arithmetic rigidity has to enter, playing the role that weights play for curves (Section 16).

8.9 Erratum on words

An earlier draft, and the founding session, called $-\log E$ the entanglement Hamiltonian. That is wrong: the entanglement Hamiltonian of an MPS is $-\log$ of a reduced density matrix built from the fixed points of E alone, is Hermitian by construction, and does not see the subleading spectrum. The operator whose spectrum contains the zeros is the transfer generator, non-Hermitian in general, whose eigenvalue phases are the oscillation frequencies of the ring norms. observation 8.7 states the correct relation; the route is listed among the dead ones in Section 26.

8.10 Summary

A zeta function is one partition function with two Fock-space descriptions: a gas of rings (cycles, closed walks, prime powers) and a gas of modes (eigenvalues of a small transfer matrix); the trace identity links them. For the primes the rings have lengths $\log n$ and weights $\Lambda(n)$, the modes are the zeros of ζ plus the pole, and the functional equation pairs the modes' decay rates as $(\sigma, 1 - \sigma)$. The Riemann hypothesis says every pair is degenerate at $\frac{1}{2}$: the transfer semigroup has one correlation length, $\ell_k^c = 2$, so the ring counts fluctuate by no more than the square root of their size, so the primes are as regular as coin tosses. That semigroup exists explicitly, its scattering symbol is the phase of the Bost–Connes system at its critical temperature, and RH is the statement that this channel is Ramanujan.

9 The Riemann Channel: Bost–Connes at Critical Temperature, Compressed

This section is the lab-book version of `notes/riemann-channel-note.md`, a scratch note of the founding session. It assembles one object out of three standard pieces: Lax–Phillips scattering on the modular surface, the Sz.-Nagy–Foias functional model of an inner function, and the Bost–Connes system at its phase transition. Nothing here is new mathematics; the content is the identification of the three, and the numerical checks that it is stated correctly. Everything below is *equivalent to*, never a proof of, the Riemann hypothesis. Several statements are theorem-level standard theory (Faddeev–Pavlov, Lax–Phillips, Adamyan–Arov, Khinchin) for which this repo holds no local source and no written-out argument; they carry the tag `sketched` for that reason, not because their truth is in doubt.

9.1 Setting

Let τ be the Mellin variable on the dilation group $\mathbb{R}_+^\times$, so that dilation by e^t acts on $L^2(\mathbb{R}_+^\times)$ as the multiplier $e^{it\tau}$. With the completed zeta

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s),$$

the scattering symbol of the modular surface in this variable (Definition 4.1) is

$$S(\tau) = \frac{\xi(1-2i\tau)}{\xi(1+2i\tau)}.$$

Write a nontrivial zero as $\rho_n = \sigma_n + i\gamma_n$, so that the Riemann hypothesis is $\sigma_n = \frac{1}{2}$ for all n (Section 2).

9.2 The symbol is inner

Proposition 9.1 (The scattering symbol is inner, with the zeros as its zeros). *[sketched]* *S is unimodular on the real line, analytic and contractive in the lower half plane $\text{Im } \tau < 0$, and its zeros there are exactly*

$$\tau_n = \frac{\gamma_n}{2} - i \frac{\sigma_n}{2}, \quad \rho_n = \sigma_n + i\gamma_n \text{ a nontrivial zero of } \zeta.$$

The note’s argument: ξ is real on the real axis and satisfies $\xi(s) = \xi(1-s)$, which gives $|S| = 1$ for real τ ; for $\text{Im } \tau < 0$ the denominator’s argument $1+2i\tau$ has real part greater than 1, inside the Euler product, so S is analytic there. Contractivity in the interior is the Phragmén–Lindelöf step, which the note does not write out (step 3 of `HANDOFF.md`); for the modular surface it is classical [24, 52]. The zeros are a substitution: $\xi(1-2i\tau) = 0$ iff $1-2i\tau$ is a nontrivial zero.

9.3 The compressed semigroup

Since S is inner, the model space $K_S = H^2 \ominus SH^2$ (Definition 4.2) carries the compressed dilation semigroup $Z(t) = P_{K_S} e^{it\tau}|_{K_S}$, $t > 0$. By Adamyan–Arov this is unitarily equivalent to the Lax–Phillips semigroup of the modular surface with the cusp forms removed; the note states this without a written argument, and no local source exists for it here.

Proposition 9.2 (One mode per zero; RH as a single decay rate).

[sketched]

For each zero λ of S in the lower half plane the reproducing kernel $k_\lambda(\tau) = (\tau - \bar{\lambda})^{-1}$ lies in K_S , and $Z(t)^* k_\lambda = e^{-it\bar{\lambda}} k_\lambda$. With $\lambda = \gamma_n/2 - i\sigma_n/2$ the eigenvalue is $\exp(-\sigma_n t/2 - i\gamma_n t/2)$: one mode per zero, decay rate $\sigma_n/2$, frequency $\gamma_n/2$. Hence the Riemann hypothesis holds if and only if every mode of $Z(t)$ decays at the single rate $1/4$.

The kernel is orthogonal to SH^2 precisely because $S(\lambda) = 0$. The last sentence is the Faddeev–Pavlov / Lax–Phillips reading of RH [24, 52]; it is the “one correlation length” criterion of Section 8 (observation 8.5) on an explicit Hilbert space, with K_S in the role of the bond space. The completely positive lift is immediate and uninformative: $\Phi_t(X) = Z(t)^* X Z(t)$ on $B(K_S)$ is the Riemann channel of Definition 4.3, a sub-unital semigroup with one Kraus operator, whose Stinespring dilation is the automorphic wave group and whose environment is the pair of incoming and outgoing subspaces, that is, the cusp. Whether a less trivial Kraus decomposition exists is conjecture 9.10.

9.4 Where the primes are

Two placements of the primes appear, and the note is explicit that they are different objects. The first is a dilation-covariant Markov semigroup on $L^2(\mathbb{R}_+^\times)$ in Holevo form, with the von Mangoldt jump measure $\nu_\sigma = \sum_{p,k} \frac{p^{-k\sigma}}{k} \delta_{k \log p}$ and symbol

$$\left(\frac{\zeta(\sigma - i\tau)}{\zeta(\sigma)} \right)^t = \exp \left(t \sum_{p,k} \frac{p^{-k\sigma}}{k} (e^{ik\tau \log p} - 1) \right),$$

a genuine Markov semigroup for $\sigma > 1$ by Khinchin’s theorem that the zeta distribution is compound Poisson (standard; no local source). It has no zeros for $\sigma > 1$, and at $\sigma = \frac{1}{2}$ it is not a semigroup at all, the symbol being unbounded. The Bost–Connes phase transition $\beta = 1$ is the finite- versus infinite-activity threshold of ν_σ [9].

The second is $Z(t)$ on K_S : the zeros as resonances, no jump structure, the primes entering only through S . The bridge between the two is the boundary phase.

Proposition 9.3 (The symbol is the phase of the critical Lévy exponent). [sketched] For real τ , $S(\tau) = \exp(-2i \arg \xi(1 + 2i\tau))$, and

$$-\arg \zeta(1 + 2i\tau) = \sum_p \sum_{k \geq 1} \frac{p^{-k}}{k} \sin(2k\tau \log p),$$

which is the imaginary part of the Bost–Connes Lévy exponent $\kappa_\sigma(\tau') = \log \zeta(\sigma - i\tau') - \log \zeta(\sigma)$ at the critical temperature $\sigma = 1$, $\tau' = 2\tau$. Hence S is the phase of the critical Lévy exponent times the Gamma phase.

The real part of κ_1 diverges, since $\sum_p p^{-1} = \infty$; that divergence is the Bost–Connes phase transition. The imaginary part converges conditionally, at prime-number-theorem strength, i.e. by $\zeta(1 + it) \neq 0$. The bridge the note proposes: take the Bost–Connes process at $\beta = 1$, discard the divergent real part of its Lévy exponent, keep the phase, and use that unimodular function as the characteristic function of a contraction semigroup. Symbolically, Lax–Phillips = (Bost–Connes at $\beta = 1$) + (Sz.-Nagy–Foias), at the level of boundary phases.

9.5 Numerical check of the symbol

Numerical result 9.4 (Unimodularity, zeros, innerness and the prime phase). [numerical]
`scripts/scat.py` (`outputs/scat.txt`, mpmath at 20 digits) reports $||S| - 1| = 0$ to machine precision at $\tau = 0.3, 2.0, 7.5, 21.3$; a root search from $\gamma_n/2 - \frac{1}{4}i$ converging for $n = 1, \dots, 6$ to $\gamma_n/2 - 0.250000i$ with $|S| \leq 1.5 \cdot 10^{-20}$; $|S| = 0.9348, 0.5393, 0.8344, 0.2718$ at the interior points $\tau = 3 - 0.6i, 10 - i, 0.5 - 2i, 30 - 0.3i$; and Proposition 9.3 to $1.3 \cdot 10^{-3}$ or better at five points:

τ	exact	prime sum + tail	residual
0.7	+0.80572	+0.80469	$-1.0 \cdot 10^{-3}$
1.9	-0.03879	-0.03854	$+2.5 \cdot 10^{-4}$
4.2	-0.17233	-0.17306	$-7.3 \cdot 10^{-4}$
9.0	+0.06468	+0.06594	$+1.3 \cdot 10^{-3}$
15.5	-0.06432	-0.06409	$+2.3 \cdot 10^{-4}$

The middle column is the prime sum over $p < 3 \cdot 10^6$ truncated at $k \leq 6$ plus the prime-number-theorem tail $\pi/2 - \text{Si}(2\tau \log X)$, which is what makes the residual 10^{-3} rather than 10^{-2} ; `outputs/scat.txt` archives the bare prime sums, before the tail, at three of the five points (0.82380, -0.05607, -0.18043). The innerness numbers are a check at four points, not a proof.

9.6 Ring norms: the trace of the semigroup is the prime measure

Distributionally $\text{Tr } Z(t) = \sum_n e^{-t\sigma_n/2 + it\gamma_n/2}$, under RH $e^{-t/4} \sum_n e^{it\gamma_n/2}$. The Weil explicit formula (Definition 4.4) turns that zero sum into a prime sum: in the ring-length variable $L = t/2$, with window $\hat{g}$ and transform g ,

$$\sum_{\gamma} \hat{g}(\gamma) = \hat{g}\left(\frac{i}{2}\right) + \hat{g}\left(-\frac{i}{2}\right) - g(0) \log \pi + \frac{1}{2\pi} \int \hat{g}(r) \text{Re}(\Gamma'/\Gamma)\left(\frac{1}{4} + \frac{ir}{2}\right) dr - 2 \sum_n \frac{\Lambda(n)}{\sqrt{n}} g(\log n).$$

Proposition 9.5 (The primes are the ring spectrum of the Riemann channel). [proved-conditional]
For $t > 0$ the distributional trace of Definition 5.2 is

$$\text{Tr}_{\text{dist}} Z(t) = 1 - 2P_+(t) - \frac{e^{-t/2}}{e^t - 1} = 1 - 2 \sum_{n \geq 2} \frac{\Lambda(n)}{n} \delta(t - 2 \log n) - \sum_{k \geq 1} e^{-(k+1/2)t} :$$

the primes are dips of weight $-2\Lambda(n)/n$ at the ring lengths $t = 2 \log n$, the constant 1 is the pole, the smooth term the archimedean ladder.

Restated 2026-09-12 night from the prover's T4.1; the founding version had weight $+\Lambda(n)n^{-1/2}$ and no pole term (see Proposition 10.20). In the MPS dictionary of Section 8: rings of length $\log n$ with weight $\Lambda(n)$, a transfer generator with eigenvalues $-\frac{1}{2} \pm i\gamma_n$, and a single correlation length iff RH. The primes are the ring spectrum, the zeros the transfer spectrum, and one identity, the explicit formula, links them (Definition 3.2).

Observation 9.6 (Sign and factor of the prime weight). [sketched-conditional] In the convention of the explicit formula displayed above, the prime part of the centred trace of $Z(t)$ carries weight $-2\Lambda(n)n^{-1/2}$ at $t = \pm 2 \log n$ (dips), and the positive-time uncentred prime part is $-2\Lambda(n)/n$. Proposition 9.5 was right about the support and off by the factor -2 in the weight; it has since been restated in the form printed above.

Recorded 2026-09-12 from the Selberg review’s action item (b) and, independently, from the codex prover’s T6 in `notes/weil-positivity/astra-proofs.md`; the Weil positivity review checks it as its item X3 (Section 18).

Numerical result 9.7 (Windowed zero sum against the full explicit formula). [numerical]
`scripts/ringnorm.py` (`outputs/ringnorm.txt`) sums the first 3000 zeros ($\gamma_{\max} = 3533.3$) against the Gaussian window $\hat{g}(r) = \cos(rL)e^{-r^2/2G_w^2}$ with $G_w = 1009.5$, for which the truncation weight at $\gamma_{\max}$ is $2.2 \cdot 10^{-3}$. The numeric side is $F(L) = \sum_{\gamma>0} e^{-\gamma^2/2G_w^2} \cos(\gamma L)$ on $L \in [0.3, 3.3]$; the prediction is every term of the explicit formula above, archimedean digamma integral included. The prime terms are Gaussian dips of depth $\frac{G_w}{2\sqrt{2\pi}} \Lambda(n)n^{-1/2}$ at $L = \log n$:

n	$\log n$	$F(L_n)$	prediction	ratio	$\Lambda(n)/\sqrt{n}$
2	0.6931	−97.95	−98.13	0.998	0.490
3	1.0986	−126.83	−127.02	0.999	0.634
4	1.3863	−68.77	−68.94	0.998	0.347
5	1.6094	−143.66	−143.86	0.999	0.720
7	1.9459	−146.71	−146.92	0.999	0.736
8	2.0794	−47.87	−48.03	0.997	0.245
9	2.1972	−72.19	−72.36	0.998	0.366
11	2.3979	−143.86	−144.07	0.999	0.723
13	2.5649	−141.21	−141.41	0.999	0.711
16	2.7726	−32.88	−33.03	0.995	0.173
27	3.2958	−39.93	−40.09	0.996	0.211

The maximum of $|F - \text{prediction}|$ over the whole interval is 0.20, against dip depths up to 147; away from the dips it is 0.18, so the residual is flat and is the truncation of the zero sum, not a mismatch of terms. Depths scale as $\Lambda(n)/\sqrt{n}$, prime powers included. The archived output carries four further rows ($n = 17, 19, 23, 25$) with the same ratios; the table above is the note’s selection.

9.7 The Bost–Connes phase operators as an MPO over the prime chain

Item 2 of `HANDOFF.md` concerns the Bost–Connes system itself rather than its compression. Its Hilbert space factorises over the primes, one oscillator per prime (Definition 4.6); the Gibbs state at inverse temperature β is a product state with local weight $(1 - p^{-\beta})p^{-\beta k}$ at occupation k , and the isometries m_n act locally.

Proposition 9.8 (The Bost–Connes phase operator is a prime-chain MPO). [sketched]
The Bost–Connes phase operator $e(a/b)$ is a matrix product operator over the prime chain with bond space $\mathbb{Z}/b$, the local transfer map being $r \mapsto p^k r$, that is, Shor’s map U_{p^k} at the site p . In the Dirichlet character basis the transfer eigenvalues are $L(\beta, \bar{\chi})/\zeta(\beta)$.

Extremal KMS_β states differ only by the boundary residue of the chain, and $\beta \rightarrow 1^+$ kills every nontrivial character sector: the symmetry breaking of [9] read as a boundary effect. Shor’s unitary U_a is the level- N Galois symmetry of the system and a snapshot of the dilation flow.

Numerical result 9.9 (MPO contraction against Hurwitz zeta and Dirichlet L -values). [numerical]
`scripts/bcmpto.py` (`outputs/bcmpto.txt`) contracts the chain over all primes below $2 \cdot 10^5$ (occupations $k \leq 60$) and compares with the Hurwitz-zeta closed form $\zeta(\beta)^{-1} b^{-\beta} \sum_r e^{2\pi i ar/b} \zeta(\beta, r/b)$:

$a/b, \beta$	MPO	closed form	diff
1/3, 2.0	$-0.333333 + 0.411340i$	$-0.333333 + 0.411340i$	$1.6 \cdot 10^{-7}$
1/4, 2.0	$-0.125000 + 0.556841i$	$-0.125000 + 0.556840i$	$2.1 \cdot 10^{-7}$
2/5, 1.5	$-0.263990 + 0.143625i$	$-0.263949 + 0.143579i$	$6.1 \cdot 10^{-5}$
1/6, 1.7	$+0.103130 + 0.502210i$	$+0.103130 + 0.502200i$	$1.0 \cdot 10^{-5}$

For $b = 5$, $\beta = 1.5$ the contracted transfer operator on the units is diagonal in the character basis to a spread of $2.6 \cdot 10^{-14}$, with eigenvalues 0.910557 , $0.354022 \mp 0.067781i$, 0.225025 against `mpmath` $L(\beta, \chi)/\zeta(\beta)$ values 0.910557 , $0.353909 \pm 0.067760i$, 0.224953 : agreement to $1.1 \cdot 10^{-4}$, with the expected conjugation between the two lists.

The residuals are the truncation of the Euler product at $2 \cdot 10^5$ and grow as β approaches 1, as they must. The relevance here is structural: the bond space of the phase operators is $\mathbb{Z}/b$, finite and abelian, whereas that of the compressed semigroup is K_S , and the note offers no map between them.

9.8 What is not established

RH itself is not touched; every statement above is an equivalent form of the uniform-rate statement. The innerness argument is not written out (Proposition 9.1), the Adamyan–Arov identification is quoted without a local source, and the bridge is a statement about boundary phases, not about the operators. No identification with the $\beta > 1$ constructions of the parent repository is claimed: those sit above the phase transition, the object here is the $\beta = 1$ phase compressed to a co-invariant subspace.

Conjecture 9.10 (Stinespring form with prime jumps). *[open] There is a Stinespring form of the compressed semigroup $Z(t)$ on K_S whose Kraus (jump) operators are the prime dilations.*

The single-Kraus lift exists trivially and says nothing; the Holevo-form covariant Lindbladian with jump measure ν_σ lives on the full regular representation and is not the compressed object. Whether a jump-type generator on K_S reproduces $Z(t)$ is the open question this note isolates, and its physical index would be the Stinespring environment. A negative answer with a reason is a result; the route is listed in Section 26.

10 The Phantasm I: What the Prime Measure Forces

Phantasm is the name used in this book for the object the Bost–Connes reframe chases (HANDOFF.md, central priority of 2026-09-12): a continuous matrix product state whose one-dimensional space is dilation time, whose bond carries a “Riemann Lindbladian” with the pole of ζ at $s = 1$ as fixed point and the zeros as relaxation modes, and whose ring norm is the prime measure. Because the ring norms of an ordinary cMPS are sums of squares (Theorem 19.3) while the zeros enter the explicit formula with a minus sign, the bond was expected to be $\mathbb{Z}_2$ -graded, the zeros odd, the pole even, and the ring norm a supertrace. This shard and the next record what can be proved about that object from the data at hand: what is forced, what cannot exist, and what exists. The proofs are the codex prover’s (notes/riemann-cmps/astra-proofs.md, Lament style, labels T1–T8), reviewed adversarially by an Opus reviewer (notes/reviews/riemann-cmps-2026-09-12.md); the brief that drafted the statements, several of which the prover corrected, is notes/riemann-cmps/astra-brief.md.

10.1 Prior art: the grading is Deninger’s, the sign is Connes’s

Cited fact 10.1 (Deninger’s graded dynamical cohomology). [cited] In Deninger’s conjectural picture [18] the completed zeta is an alternating product of regularised determinants over H^0 , H^1 , H^2 of a flow Θ : $H^0 = \mathbb{R}$ with $\Theta = 0$; H^1 infinite dimensional, “the spectrum of Θ consisting of the non-trivial zeroes ρ of $\zeta_K(s)$ with their multiplicities”; $H^2 \cong \mathbb{R}$ “but with $\Theta = \text{id}$ ”; and on $\mathbb{R}^{>0}$ the Lefschetz formula reads $\sum_i (-1)^i \text{Tr}(\phi^* | H^i) = 1 - \sum_{\rho} e^{t\rho} + e^t$, equal to the prime comb plus archimedean terms. Verbatim: $H^1_{\text{dyn}}(\overline{\text{spec } \Theta}, \mathbb{R})$ is infinite dimensional, the spectrum of Θ consisting of the non-trivial zeroes ρ of $\zeta_K(s)$ with their multiplicities, (math/0505354:main.tex:323); $H^2_{\text{dyn}}(\overline{\text{spec } \Theta}, \mathbb{R}) \cong \mathbb{R}$ but with $\Theta = \text{id}$. (math/0505354:main.tex:323) $\left(\sum_{i=0}^{\infty} (-1)^i \text{Tr}(\phi^* | H^i_{\text{dyn}}(\overline{\text{spec } \Theta}, \mathbb{R})) \right) = 1 - \sum_{\rho} e^{t\rho} + e^t$ (math/0505354:main.tex:878)

Cited fact 10.2 (Connes: the zeros as an absorption spectrum). [cited] Connes [11]: “the spectral interpretation of the zeros of the Riemann zeta function should be as an absorption spectrum rather than as an emission spectrum, to borrow the language of spectroscopy”; the minus sign is that of H^1 in the Lefschetz formula. Verbatim: the spectral interpretation of the zeros of the Riemann zeta function should be as an absorption spectrum rather than as an emission spectrum, to borrow the language of spectroscopy. (math/9811068:main.tex:579–581).

So the forced part of the Phantasm, a graded operator with the pole even and the zeros odd whose Lefschetz trace is the explicit formula, is Deninger’s programme, and the sign is Connes’s. What this book adds is the reading through ring norms and transfer operators: which data are forced by the ring norms alone, which realisations cannot exist, and which exist concretely.

10.2 Rigidity: the supertrace determines the graded spectrum

Lemma 10.3 (Existence of the supertrace distribution). [proved] For a graded spectral datum (Definition 5.1) the series $\sum \epsilon m e^{\lambda t}$ converges absolutely after pairing with every test function in $C_c^\infty(0, \infty)$ and defines a distribution there, independently of the order of summation.

The estimate integrates by parts in the full complex parameter λ , not only in its imaginary part (the draft’s version was corrected there).

Theorem 10.4 (Rigidity up to cancelling pairs). *[proved] Two graded spectral data with the same supertrace distribution on $(0, \infty)$ have the same net multiplicity $\nu_{\text{net}}(\lambda)$ at every $\lambda \in \mathbb{C}$.*

The proof pairs the supertrace with cut-off versions of $t^k e^{-st}$, k large, the cutoffs nonnegative and monotone in both parameters (which is what part (i) of Proposition 10.12 later needs), shows the limit is $k! \sum_{\lambda} \nu_{\text{net}}(\lambda)(s - \lambda)^{-k-1}$, meromorphic on $\mathbb{C}$ with principal part $k! \nu_{\text{net}}(\lambda)(s - \lambda)^{-k-1}$ at each λ , and compares principal parts. Ring norms therefore see exactly the net graded spectrum and nothing else: not Jordan blocks, not the Hilbert-space geometry, not the operator domain.

Assumption 10.5 (Explicit formula in trace form). *[assumed] For $u > 0$, as distributions on $(0, \infty)$, $\sum_{\rho} e^{\rho u} = e^u - \text{comb}(u) - 1/(e^{2u} - 1)$, the zero sum in symmetric order; equivalently the x -derivative of $\psi_0(x) = x - \sum_{\rho} x^{\rho}/\rho - \log 2\pi - \frac{1}{2} \log(1 - x^{-2})$ at $x = e^u$.*

Assumption 10.6 (Zero count and location). *[assumed] $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T)$; every nontrivial zero has $0 < \text{Re } \rho < 1$ and $\text{Im } \rho \neq 0$; the zero multiset is locally finite and invariant, with multiplicity, under conjugation and $\rho \mapsto 1 - \rho$. No RH input.*

Corollary 10.7 (The forced net spectrum). *[proved-conditional] A graded datum whose supertrace on $(0, \infty)$ is $\text{comb}(u)$ has $\nu_{\text{net}}(1) = 1$, $\nu_{\text{net}}(\rho) = -m_{\rho}$ at each nontrivial zero, $\nu_{\text{net}}(-2k) = -1$ for $k \geq 1$, and $\nu_{\text{net}} = 0$ elsewhere. One whose supertrace is $2P_+(t)$ has $\nu_{\text{net}}(0) = 1$, $\nu_{\text{net}}(-\bar{\rho}/2) = -m_{\rho}$, $\nu_{\text{net}}(-(k + \frac{1}{2})) = -1$, and 0 elsewhere.*

The second datum is the first transported by $\lambda \mapsto (\lambda - 1)/2$ and the symmetry $\rho \mapsto 1 - \rho$, with the half-time Jacobian $\text{comb}(t/2) = 2 \sum \Lambda(n) \delta(t - 2 \log n)$; the archimedean term $1/(e^{2u} - 1) = \sum_{k \geq 1} e^{-2ku}$ becomes the odd ladder $-(k + \frac{1}{2})$.

10.3 Positivity forces the parities

Assumption 10.8 (Positive distributions). *[assumed] A distribution on $(0, \infty)$ that is nonnegative on nonnegative test functions is integration against a unique positive Radon measure, and conversely.*

Theorem 10.9 (Finite parity changes). *[proved-conditional] Assign one parity per value of $\{1\} \cup \{\rho\} \cup \{-2k\}$ and let F_Z be the nontrivial zeros whose parity differs from the reference (pole even, everything else odd), F_T the flipped trivial zeros, b the pole flag. If F_Z is finite, the supertrace is a positive distribution iff $b = 0$ and $F_Z = \emptyset$, while F_T may be any set, even infinite. The supertrace is real iff the flipped set is conjugation-invariant.*

The flipped part is then a real-analytic function; positivity of comb plus a continuous function forces the function nonnegative; a flipped pole gives $-2e^u$, and a flipped set of zeros gives a leading trigonometric polynomial with zero mean and positive mean square, which is negative on an unbounded set. The draft's “a cosine is negative somewhere” was replaced by this argument on the whole leading-frequency sum.

Assumption 10.10 (Landau's theorem, Laplace form). *[assumed] The Laplace transform of a positive measure on $(0, \infty)$ with finite abscissa of convergence σ_c has a singularity at $s = \sigma_c$.*

Cited fact 10.11 (Landau's theorem, Dirichlet form). *[cited] A Dirichlet series with nonnegative coefficients does not extend holomorphically to a neighbourhood of its abscissa of absolute convergence [63]. Verbatim: Suppose that $f(s) = \sum a_n n^{-s}$ has abscissa of absolute convergence equal to 0. If $a_n \in \mathbb{R}$, $a_n \geq 0$ for all n then f does not extend holomorphically to a neighborhood of $s = 0$. (1009.0228:main.tex:177).*

Proposition 10.12 (Infinite flips: the pole, and the zeros under domination). *[proved-conditional]* With F_Z possibly infinite: (i) positivity of the supertrace alone forces $b = 0$, the pole even, by monotone convergence of the cutoff pairings, with no domination premise and no Landau input; (ii) if moreover the supertrace dominates the comb as measures then $F_Z = \emptyset$. In both parts F_T remains arbitrary.

Part (i) is the reviewer’s strengthening of the prover’s statement; part (ii) cannot be freed of its premise by the same route, because for infinite F_Z the flipped subseries is only a distribution and can carry negative mass at the prime-power atoms.

Observation 10.13 (The unrestricted infinite case is open). *[sketched]* Under positivity alone, with infinitely many zero parities flipped, the difference between the supertrace and the comb is a signed measure whose atoms at the prime powers can be negative, so the Laplace-transform argument does not close; whether positivity alone forces every parity is open here. Exact equality to the prime measure is rigid by Theorem 10.4 regardless.

10.4 The no-go: numerators are odd

Cited fact 10.14 (Lidskii’s theorem). *[cited]* The trace of a trace-class operator on a Hilbert space coincides with its spectral trace, the sum of its eigenvalues counted with algebraic multiplicity [66, 68]. Verbatim: V.B. Lidskii [2], in 1959, proved his famous theorem on the coincidence of the trace of the S_1 -operator in an (infinite dimensional) Hilbert space with its spectral trace $\sum_{k=1}^{\infty} \mu_k(T)$. (1105.2914:main.tex:119-121).

Theorem 10.15 (No ungraded realisation of a numerator). *[proved]* Let $N_n = \sum_i b_i^n - \sum_j \alpha_j^n$ with nonzero b_i, α_j , the two multisets disjoint after cancellation and at least one α_j . Then no finite matrix and no trace-class operator on a separable Hilbert space has $\text{Tr } E^n = N_n$ for all $n \geq 1$.

At $u = 1/\alpha_j$ the generating function $\sum_n N_n u^n$ has residue $+m_j/\alpha_j$, whereas $\sum_\lambda m_E(\lambda) \lambda u / (1 - \lambda u)$ has residue $-m_E(\alpha_j)/\alpha_j$ with $m_E \geq 0$; a numerator root needs a negative multiplicity.

Proposition 10.16 (Trace and supertrace criteria). *[proved]* (N_n) is the trace sequence of a trace-class operator iff $\exp(\sum_n N_n u^n / n) = \prod_i (1 - \lambda_i u)^{-1}$ with $\sum |\lambda_i| < \infty$; it is the supertrace sequence of a finite graded matrix iff that germ is a rational function with value 1 at 0, the even net spectrum being the reciprocal poles and the odd net spectrum the reciprocal zeros after cancellation. Eigenvalue 0 and nilpotent blocks are invisible to all positive powers.

Assumption 10.17 (Weil’s theorem for curves). *[assumed]* For a smooth projective geometrically connected curve C of genus g over $\mathbb{F}_q$ there are $2g$ nonzero algebraic numbers α_j , $|\alpha_j| = \sqrt{q}$, the Frobenius eigenvalues on H^1 , with $\#C(\mathbb{F}_{q^n}) = 1 + q^n - \sum_j \alpha_j^n$.

Corollary 10.18 (Curves need the grading; graphs do not). *[proved-conditional]* For a curve of genus $g \geq 1$ the counts N_n are not the trace sequence of any finite matrix or trace-class operator, hence not the ring norms $\text{Tr}(\sum_s A_s \otimes \overline{A_s})^n$ of any translation-invariant bosonic MPS with a fixed local tensor; they are the supertrace sequence of $\text{diag}(q, 1)_+ \oplus (\text{Fr}|H^1)_-$, and the odd net spectrum is forced to be the Frobenius eigenvalues. Non-backtracking graph counts $\text{Tr } T^\ell$ are ordinary trace sequences, and the Ihara zeta $\det(1 - uT)^{-1}$ has no finite zeros.

This is the precise content of “the sign” of Section 16: a zeta function with a numerator has its interesting eigenvalues as negative net multiplicities, and the grading is the only way to write them as a trace. It is also why the quadratic Artin–Schreier transfer matrix of Section 14, whose trace is an exponential sum, is a complex partition-function transfer and not the norm transfer of a state (a point the prover insisted on): the honest state, the uniform superposition over the points of the curve, has no fixed-tensor MPS with those norms.

10.5 What exists: the compressed semigroup is the odd sector

Assumption 10.19 (Lax–Phillips modes of the compressed semigroup). [assumed] $Z(t) = e^{tB}$ is a strongly continuous contraction semigroup on K_S whose generator has pure point spectrum $\{-\bar{\rho}/2\}$, one eigenvector per zero with multiplicity m_ρ , and $Z(t) \rightarrow 0$ strongly. (The book’s own sketch of this standard input is Proposition 9.2.)

Proposition 10.20 (The Riemann graded generator). [proved-conditional] On $\mathfrak{B}$ (Definition 5.3), $e^{tG} = 1 \oplus Z(t) \oplus \text{diag}(e^{-(k+1/2)t})$ is a strongly continuous contraction semigroup whose fixed vectors are exactly the even line, and for $t > 0$

$$\text{str } e^{tG} = 1 - \text{Tr}_{\text{dist}} Z(t) - \sum_{k \geq 1} e^{-(k+1/2)t} = 2P_+(t) = 2 \sum_{n \geq 2} \frac{\Lambda(n)}{n} \delta(t - 2 \log n).$$

The single-particle graded spectrum is thus forced (Corollary 10.7) and realised. Its odd K_S part is the Riemann channel of Section 9; the missing even fixed line is the pole; the ladder is the archimedean place. The uncentred trace of $Z(t)$ alone is restated as Proposition 9.5.

Proposition 10.21 (The ladder parity is free; all-even is impossible). [proved-conditional] Over the even line plus odd K_S alone the supertrace is $2P_+ + e^{-t/2}/(e^t - 1)$; with the ladder counted even it is $2P_+ + 2e^{-t/2}/(e^t - 1)$; both are positive, so positivity does not fix the ladder parity. Exact equality to $2P_+$ forces the net spectrum of G , and no all-even graded datum has supertrace $2P_+$ or comb.

Observation 10.22 (Scope of the interpretation). [sketched] RH is the statement that every odd K_S mode of G has real part $-\frac{1}{4}$; the even line represents the pole of ζ (the constant function on the modular surface, the residue of the Eisenstein series at $s = 1$), an identification not constructed here; $Z(t)$ is not trace class for $t > 0$; and the identity above is a spectral identity, not a cMPS tensor whose Hilbert norm is the prime measure.

The numerical check of the supertrace identity, with 3000 zeros, is numerical result 11.19 in the next shard.

11 The Phantasm II: Vacuum Decay, the Prime Chain, and the Two Halves

Section 10 realised the forced graded spectrum by a Hilbert-space semigroup. The reframe asks for more: a Lindbladian on operators, with a fixed point that is a state and the zeros as relaxation modes. This shard records the two Lindbladians that can actually be built, and why each is only half of what is wanted.

11.1 Prior art on graded bonds

Cited fact 11.1 (Fermionic matrix product states contract by a supertrace). [cited] Bultinck, Williamson, Haegeman and Verstraete [10] contract fermionic MPS with the graded contraction $\mathcal{C}(|i\rangle \otimes |j\rangle) = (-1)^{|i|} \delta_{ij}$, so a closed ring of a $\mathbb{Z}_2$ -graded bond is a supertrace, with a parity insertion in the odd sector. Verbatim: `\mathcal{C}(\ket{i} \otimes \bra{j}) = (-1)^{|i||j|} \mathcal{C}(\bra{j} \otimes \ket{i}) = (-1)^{|i|} \delta_{i,j}`. (1610.07849:FermionicMPS.tex)

Cited fact 11.2 (The fermionic cMPS already has the graded bond). [cited] Haegeman, Cirac, Osborne and Verstraete [30] give the fermionic cMPS: a $\mathbb{Z}_2$ -graded bond with block-diagonal Q , block-off-diagonal R for fermionic species, and parity-graded boundary matrices. Verbatim: `R_{\alpha} = \begin{bmatrix} 0_{D^{(+)}} \times D^{(+)} & R_{\alpha}^{(+)} \\ R_{\alpha}^{(-)} & 0_{D^{(-)}} \times D^{(-)} \end{bmatrix} \quad \text{(particle species } \alpha \text{ is fermionic)}`. (1211.3935:calculus.tex:327).

Cited fact 11.3 (The transfer fixed point is the half-line reduced state). [cited] Verstraete and Cirac [75]: the eigenvalues of the fixed point of the cMPS transfer Lindbladian are the squared Schmidt coefficients of a bipartition. Verbatim: the eigenvalues of the matrix $\rho(x)$ would exactly correspond to the squares of the Schmidt coefficients when considering a bipartition at site x (1002.1824:mpsQFT4.tex:81).

So the graded bond with a supertrace ring contraction is not new as a tensor-network construction; the arithmetic reading is what this book proposes.

11.2 Vacuum decay: the zeros as odd coherences, the vacuum as fixed point

Theorem 11.4 (The finite vacuum-decay model). [proved] For finite-dimensional H , B with $-(B + B^\dagger) = |j\rangle\langle j|$ and $e^{tB} \rightarrow 0$, the vacuum-decay Lindbladian $\mathcal{L}_{\text{vac}}$ of Definition 5.4 is a trace-preserving GKLS generator with unique stationary density Ω_{vac} ; in the block decomposition of $B(\mathbb{C} \oplus H)$ it is triangular, the coherence blocks evolve exactly by e^{tB} , and

$$\text{spec } \mathcal{L}_{\text{vac}} = \{0\} \uplus \text{spec } B \uplus \overline{\text{spec } B} \uplus \{b_i + \bar{b}_j\},$$

ordered pairs with multiplicities; the coherences are odd and the populations even.

Explicitly $e^{t\mathcal{L}_{\text{vac}}}$ is conjugation by $1 \oplus e^{tB}$ plus the completely positive feed $X \mapsto \text{Tr}[(1 - e^{tB^\dagger} e^{tB})X] \Omega_{\text{vac}}$.

Corollary 11.5 (Traces of the vacuum-decay semigroup). [proved] $\text{Tr } e^{t\mathcal{L}_{\text{vac}}} = |1 + \text{Tr } e^{tB}|^2$ and $\text{str } e^{t\mathcal{L}_{\text{vac}}} = |1 - \text{Tr } e^{tB}|^2$: the vacuum-decay transfer carries both coherence lists and all pair sums, and is not the single-particle generator $\mathbf{G}$.

Proposition 11.6 (The absorbing-vacuum channel). [proved-conditional] For any strongly continuous contraction semigroup $Z_t \rightarrow 0$ (strongly) on a separable Hilbert space, the

absorbing-vacuum channel of Definition 5.4 is a strongly continuous CPTP semigroup on trace-class operators on $\mathbb{C} \oplus H$ with unique stationary density Ω_{vac} and exact odd coherence evolution by Z_t . For $H = K_S$ it realises the Riemann channel as its odd coherence sector.

Observation 11.7 (Zeros in the coherences, pair sums in the populations). [sketched] For $H = K_S$ each zero appears once as an odd coherence mode $-\bar{\rho}/2$ and once as its conjugate (odd multiplicity $2m_\rho$), the even populations carry the pair sums $-(\bar{\rho}_n + \rho_m)/2$, and their formal trace $|\text{Tr}_{\text{dist}} Z(t)|^2$ is not a defined product of distributions; no pair-correlation statement follows.

Proposition 11.8 (The vacuum is absorbing, and the entanglement is trivial). [proved]
The jump reinserts into the vacuum, but the renewal integral $\int_0^T e^{t\bar{B}} \Omega_{\text{vac}} e^{tB^\dagger} dt = T \Omega_{\text{vac}}$ diverges: the vacuum is absorbing, not a rebound state of finite holding time, so this model does not solve the renewal equation of the SHW picture with a rebound inside K_S . The canonical cMPS selected by the stationary bond density has a single nonzero Schmidt weight: its half-line entanglement is trivial.

This is the first half: side B without side A. The construction has the right relaxation modes and no entanglement.

11.3 The prime chain: the Gibbs fixed point, and no zeros

Cited fact 11.9 (Bost–Connes KMS classification). [cited]

For $0 < \beta \leq 1$ the Bost–Connes system has a unique KMS_β state; for $\beta > 1$ the extremal KMS states are the Gibbs states with partition function $\zeta(\beta)$, labelled by embeddings of $\mathbb{Q}^{\text{ab}}$ [13, 9]. Verbatim: For $0 < \beta \leq 1$ there exists a unique $\{\text{rm KMS}\}_{\beta}$ state φ_{β} for the above system. (math/0404128:houcheschapter1final7.tex:1820–182

Proposition 11.10 (Normal KMS states are the Gibbs densities, only above the transition). [proved]
For the dynamics $\text{Ad}(N^{\text{it}})$ on $B(\ell^2(\mathbb{N}))$ a normal KMS_β state exists iff $\beta > 1$, and then it is ϱ_β^{G} ; the spectrum of $-\log \varrho_\beta^{\text{G}}$ is $\{\beta \log n + \log \zeta(\beta)\}$; and TFD_β factorises over the prime sites, with bond dimension one along the prime chain. The unique critical KMS state of the Bost–Connes algebra is not a normal density in this representation.

Lemma 11.11 (M/M/1 spectrum). [proved] *The birth–death generator on $\mathbb{N}_0$ with constant up-rate λ and down-rate $\lambda' > \lambda$, on L^2 of its geometric stationary law, has spectrum $\{0\} \cup [-(\sqrt{\lambda} + \sqrt{\lambda'})^2, -(\sqrt{\lambda} - \sqrt{\lambda'})^2]$, the zero simple.*

Proposition 11.12 (The classical prime-chain dynamics). [proved] *The diagonal part of $\mathcal{L}_\beta^{\text{pc}}$ ($\sum_p r_p < \infty$, $\beta > 1$) is a nonexplosive product of independent birth–death chains, one per prime, with unique stationary law $n^{-\beta}/\zeta(\beta)$; its self-adjoint generator on L^2 of that law has spectrum the closure of $\{0\}$ and the finite Minkowski sums of the bands $[-r_p(p^{\beta/2} + 1)^2, -r_p(p^{\beta/2} - 1)^2]$. It is real, and contains no zero of ζ .*

Proposition 11.13 (Off-diagonal sectors). [proved] *If $\sum_p r_p p^\beta < \infty$, $\mathcal{L}_\beta^{\text{pc}}$ generates a bounded CPTP semigroup with ϱ_β^{G} stationary, and on the weighted Hilbert space with norm $\|\varrho_\beta^{\text{G}-1/4} x \varrho_\beta^{\text{G}-1/4}\|_{\text{HS}}$ it is bounded, self-adjoint and nonpositive, off-diagonal sectors included: its spectrum is real.*

Corollary 11.14 (Delocalisation at the transition). [proved] *As $\beta \downarrow 1$ the Gibbs density has no trace-norm limit and its mass on every finite-rank projection tends to zero; for $0 < \beta \leq 1$ the prime occupation process has no stationary probability on finite-support configurations. This is the precise content available for the Bost–Connes transition; it is not a proof of ergodicity breaking of a Riemann Lindbladian.*

This is the second half: side A without side B. The stationary state is right and the spectrum is blind to the zeros, as every prime-by-prime construction is (observation 26.5).

11.4 Entropy of the bond state and its cutoffs

Numerical result 11.15 (Entropy of the Gibbs density under cutoffs). [numerical]
`scripts/bc_entropy.py` (`outputs/bc_entropy.txt`): $\mathcal{S}_{\text{ent}}(\varrho_\beta^G) = \log \zeta(\beta) - \beta \zeta'(\beta)/\zeta(\beta)$ against the prime-product sum; with the prime cutoff $p \leq P$ at $\beta = 1$ the entropy follows $\log P + \log \log P$ to 10^{-3} at $P = 10^6$, grows like $\beta P^{1-\beta}/(1-\beta)$ for $\beta < 1$ and saturates for $\beta > 1$; with the bond cutoff $n \leq N$ at $\beta = 1$ the ratio $\mathcal{S}_{\text{ent}}/\log N$ falls towards $\frac{1}{2}$ (0.657 at $N = 10^7$) against 1 for $\beta < 1$ and 0 for $\beta > 1$; the number of entanglement levels below E is $\lfloor e^E \rfloor$. Constants fixed by the review: $\mathcal{S}_{\text{ent}} = 1/(\beta - 1) + \log(1/(\beta - 1)) + (1 - \gamma_E) + o(1)$; the prime-cutoff constant is exactly 0, the two Mertens constants cancelling; the bond-cutoff constant is $-\gamma_E/2$.

Observation 11.16 (The entanglement Hamiltonian is free and Hagedorn). [sketched]
 If the Phantasm's bond fixed point is ϱ_β^G , then by cited fact 11.3 its entanglement Hamiltonian is the Bost–Connes Hamiltonian $\beta \log N$, free over the primes with single-particle levels $\log p$; the entanglement entropy is finite for $\beta > 1$, grows like $\log P$ (prime cutoff) or $\frac{1}{2} \log N$ (bond cutoff) at $\beta = 1$, and like a power of the cutoff for $\beta < 1$: gapped, critical, and not a cMPS at all. The level density e^E is Hagedorn growth with $\beta_H = 1$ (Julia's reading of the transition [40]), not Cardy growth, so the finite-entanglement scaling law of critical CFTs does not apply to this state.

11.5 The two halves, and the guiding observation

Observation 11.17 (Complementary halves). [sketched] The vacuum-decay channel has the zeros as odd relaxation modes (each zero twice, as a coherence and its conjugate, plus an even pair-sum sector) but a pure stationary state, hence trivial entanglement. The prime-chain Gibbs Lindbladian, for $\beta > 1$, has ϱ_β^G as stationary state, with entanglement spectrum $\beta \log n + \log \zeta(\beta)$, proportional to the ring lengths $2 \log n$, but a real spectrum with no zeros: unconditionally for its classical part, and on the weighted space of Proposition 11.13 for the full generator when $\sum r_p p^\beta < \infty$. No construction in this book has both, and the Phantasm would need both (conjecture 26.3).

Observation 11.18 (The grading is trivial versus nontrivial characters). [sketched-conditional]
 In the Artin–Schreier super-transfer matrix (Theorem 15.9) the even block is the trivial additive-character sector, the de Bruijn count q^n (the pole at $u = 1/q$), together with the point at infinity (the pole at $u = 1$, not a character sector), and the odd block is the direct sum of the corrected Frobenius blocks F_{ag} over the nontrivial characters, which carries the zeros as its net spectrum. By analogy, a Galois-symmetric Riemann bond decomposed under $\hat{\mathbb{Z}}^\times$ would have as even sector the pole of ζ , the trivial Dirichlet character being the only one with a pole, and as odd sector the zeros of every $L(s, \chi)$: the graded object would compute the zeta of the cover, the Dedekind zeta of the cyclotomic field at level N , not ζ alone. This is consistent with the character-basis transfer eigenvalues $L(\beta, \bar{\chi})/\zeta(\beta)$ of Proposition 9.8, which by Proposition 11.12 is nevertheless not that object, and it is where Connes's trace formula lands (all L -functions with Grössencharakter). It is an analogy, not a theorem; its finite-level test is conjecture 26.4.

Numerical result 11.19 (Supertrace, vacuum decay and prime-chain bands, checked). [numerical]
`scripts/riemann_graded_trace.py` (`outputs/riemann_graded_trace.txt`, 42/42 checks): the windowed supertrace of G with 3000 zeros against $2P_+$ has ratio $1 - 5 \cdot 10^{-4}$ at twelve prime powers and is positive to within truncation (minimum -0.041 against peaks of 147); both ladder alternatives match and are positive; flipping the first zero pair to even gives -2.88 ; the finite

vacuum-decay spectral multiset matches Theorem 11.4 to $7 \cdot 10^{-14}$ for $D = 3, 5, 8$ with CPTP and coherence checks; single-prime birth–death spectra lie in the $M/M/1$ bands and the product gap equals the $p = 2$ gap.

Observation 11.20 (Established scope). [sketched] Established: the forced graded net spectrum and its realisation by a graded contraction semigroup with one even fixed line; the no-go for ungraded trace-class realisations; a genuine absorbing-vacuum channel with the Riemann channel as its odd coherences; the normal-Gibbs classification and the prime-chain spectra. Not established: unrestricted infinite parity rigidity, the analytic realisation of the Lax–Phillips input, a cMPS whose Hilbert norm is the prime measure, and a stationary entangled bond with the critical arithmetic state.

12 The Phantasm III: Stationarity, Symmetry, and Generator Freedom

TJO's specification is an unknown Lindbladian with the BC state stationary, Galois symmetry, and compatible adelic symplectic structure. The no-event generator and the jumps are both unknown. This section completes a finite constraint calculation, not the construction of the adelic bond. Detailed arguments B0–B6 are in `notes/bc-symmetry-generators.md`; source loci are in `notes/extract/bc-symmetry-sources.md`. These arguments have not had an independent review and therefore carry **sketched**, including the statements with complete written derivations. Numerical evidence is separately tagged. No claim of novelty is made for the general GKLS machinery.

12.1 The state on the finite phase algebra

Proposition 12.1 (Residue probabilities).

[sketched] For the state of Definition 7.1,

$$w_\beta(0) = p^{-\beta}, \quad w_\beta(x) = \frac{1 - p^{-\beta}}{p - 1} \quad (x \neq 0).$$

At criticality the phase marginal is uniform at every integer level N , compatible under reduction, and $\varrho_{p,1}^{\text{res}} = I/p$.

This is finite Fourier inversion of the BC KMS formula [13]; for $\beta > 1$ it also follows by averaging the residue-class Gibbs sums over the Galois action. A single extremal state is generally not invariant. Uniform phase probabilities fix the diagonal of a matrix extension, not its off-diagonal entries. The full critical BC state is not a normal density in the standard representation (Proposition 11.10).

Proposition 12.2 (Finite representations collapse the BC phases).

[sketched]

Every finite-dimensional unital $$ -representation of the full BC algebra sends all $e(r)$, $r \in \mathbb{Q}/\mathbb{Z}$, to I . Consequently it cannot realise the nontrivial critical finite phase marginal.*

Indeed, $\mathbf{m}_n^* \mathbf{m}_n = I$ makes $\mathbf{m}_n$ unitary in finite dimension. The BC relation gives $I = \mathbf{m}_n \mathbf{m}_n^* = n^{-1} \sum_{k=0}^{n-1} e(k/n)$, the spectral projection onto the eigenvalue 1 of $e(1/n)$. Thus $e(1/n) = I$ for every n . This excludes an exact finite representation; it does not exclude restrictions, CP compressions, or asymptotic models.

Proposition 12.3 (Invariant state extensions).

[sketched] A unique stationary density of a

covariant semigroup is invariant under its symmetry group. The diagonal extension $\varrho_{p,\beta}^{\text{res}}$ is Galois invariant, and is Weil invariant iff $\beta = 1$. The same phase marginal has a unique Weil-invariant matrix extension, when it is positive:

$$\varrho_{\text{b}} = aI + (r - a)\mathbf{P}, \quad r = p^{-\beta}, \quad a = \frac{1 - r}{p - 1}.$$

Its parity-block eigenvalues are r and $2a - r$, so positivity holds iff $r \leq 2/(p+1)$. Full Weyl invariance forces I/p .

The Weil-invariant operator space is $\text{span}\{I, \mathbf{P}\}$: the symplectic group is transitive on the nonzero Weyl labels and $\sum_v X_v = p\mathbf{P}$. The diagonal fixes the two coefficients. These coherent extensions are explicit state-extension examples, not an identification with the full BC bond. Above the transition, covariance can instead act on a family of stationary extremal states; uniqueness is the premise of the first assertion.

12.2 The complete finite feasibility problem

Theorem 12.4 (A linear section of the conditional Choi cone). *[sketched]* The cone $\mathfrak{C}_U(\varrho_b)$
consists exactly of complex-linear maps L with

$$\begin{aligned} L(X^*) &= L(X)^*, \quad \text{Tr } L(X) = 0, \quad L(\varrho_b) = 0, \\ [L, \text{Ad}(U_g)] &= 0 \quad \text{for every } g, \quad QC_LQ \geq 0. \end{aligned}$$

Equivalently, in an orthonormal traceless operator basis F_i ,

$$L(X) = -i[H, X] + \sum_{i,j} A_{ij} (F_i X F_j^* - \tfrac{1}{2} \{F_j^* F_i, X\}), \quad A \geq 0, \quad H = H^*.$$

With $\text{Tr } H = 0$, covariance requires H invariant and A commuting with the induced representation on traceless operators. If that representation is $\bigoplus_\lambda V_\lambda \otimes \mathbb{C}^{m_\lambda}$, then $A = \bigoplus_\lambda I_{V_\lambda} \otimes A_\lambda$, $A_\lambda \geq 0$; stationarity supplies further linear equations. Every invariant faithful ϱ_b has an interior feasible generator $X \mapsto \varrho_b \text{Tr } X - X$.

Necessity follows by differentiating a positive channel Choi matrix on the complement of the maximally entangled vector. For sufficiency, the positive matrix QC_LQ supplies the traceless Kraus part; the remainder is the Choi matrix of $KX + XK^*$, and trace annihilation fixes its Hermitian part. This yields the bounded GKLS form recalled in [67]. The reset generator has conditional Choi matrix $Q(I \otimes \varrho_b)Q$, strictly positive on the range of Q .

Theorem 12.5 (Dimensions at the critical tracial extension). *[sketched]* For odd prime p , at stationary density I/p , the real dimensions of the generator cones' linear spans are

<i>commuting actions</i>	<i>dimension, including time scale</i>
<i>Galois torus</i>	$(p-1)(p+1)^2$
<i>full linear Weil action</i>	$2p-2$
<i>Weyl translations and Galois torus</i>	$p+1$
<i>Weyl translations and full Weil action</i>	1.

In Weyl coordinates, trace annihilation and stationarity remove the identity row and column. The first two dimensions count simultaneous group orbits on ordered pairs of nonzero vectors. The torus acts freely; the symplectic orbits are $w = av$ ($a \neq 0$) and $\det(v, w) = b$ ($b \neq 0$), giving $2(p-1)$. Since $-I$ belongs to both groups, Hermiticity makes the orbit coefficients real. The depolarizing generator is an interior point of the conditional Choi cone, so positivity does not reduce these dimensions. Orbit coefficients are not independent nonnegative rates in the first two rows.

Proposition 12.6 (The cost of extra Weyl covariance). *[sketched]*
All Weyl-covariant generators are $L(X) = \sum_{v \neq 0} c_v (X_v X X_v^ - X)$, $c_v \geq 0$. Galois covariance makes the rates constant on the two punctured axes and the $p-1$ sets $xy = b \neq 0$. In particular $c_v = c_{-v}$, making every Weyl eigenvalue real and nonpositive. Full symplectic covariance makes all the rates equal and gives exactly $L = \gamma(\mathcal{D}_p - \text{id})$, $\gamma \geq 0$.*

This extra commuting symmetry removes the oscillatory zero-like modes in this model. It is not a consequence of merely requiring a symplectic representation or an adelic structure.

12.3 Odd spectral freedom and coupling freedom

Proposition 12.7 (A symplectic odd-sector counterexample). *[sketched] Let a be a primitive root modulo p and set $R_a(I) = 0$, $R_a(X_v) = X_{av}$ for $v \neq 0$. Then*

$$L = \mathcal{D}_p - \text{id} + \varepsilon R_a, \quad \varepsilon = \frac{1}{4p^2}$$

is a CPTP generator with unique stationary state I/p and full Weil, Galois and parity covariance. For $p = 7$ its odd eigenvalues are $-1 - \varepsilon$ and $-1 + \varepsilon/2 \pm i\sqrt{3}\varepsilon/2$, each with multiplicity 8. The odd decay rates are unequal.

The map R_a is Hermiticity preserving and equivariant, but need not itself be positive. Its Choi operator norm is bounded by its Frobenius norm $\sqrt{p^2 - 1} < p$, whereas the depolarizer's conditional Choi matrix is Q/p ; the specified perturbation therefore remains strictly positive on the range of Q . At $p = 7$, scalar multiplication has eight cycles of length six, and the odd sector selects the roots whose cube is -1 . Thus the counterexample does not fit any input zero positions.

Proposition 12.8 (Couplings invisible to the one-prime restrictions). *[sketched] For distinct primes p, q , fixed positive rates a, b , and $0 \leq c \leq \min(a, b)$, let*

$$L_c = (a - c)(\mathcal{D}_p - \text{id}) \otimes \text{id} + (b - c)\text{id} \otimes (\mathcal{D}_q - \text{id}) \\ + c(\mathcal{D}_p \otimes \mathcal{D}_q - \text{id}).$$

These CPTP generators have the unique stationary state $I/(pq)$ and full local Weyl/Weil covariance. Their restrictions on the one-prime observable algebras are independent of c , with rates a and b , while the rate on the joint traceless sector is $a + b - c$.

Each summand is a nonnegative multiple of a channel minus the identity. The four tensor sectors give the eigenvalues $0, -a, -b, -a - b + c$ directly. This supplies compatible two-prime residue dynamics under CRT with a variable coupling; it is not an infinite adelic construction.

Numerical result 12.9 (Finite arithmetic symmetry checks). *[numerical] The script `scripts/bc_symmetry_generators.py` passes 427 checks; its output is `outputs/bc_symmetry_generators.txt`. The checks cover $p = 3, 5, 7$: BC residue probabilities against Hurwitz sums, invariant matrix extensions, orbit dimensions against independent Burnside counts, conditional and finite-time Choi positivity, trace preservation, stationarity and uniqueness, covariance, Egorov, odd-sector spectra, and two-prime restrictions. The four cone dimensions are respectively $(32, 4, 4, 1)$, $(144, 8, 6, 1)$, and $(384, 12, 8, 1)$.*

12.4 What is still missing

The finite inverse problem is now a concrete positive-semidefinite cone. The missing input is a representation or CP comparison map connecting the full BC algebra, including its prime isometries, to the scattering bond, with compatible local Weil data. It must select the permitted commuting symmetries and constrain the generator blocks and the inter-prime couplings. The $\Gamma_0(N)$ character/scattering computation of conjecture 26.4 remains open. Prime-power levels, the real place, metaplectic compatibility and the critical operator-algebra limit have not been constructed. A chosen graded transfer spectrum alone does not identify a physical cMPS norm.

The earlier reset test $-(B\rho_b + \rho_b B^*) \geq 0$ is a criterion for a fixed finite no-event generator B and scalar reset, when its exit flux is positive. It is not a no-go criterion for the full cone in Theorem 12.4, where B and all jumps may vary.

13 Weil–LPS Channels: Exactly Ramanujan Quantum Expanders

This is the finite model of the Riemann channel of Section 9 with every ingredient present and every claim checkable: commuting dilations indexed by primes, an arithmetic quotient supplying the expansion, and the Weil representation as the Hilbert space. The construction was proposed in the founding session’s handoff and computed on 2026-09-11; everything below was computed, and every construction step carries its own numerical check. The object itself is Definition 4.11, its ingredients Definition 4.9 and Definition 4.10.

13.1 The object

Fix an odd prime p and an LPS parameter $q \equiv 1 \pmod{4}$ with q a square mod p . The Weil intertwiner exists only for symplectic (determinant one) matrices, and q being a square is what lets the norm- q quaternions land in $\mathrm{SL}_2(\mathbb{F}_p)$ after scaling. The admissible pairs computed are these.

p	q	$ \mathrm{PSL}_2(\mathbb{F}_p) $	Weil blocks $M_{(p+1)/2}, M_{(p-1)/2}$
5	29	60	M_3, M_2
7	29	168	M_4, M_3
11	5	660	M_6, M_5
13	17, 29	1092	M_7, M_6
17	13	2448	M_9, M_8
19	5, 17	3420	M_{10}, M_9
29	5, 13	12180	M_{15}, M_{14}

Step 1, generators. The $q+1$ integer quaternions $\alpha = a_0 + a_1i + a_2j + a_3k$ of norm q with $a_0 > 0$ odd and a_1, a_2, a_3 even. Jacobi’s four-square theorem gives exactly $q+1$ of them; the enumeration was checked against the divisor sum $\sum_{d|n} d$ for the norms q, q^2 and q_1q_2 used below. The same $p=5$ quaternions appear in Appendix C of [38].

Step 2, images in $\mathrm{SL}_2(\mathbb{F}_p)$. Split the Hamilton quaternions over $\mathbb{F}_p$ by

$$\mathbf{I} = \begin{pmatrix} x & y \\ y & -x \end{pmatrix}, \quad \mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad x^2 + y^2 + 1 \equiv 0 \pmod{p}.$$

Then $\alpha \mapsto a_0 + a_1\mathbf{I} + a_2\mathbf{J} + a_3\mathbf{IJ}$ is a ring homomorphism whose determinant is the quaternion norm. Dividing by a square root s_q of q mod p gives $g_\alpha \in \mathrm{SL}_2(\mathbb{F}_p)$. Checked: $\det g_\alpha = 1$, $g_{\bar{\alpha}} = g_\alpha^{-1}$, and all $q+1$ images distinct.

Step 3, the Weil representation, built and checked numerically. On $\ell^2(\mathbb{F}_p)$ take the clock and shift operators Z, X and the Heisenberg–Weyl operators $T(u, v) = X^u Z^v$. The Fourier transform $\mathcal{F}$, the chirp M and a dilation D each satisfy an intertwining relation of the form

$$W(g)T(u, v)W(g)^\dagger \propto T(g(u, v)),$$

with a symplectic g that the code *reads off* rather than assumes. A breadth-first search over $\mathrm{SL}_2(\mathbb{F}_p)$ in these three generators assigns a unitary $W(g)$ to every group element, projectively; for $p=13$ the table has the expected $p(p^2-1) = 2184$ entries. The relation above was then verified directly for each LPS generator g_α , with error at most $4 \cdot 10^{-14}$ in every case ($7.2 \cdot 10^{-15}$ for $(p, q) = (13, 17)$, $7.0 \cdot 10^{-15}$ for $(13, 29)$). Since only $\mathrm{Ad} W$ enters the channel, the projective phases are irrelevant.

Step 4, the two irreducible blocks. $W(-1)$ is the parity operator, so W splits into even and odd functions, of dimensions $(p \pm 1)/2$. Each block $W_{\pm}$ is irreducible and -1 acts on it as a scalar, so it is a representation of $\text{PSL}_2(\mathbb{F}_p)$ up to phase. This matters: on the full M_p the parity operator would be a second fixed point of the channel, and Harrow’s theorem needs irreducibility. The channels are

$$\Phi_q^{\pm}(X) = \frac{1}{q+1} \sum_{\alpha} W_{\pm}(g_{\alpha}) X W_{\pm}(g_{\alpha})^{\dagger} \quad \text{on } M_{(p \pm 1)/2},$$

that is, $D = q + 1$ unitary Kraus operators in the normalisation of Section 2. Block unitarity and Hilbert–Schmidt self-adjointness of the superoperator were checked at 10^{-14} (for (13, 17): deviations $2.7 \cdot 10^{-15}$ and $1.8 \cdot 10^{-15}$).

13.2 Exact Ramanujan behaviour

Proposition 13.1 (Weil–LPS channels are exactly Ramanujan). *[sketched]* For every admissible pair $(p; q)$, each block channel $\Phi_q^{\pm}$ is a Ramanujan quantum expander in the sense of Definition 3.17: $\lambda_2(\Phi_q^{\pm}) \leq \lambda_H$, equivalently $(q+1)\lambda_2 \leq 2\sqrt{q}$ in adjacency units.

The argument is two lines. Harrow’s transfer inequality (cited fact 23.5, [31]) says that for the channel built from an irreducible representation of a group and a symmetric generating set, the second singular value of the channel is at most the second singular value of the transition matrix of the Cayley graph on that generating set. Here the group is $\text{PSL}_2(\mathbb{F}_p)$, the generating set is the $q+1$ images g_{α} of Definition 4.9, and the representation is $W_{\pm}$, irreducible by Step 4; so $(q+1)\lambda_2(\Phi_q^{\pm})$ is bounded by the largest modulus of a nontrivial eigenvalue of the LPS Cayley graph A_q . That graph is Ramanujan, $|\lambda| \leq 2\sqrt{q}$, by Lubotzky, Phillips and Sarnak [56], and $2\sqrt{q} = 2\sqrt{D-1}$ is exactly Hastings’ bound λ_H in adjacency units [33]. The status is *sketched* rather than *proved* because the Lubotzky–Phillips–Sarnak theorem has no byte-verified local source in **refs/**, and because the irreducibility of $W_{\pm}$ and the projective-phase bookkeeping of Step 4 are verified numerically here rather than cited. Iyer, Jain, Jordan and Somma [38] state the $\text{SU}(2)$ version of the same mechanism and call explicit Ramanujan quantum expanders a longstanding open problem; see Section 23.

Numerical result 13.2 (Ramanujan, Harrow containment, Hecke, commutation). *[numerical]* For $(p; q)$ in $\{(5; 29), (7; 29), (11; 5), (13; 17, 29), (17; 13), (19; 5, 17), (29; 5, 13)\}$ the channels $\Phi_q^{\pm}$ on the even and odd Weil blocks satisfy the Ramanujan bound of Definition 3.17 and have a unique fixed point; Harrow containment, the Hecke relation of Definition 4.12 and commutation across q hold to 10^{-14} .

Every entry of the table below is on the correct side of the bound $2\sqrt{q}$, in adjacency units.

p	q	Cayley A_q		Φ_q^{+}		Φ_q^{-}		$2\sqrt{q}$
		λ_2	$\lambda_{\min}$	$(q+1)\lambda_2$	$(q+1)\lambda_{\min}$	$(q+1)\lambda_2$	$(q+1)\lambda_{\min}$	
5	29	6.000	−8.000	6.000	−2.000	2.000	2.000	10.770
7	29	8.000	−6.000	6.000	−6.000	2.000	−6.000	10.770
11	5	4.372	−4.024	3.562	−4.024	3.406	−4.024	4.472
13	17	6.928	−7.851	6.000	−6.685	5.555	−5.555	8.246
13	29	9.932	−9.208	9.000	−9.000	9.000	−9.000	10.770
17	13	6.106	−7.090	6.000	−6.828	6.000	−6.828	7.211
19	5	4.275	−4.274	4.243	−4.000	4.243	−4.000	4.472
19	17	7.671	−7.638	7.128	−7.638	7.128	−7.638	8.246
29	5	4.442	−4.411	4.442	−4.049	4.442	−4.049	4.472
29	13	6.765	−6.949	6.765	−6.912	6.579	−6.912	7.211

The further checks, all passed in every case, are these.

- **Harrow containment.** Every channel eigenvalue, times $q + 1$, lies in the spectrum of the $\mathrm{PSL}_2(\mathbb{F}_p)$ Cayley graph to $5 \cdot 10^{-7}$ (for $(13, 17)$: maximum distance $4.4 \cdot 10^{-7}$ on the even block, $3.1 \cdot 10^{-7}$ on the odd). The graph spectrum was computed in double precision on up to 3420 vertices; for $p = 29$ only the extreme eigenvalues were computed, by Lanczos, which is why this is a containment check and not an equality check.
- **Unique fixed point.** The eigenvalue 1 of each block channel has multiplicity one, so the channels are quantum expanders in the sense of Definition 3.16 and not merely bounded.
- **Hecke relation.** $A_q^2 - q \mathbf{1} = A_{q^2}$ on $\mathrm{PSL}_2(\mathbb{F}_p)$, with A_{q^2} built from the $q^2 + q + 1$ normalised quaternions of norm q^2 (307 for $q = 17$, 871 for $q = 29$). Residual exactly 0. This is the tree relation “two steps without backtracking”.
- **Commutation across q .** For $(p; q_1, q_2) = (13; 17, 29), (19; 5, 17), (29; 5, 13)$ the commutator $\|[\Phi_{q_1}^\pm, \Phi_{q_2}^\pm]\|$ is at most $3 \cdot 10^{-15}$ (for $p = 13$: $5.4 \cdot 10^{-16}$ even, $4.5 \cdot 10^{-16}$ odd), and $A_{q_1} A_{q_2} = A_{q_1 q_2}$ exactly, with the $(q_1 + 1)(q_2 + 1) = 540$ quaternions of norm $q_1 q_2$.

13.3 The quantum Ihara zeta of these channels

Numerical result 13.3 (Edge spectrum and Ihara–Bass for $(p, q) = (13, 17)$). [numerical] For $(13, 17)$, odd block, the non-backtracking edge superoperator T on $M_6 \otimes \mathbb{C}^{18}$, of dimension 648, has $|\mu|$ in $\{17, \sqrt{17}, 1\}$ with multiplicities 1, 70, 577, and satisfies the Ihara–Bass identity to 10^{-14} .

The edge superoperator of Definition 3.18 was built directly, and the Ihara–Bass factorisation of Definition 3.8,

$$\det(1 - uT) = (1 - u^2)^{n^2(D-2)/2} \det(1 - u D \Phi_q^- + q u^2), \quad n = 6, D = 18,$$

holds with relative residuals $2.8 \cdot 10^{-15}$ at $u = 0.11$, $2.3 \cdot 10^{-15}$ at $u = 0.2 + 0.07i$ and $1.5 \cdot 10^{-14}$ at $u = -0.23$. The multiplicities $1 + 70 + 577 = 648$ exhaust the edge space, and $\sqrt{q} = 4.12311$. So the quantum Ihara zeta ζ_Φ of this channel satisfies the RH analogue of Definition 3.4 on the nose: every nontrivial zero sits on the critical circle. For every other case the same conclusion follows from the channel spectrum through the quadratic relation $\mu^2 - \lambda \mu + q = 0$ that links an edge eigenvalue to an adjacency eigenvalue λ , without building T .

13.4 What the joint spectrum looks like

For $(13; 17, 29)$ the even block has 19 distinct joint eigenvalue pairs (a_{17}, a_{29}) of the commuting pair Φ_{17}^+, Φ_{29}^+ . Examples, in adjacency units.

(a_{17}, a_{29})	multiplicity	remark
$(18, 30)$	1	trivial: $q + 1$
$(6, 2)$	2	integers
$(3, -1), (3, 3), (3, 9)$	3, 3, 2	integers
$(-5, -5)$	6	integers
$(\frac{3+\sqrt{17}}{2}, -1 - \sqrt{17})$	3	quadratic, field $\mathbb{Q}(\sqrt{17})$
$(5.6847, -2), (5.5549, -0.3344), (1.7730, 5.1119), \dots$	3 each	higher degree

That the eigenvalues are algebraic integers is not a test: the Cayley adjacency matrix A_q is an integer matrix by construction. What is visible is the structure a Hecke identification would predict, namely joint eigenvalue pairs, small-degree algebraic integers, multiplicities equal to the multiplicity of the corresponding irreducible representation of $\mathrm{PSL}_2(\mathbb{F}_p)$ in $W_\pm \otimes \overline{W}_\pm$, and every pair inside the box $[-2\sqrt{q_1}, 2\sqrt{q_1}] \times [-2\sqrt{q_2}, 2\sqrt{q_2}]$.

Conjecture 13.4 (Hecke identification of the Weil–LPS spectra). *[open] The joint spectra of the commuting Weil–LPS channels $\Phi_q^\pm$ are Hecke eigenvalues $a_q(f)$ of weight-2 forms attached to the quaternion algebra ramified at $\{2, \infty\}$, with level determined by p , via Jacquet–Langlands.*

The identification was *not* done here; it needs LMFDB and the Jacquet–Langlands bookkeeping.

13.5 Reading

Riemann channel	Weil–LPS channel at p
commuting dilations by primes	commuting Hecke channels Φ_q , one per LPS prime q
arithmetic quotient supplying expansion	$\mathrm{PSL}_2(\mathbb{F}_p)$ as quotient of the $(q+1)$ -regular tree
Weil representation	$W_\pm$ on $\ell^2(\mathbb{F}_p)$, even and odd
joint spectrum = zeros	joint spectrum = Hecke eigenvalues
RH: all modes at $\mathrm{Re} = 1/2$	Ramanujan: all modes at $ \mu = \sqrt{q}$
source of the bound: unknown	source of the bound: Deligne, via LPS

The last row is the honest one. These channels are exactly Ramanujan because Deligne proved the Weil conjectures (Section 16), not because they are channels. What the computation establishes is that the object proposed in the founding session’s handoff exists, is cheap to build, is self-checking at every step, and lands on the parent campaign’s Hilbert space with the parent campaign’s representation.

13.6 Not established

- The identification of the joint spectrum with specific modular forms in LMFDB (conjecture 13.4).
- Any statement about $p \rightarrow \infty$, or about assembling the Φ_q over all p into an object whose rings are the primes; that is the parent repository’s global question in a new guise, recorded as conjecture 26.2 in Section 26.
- Anything about the Riemann hypothesis itself.

14 A Matrix-Product Formulation of the Weil Conjectures for Quadratic Artin–Schreier Curves

The question the founding session put to this family is narrow and answerable: is there an honest matrix product state behind the Weil conjectures for curves, meaning a *fixed* local tensor whose n -site ring contraction is the point count N_n ? In the vocabulary of this book the ingredients line up: $N_n = \#C(\mathbb{F}_{q^n})$ is a ring norm, the Frobenius eigenvalues α_i appearing in $Z_C(u)$ are a transfer spectrum, and the Riemann hypothesis for the curve is the RH analogue of Definition 3.4. What is at issue is only whether the defining equation of the curve is a *local* constraint on the ring. For the quadratic Artin–Schreier family the answer is yes, and the answer stops exactly at degree three.

14.1 The shift picture

By Definition 4.14 the field $\mathbb{F}_{q^n}$ carries a normal basis $\{\theta, \theta^q, \dots, \theta^{q^{n-1}}\}$ in which Frobenius $x \mapsto x^q$ is the cyclic shift of coordinates. An element of $\mathbb{F}_{q^n}$ is then a ring of n sites with alphabet $\mathbb{F}_q$, Frobenius is translation by one site, and points over $\mathbb{F}_{q^n}$ are ring configurations. This is the matrix-product setting of Definition 3.12 with no further choices to make.

For the Artin–Schreier curve $y^q - y = g(x)$ of Definition 4.13, Hilbert 90 reduces the affine count to exponential sums,

$$N_n = q^n + \sum_{a \in \mathbb{F}_q^\times} \mathcal{S}_n(ag), \quad \mathcal{S}_n(g) = \sum_{x \in \mathbb{F}_{q^n}} e_q(\text{tr}_{q^n/q} g(x)), \quad (1)$$

so everything is carried by the $\mathcal{S}_n$. Writing the exponent m of a monomial x^m in base q , $m = \sum_j m_j q^j$, one has $x^m = \prod_j (x^{q^j})^{m_j}$: a product of shifted copies of x , whose range is the number of base- q digits. In a self-dual normal basis, which exists for n odd, the trace form is the dot product, and so for the quadratic family $g(x) = \sum_{j=0}^J a_j x^{1+q^j}$,

$$\text{tr}_{q^n/q} g(x) = \sum_{j=0}^J a_j \sum_i x_i x_{i+j}, \quad (2)$$

a translation-invariant quadratic spin model of range J on the ring. Its partition function is $\text{Tr } E^n$ for the transfer matrix

$$E_{(s_1, \dots, s_J) \rightarrow (s_2, \dots, s_J, x)} = e_q \left(a_0 x^2 + \sum_{j=1}^J a_j s_{J+1-j} x \right), \quad (3)$$

of size q^J , with physical index $\mathbb{F}_q$ and bond dimension q^J . The spins s_i are the J most recent site values; the map reads in the newest value x and forgets the oldest spin s_1 .

Proposition 14.1 (Transfer matrix for the quadratic family: the two forms). [*proved-conditional*] For $g(x) = \sum_{j=0}^J a_j x^{1+q^j}$ and every $n \geq 1$, including $n \leq J$: $\text{Tr } E_g^n = \sum_{x \in \mathbb{F}_q^n} e_q(Q_g^\circ(x))$ with Q_g° the ring form of matrix $P_g(\mathbb{S})$, and $\mathcal{S}_n(g) = \sum_x e_q(Q_g^{\text{tr}}(x))$ with Q_g^{tr} the trace form of matrix $CP_g(\mathbb{S})$ in normal-basis coordinates (Definition 5.7). The two forms have the same radical $\mathcal{R}_n$, so $|\mathcal{S}_n(g)| = |\text{Tr } E_g^n|$ and $\log_q(|\mathcal{S}_n|^2/q^n) = d_n$ is a nonnegative integer.

Two remarks on the statement, both corrected on 2026-09-12 night. First, the founding session’s reading of the even- n sign flip as the normalisation $\alpha_i = -\lambda_i(E)$, “uniformly in n ”, is false in general: it holds exactly when $P_g(-1) \neq 0$, and the exact law $\mathcal{S}_n = (-1)^{n-1} \delta_n \text{Tr } E_g^n$ with the radical sign

δ_n is Theorem 15.3 of Section 15, where the corrected Frobenius block is built. Second, a self-dual normal basis exists only for odd n ; in a general normal basis the trace form is $C P_g(\mathbb{S})$, not $P_g(\mathbb{S})$, and the discrepancy between the two Gauss sums is precisely the discriminant of C on the complement of the radical, which is Stickelberger's sign $(-1)^{n-1}$ corrected on the radical.

14.2 What was computed

The script brute-forces $\mathcal{S}_n$ in $\mathbb{F}_{q^n}$ in a polynomial basis, with the modulus verified irreducible, and compares with $\text{Tr } E^n$.

q	$a = (a_0, \dots, a_J)$	$\dim E$	n tested	$\mathcal{S}_n$ against $\text{Tr } E^n$
3	(0, 1)	3	2–7	equal for odd n , sign flip for even n
3	(1, 1)	3	3, 5, 7	equal
5	(0, 1)	5	2–5	equal for odd n , sign flip for even n
3	(0, 1, 1)	9	3, 5, 7	equal
3	(0, 0, 1)	9	3, 5, 7	equal

Numerical result 14.2 (Transfer identity and unitarity, checked). [numerical] For the five (q, a) of the table above and all n listed, the brute-force exponential sum agrees with $\text{Tr } E^n$ up to the sign convention $\alpha_i = -\lambda_i(E)$. For eight (q, a) with $q \in \{3, 5, 7\}$ and $\dim E$ up to 49, every eigenvalue of E has modulus $\sqrt{q}$ and $\|EE^\dagger - qI\| \leq 5.2 \times 10^{-15}$.

The ratios $|\mathcal{S}_n|/q^{n/2}$ are 1 at every tested n except $n = 4$, where they are 3 for $q = 3$ and 5 for $q = 5$: the sums saturate the Weil bound $|\mathcal{S}_n| \leq (\deg g - 1) q^{n/2}$, the maximal-curve phenomenon. The degree of L is $q^J = \deg g - 1$, matching Weil's count.

14.3 RH is unitarity

Proposition 14.3 (Scaled unitarity of the transfer matrix). [proved] *If $a_J \neq 0$ then $EE^\dagger = qI$ on the q^J -dimensional bond (the scalar quadratic Gauss sum when $J = 0$). Hence $E/\sqrt{q}$ is unitary, every eigenvalue of E is nonzero, semisimple and of modulus $\sqrt{q}$, and, through the corrected Frobenius block of Theorem 15.7, the Weil bound $|\alpha_i| = \sqrt{q}$ holds for the whole quadratic Artin–Schreier family.*

The note's argument is one line. In $(EE^\dagger)_{s's''}$ the oldest spin s_1 , which is the index summed over, appears only through the linear term $a_J s_1(x' - x'')$ of (3); provided $a_J \neq 0$ the sum over s_1 is therefore $q \delta_{x'x''}$. The hypothesis $a_J \neq 0$ is not cosmetic: it is what makes J the true range, and without it the same argument applies at the largest j with $a_j \neq 0$ and a smaller transfer matrix. This is the mechanism already seen for a finite permutation in observation 8.2: the RH analogue is manifest because the transfer matrix is a scaled unitary. In the same terms, rationality of $Z_C(u)$ holds because E is finite, and the note records that the functional equation is the involution $E \mapsto q E^{-\dagger}$, which scaled unitarity makes trivial. The proof with the output-row index bookkeeping of (3) made explicit is the prover's T7.6 (Section 15); the even- n sign is settled there as well.

14.4 Where it stops, and why that is the right place

Take a non-quadratic exponent, say $g(x) = x^3$ over $\mathbb{F}_q$ with $q \geq 5$. The base- q expansion of 3 has a single digit, so x^3 is an on-site cubic; but in normal-basis coordinates $\text{tr}_{q^n/q}(x^3) = \sum_{ijk} c_{ijk} x_i x_j x_k$, with the multiplication structure constants c_{ijk} of the basis, which are non-local and depend on n . A self-dual normal basis makes the *bilinear* trace form local; it says nothing about the trilinear one.

g	locality in the shift basis	transfer matrix	RH
quadratic, $\sum_j a_j x^{1+q^j}$	local, range J	$q^J \times q^J$, $\sqrt{q}$ -unitary	manifest
degree ≥ 3 in any digit	non-local, n -dependent	none of fixed size in this basis	Weil / Deligne

Observation 14.4 (The non-quadratic case and Dwork’s operator).

[sketched]

For non-quadratic g the trace form in the shift basis is non-local and depends on n , so no fixed local tensor represents the ring model. The uniform transfer operator that does exist is Dwork’s p -adic operator (Definition 4.15, [19]), which delivers rationality and, through Dwork’s duality, the functional equation, but which sees the p -adic sizes of its eigenvalues and not their complex absolute values.

Dwork’s operator has exactly the form of a coarse-graining tensor-network step, “local weight, then decimate by q ”: multiplication by a splitting function followed by $x^m \mapsto x^{m/q}$, with infinite but nuclear bond dimension, and L as a Fredholm determinant. What it cannot deliver is the RH analogue, because the Newton polygon it computes is a statement about p -adic valuations. That is the structural reason Deligne’s proof [17] lives in ℓ -adic cohomology, and the Weil/Deligne divide is taken up in Section 16. The sums that fall on the far side of the table, cubic Gauss sums and Kloosterman-type sums, are precisely the ones that needed algebraic geometry.

14.5 Not established

The note is explicit about what was not shown. No matrix product state with a fixed local tensor is known here for a non-quadratic Artin–Schreier curve, or for any curve of genus ≥ 1 that is not of this maximal type. Whether a *different* basis of $\mathbb{F}_{q^n}$, uniform in n , makes cubic traces local was neither settled nor searched: no candidate was found. And the unitarity argument of Proposition 14.3 is now proved, and the transfer unitary is identified as a Weil-representation operator in Section 15.

The boundary itself is the result worth keeping, and it is the same boundary met for graphs and for the Riemann channel: the Ramanujan bound is either manifest, because the transfer matrix is a scaled unitary, or it comes from arithmetic input that sits outside the transfer structure altogether.

15 Artin–Schreier II: the Sign Law, the Super-Transfer Matrix, and the Weil Representation

Section 14 recorded that $\mathcal{S}_n(g) = \text{Tr } E_g^n$ for odd n and guessed that the even- n sign flip was the normalisation $\alpha_i = -\lambda_i(E_g)$. That guess is false in general. The numerics lane found the failures, the orchestrator’s replacement (a conjugated law) failed in two further cases, and the codex prover found and proved the exact law; the numerics lane then confirmed it in 624 further checks. This shard is the corrected account, with the prototype now carrying, explicitly, every feature the Phantasm is supposed to have: a graded bond, a supertrace count, a fermionic sign that is a permutation sign, and a transfer unitary from a Weil representation. Throughout q is an odd prime and the notation is that of Definition 5.7.

15.1 The two quadratic forms and the exact sign law

Lemma 15.1 (Gauss sums of quadratic forms). *[proved] For a quadratic form Q on $\mathbb{F}_q^n$ of rank r with nondegenerate part Q_{nd} , $\sum_x e_q(Q(x)) = q^{n-r} \eta(\det Q_{\text{nd}}) \mathfrak{g}^r$ with $\mathfrak{g} = \sum_c e_q(c^2)$, $\mathfrak{g}^2 = \eta(-1)q$ and $\bar{\mathfrak{g}} = \eta(-1)\mathfrak{g}$; every such sum is nonzero, has $|\cdot|^2 = q^{n+(n-r)}$, and its conjugate is $\eta(-1)^r$ times itself.*

Assumption 15.2 (Normal bases). *[assumed] $\mathbb{F}_{q^n}$ has a normal basis over $\mathbb{F}_q$; for n odd it can be chosen self-dual [54, 53].*

Theorem 15.3 (The sign law). *[proved-conditional] For every $n \geq 1$, including $q \mid n$,*

$$\mathcal{S}_n(g) = (-1)^{n-1} \delta_n \text{Tr } E_g^n, \quad \delta_n = \det(\mathbb{S}|_{\mathcal{R}_n}) \in \{\pm 1\},$$

equivalently $\mathcal{S}_n(g) = (-1)^{n-1} \delta_n \eta(-1)^{n-d_n} \text{Tr } E_{-g}^n$.

The proof builds a square root of the Gram matrix over $\overline{\mathbb{F}_q}$: the conjugate-embedding matrix $A_{ri} = \theta^{q^{r+i}}$ satisfies $A^t = A$, $A^2 = C$, $A^{[q]} = \mathbb{S}A$, so $\eta(\det(C|_R)) = \det(\mathbb{S}|_R)$ for every $\mathbb{S}$ -invariant subspace R on which the identities restrict. Applied to the whole space this is Stickelberger’s theorem, $\eta(\text{disc}) = (-1)^{n-1}$, the sign of the n -cycle by which Frobenius permutes the conjugates; applied to the radical it gives the correction δ_n . Both Gauss sums have the same rank and the same power of $\mathfrak{g}$; only the discriminants of their nondegenerate parts differ, by $\det C / \det(C|_{\mathcal{R}_n})$.

Assumption 15.4 (Polynomiality of the character L -function). *[assumed] For $\deg g = q^J + 1$ and every $a \in \mathbb{F}_q^\times$, $L(ag, u) = \exp(\sum_n \mathcal{S}_n(ag) u^n / n)$ is a polynomial of degree exactly q^J with constant term 1. Root sizes are not assumed.*

Corollary 15.5 (The periodic sign and the endpoint criteria). *[proved-conditional] With $f(z) = z^J \mathcal{P}_g(z)$, v_- its order of vanishing at -1 (always even) and $\mathfrak{h}$ the period parameter: $\delta_n = 1$ for n odd and $(-1)^{\min(v_-, q^{v_q(n)})}$ for n even, so*

$$\mathcal{S}_n(g) = (1 - 2 \cdot \mathbf{1}_{2\mathfrak{h} \mid n}) \text{Tr } E_g^n.$$

The claim $\alpha_i = -\lambda_i(E_g)$ of Section 14 holds for all n iff $\mathcal{P}_g(-1) \neq 0$; the conjugated law $\mathcal{S}_n = -(-\eta(-1))^n \overline{\text{Tr } E_g^n}$ holds for all n iff $\mathcal{P}_g(-\eta(-1)) \neq 0$.

Proposition 15.6 (Both drafted laws fail). *[proved-conditional] For $q = 3$, $g = 2x^2 + x^4$, $n = 1$: $\mathcal{S}_1 = \text{Tr } E_g = 3$, while the conjugated law predicts -3 ($\mathcal{P}_g(1) = 0$). For $q = 5$, $g = x^2 + x^6$, $n = 2$: $\mathcal{S}_2 = \text{Tr } E_g^2 = 5\sqrt{5}$, while both laws predict $-5\sqrt{5}$ ($\mathcal{P}_g(-1) = 0$). The failures occur at n prime to q .*

q	a	$P_g(1)$	$P_g(-1)$	v_-	h	$\alpha = -\lambda(E)$	conjugated law
3	(0, 1)	1	2	0	1	holds	holds
3	(1, 1)	2	0	2	3	fails	holds
3	(0, 1, 1)	2	0	4	9	fails	holds
3	(2, 1)	0	1	0	1	holds	fails
5	(1, 2)	3	4	0	1	holds	holds
5	(0, 1, 1)	2	0	2	5	fails	fails
7	(1, 1)	2	0	2	7	fails	holds

15.2 The Frobenius block and the super-transfer matrix

Theorem 15.7 (The corrected Frobenius block). [proved-conditional]
 $L(g, u)^h = \prod_{\omega^{2h}=1} D_g(\omega u) / D_g(u)^h$; the reciprocal-root multiplicity of $L(g, u)$ at z is $m_F(z) = \frac{1}{h} \sum_{\omega^{2h}=1} m_E(\omega^{-1}z) - m_E(z)$, a nonnegative integer, and the diagonal $q^J \times q^J$ matrix F_g with these multiplicities satisfies $F_g F_g^\dagger = q$, $L(g, u) = \det(1 - u F_g)$ and $\mathcal{S}_n(g) = -\text{Tr } F_g^n$ for all n . If $P_g(-1) \neq 0$ one may take $F_g = -E_g$; if $P_g(-\eta(-1)) \neq 0$, $F_g = -\eta(-1)E_{-g}$.

Assumption 15.8 (Frobenius semisimplicity). [assumed] The Frobenius action on H^1 of a smooth projective curve over a finite field is semisimple over an algebraic closure of the coefficient field. Used only for operator similarity, never for net spectra.

Theorem 15.9 (The point count is a supertrace). [proved-conditional] The projective curve $y^q - y = g(x)$ has one point at infinity; $N_n = 1 + q^n + \sum_{a \neq 0} \mathcal{S}_n(ag)$ (additive Hilbert 90); $\text{Tr } E_0^n = q^n$; and the super-transfer matrix $\mathbb{E}$ of Definition 5.8 satisfies $N_n = \text{str } \mathbb{E}^n$ for all $n \geq 1$. Its odd block F has $FF^\dagger = q$, and its multiset is the $2g = (q-1)q^J$ Frobenius eigenvalues with no even-odd cancellation; under assumption 15.8 it is similar to Frobenius on H^1 . The simpler odd block $-\eta(-1) \bigoplus_a E_{ag}$ is correct iff $P_g(-\eta(-1)) \neq 0$: for $q = 3$, $g = 2x^2 + x^4$ it gives -2 against $N_1 = 10$.

The even block is the trivial-character sector and the odd block the nontrivial characters: this is the grading observation of observation 11.18. The count is a supertrace because, by Corollary 10.18, it cannot be a trace.

15.3 The transfer unitary is a Weil-representation operator

Cited fact 15.10 (Uniqueness of Weil implementers). [cited] Gurevich and Hadani [29]: the intertwiner of the Heisenberg representation with its twist by $g \in \text{Sp}$ “is determined up to scalar” by the Egorov relation. Verbatim: satisfies and in fact is determined up to scalar by the following equation (math/0610818:main.tex:422); $\rho(g)\pi(h)\rho(g)^{-1} = \pi(g \cdot h)$, \label {Egorov2} (math/0610818:main.tex:425).

Cited fact 15.11 (Character of the Weil representation). [cited] Thomas [74]: for a finite field F the trace of the Weil representation at $g \in \text{Sp}(V)$ is $|F|^{\frac{1}{2} \dim \ker(g-1)}$ times a product of Weil-index phases; in particular its modulus squared is $|F|^{\dim \ker(g-1)}$ (Howe [35]). Verbatim: $\text{Tr } \rho(g) = |F|^{\frac{1}{2} \dim \ker(g-1)} \gamma(1)^{\dim V - \dim \ker(g-1)} \gamma(\det \sigma_g)$. (math/0610644:ArxivThomasWeil3.tex:196).

Assumption 15.12 (Exact centred Weil implementers exist). [assumed] There is a genuine unitary representation W of $\text{Sp}(2J, \mathbb{F}_q)$, q odd, with $W(M)X_{u,v}W(M)^{-1} = X_{M(u,v)}$ exactly for the centred Weyl operators.

Theorem 15.13 (The symplectic transfer map). *[proved-conditional] $\mathcal{U}_g = E_g/\sqrt{q}$ factors as the Fourier transform on the retiring register (scaled by a_J), then the quadratic phases, then the cyclic relabelling, and $\mathcal{U}_g X_{u,v} \mathcal{U}_g^{-1} = X_{M_g(u,v)}$ exactly, where*

$$u'_k = u_{k+1}, \quad u'_J = -v_1/a_J, \quad v'_k = v_{k+1} + a_{J-k}u'_J \quad (k < J), \quad v'_J = a_J u_1 + \sum_{k \geq 2} a_{J+1-k} u_k + 2a_0 u'_J,$$

a symplectic map. Hence $\mathcal{U}_g = e^{i\vartheta_g} W(M_g)$.

Corollary 15.14 (Fixed spaces, the Weil bound, and the permutation sign). *[proved-conditional] $|\text{Tr } E_g^n|^2 = q^{n+\dim \ker(M_g^n - 1)}$ (from $\text{Tr Ad}(\mathcal{U}_g^n) = \text{the number of fixed Weyl labels}$), so $d_n = \dim \ker(M_g^n - 1) \leq 2J$; $|\mathcal{S}_n(g)| \leq q^J q^{n/2}$ with equality iff $M_g^n = 1$; and when $d_n = 0$, $\mathcal{S}_n(g) = (-1)^{n-1} \text{Tr } E_g^n$, the sign of the Frobenius n -cycle. For $q = 3$, $g = x^4$: $M_g = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ of order 4, $d_2 = 0$, $d_4 = 2$.*

So the maximal-curve phenomenon of Section 14 ($|\mathcal{S}_4|/q^2 = q$ for $g = x^4$) is the statement $M_g^4 = 1$, and the Weil bound for the whole quadratic family is Howe's character bound for the Weil representation. The fermionic sign of the count on a ring of n sites is the sign of the Frobenius permutation, corrected on the fixed space of M_g^n by the radical sign.

Numerical result 15.15 (The corrected account, checked). [numerical]
`scripts/artin_schreier_super.py` (`outputs/artin_schreier_super.txt`; 1969 checks, of which the 24 failures are exactly the refutation of the two drafted laws): eleven (q, g) over $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, $n \leq 9$; the exact sign law and its periodic form and endpoint criteria (624/624); the corrected Frobenius blocks and $\text{str } \mathbb{E}^n = N_n$ against direct point counts in every case; the Clifford property of $E_g/\sqrt{q}$ with symplectic matrices matching the explicit formula matrix for matrix (orders 4, 3, 30); the character formula in 258/258 cases; and Prony on the genus-3 counts recovering the Frobenius eigenvalues with coefficient -1 .

16 Deligne's Proof, Told Through Regular Graphs

Expository. Nothing here is new; the point is to isolate what is actually hard in the proof of the Riemann hypothesis over finite fields [17], using the Ihara zeta [37] of a regular graph as the running model, and to read off which ingredients this project has and which it does not.

16.1 One graph, two sides

Fix a finite connected $(q+1)$ -regular graph X . The objects are those of Section 3: non-backtracking walks and prime cycles (Definition 3.5), the count $N_\ell \geq 0$ of closed non-backtracking walks of length ℓ , the Ihara zeta with its Euler product (Definition 3.6), the Hashimoto operator with $N_\ell = \text{Tr } T^\ell$ and $\zeta_X(u) = \det(1 - uT)^{-1}$ (Definition 3.7), and its compression to the adjacency matrix A by Ihara–Bass (Definition 3.8, [2]). Side A is the orbit side, side B the small matrix (Definition 3.1); passing between them is bookkeeping.

Call ± 1 and the pair $\mu \in \{q, 1\}$ from the adjacency eigenvalue $\lambda = q+1$ the *trivial* eigenvalues of T , everything else nontrivial. Then

$$N_\ell = q^\ell + 1 + \sum_{\mu \text{ nontrivial}} \mu^\ell + (|E| - |V|)(1 + (-1)^\ell),$$

with extra terms if X is bipartite: the leading term is the growth rate of the $(q+1)$ -regular tree, and the nontrivial eigenvalues are the error. Since $\mu\mu' = q$, a pair either has both members of modulus $\sqrt{q}$ (when $|\lambda| \leq 2\sqrt{q}$, where they are complex conjugates) or is real with one member larger. So all nontrivial $|\mu| = \sqrt{q}$ iff all nontrivial $|\lambda| \leq 2\sqrt{q}$ iff the error in N_ℓ is $O(q^{\ell/2})$: the Ramanujan property (Definition 3.9) and the RH analogue (Definition 3.4) are the same statement, that closed walks spread as evenly as they possibly could.

Two examples from the founding session. K_4 has $q = 2$, adjacency spectrum $3, -1, -1, -1$, hence $\mu = (-1 \pm i\sqrt{7})/2$ with $|\mu|^2 = 2$: Ramanujan, with $N_3 = 8 + 1 + 6 \cdot \frac{5}{2} = 24$, the oriented pointed triangles. The prism $C_{16} \times K_2$ has an eigenvalue $2 \cos(\pi/8) + 1 \approx 2.848 > 2\sqrt{2}$: a real pole inside the circle and an error growing faster than $2^{\ell/2}$. A long thin graph always fails: a bottleneck gives a real μ near q , and walks pile up on one side of it.

16.2 The same statement for a curve, with one sign changed

A curve C over $\mathbb{F}_q$ has N_ℓ points over $\mathbb{F}_{q^\ell}$: again a nonnegative count, again a generating function $Z_C(u)$ with an Euler product, now over closed points, the orbits of Frobenius Fr on the points over the algebraic closure. Side A is identical in form; side B gives a $2g \times 2g$ matrix (g the genus) with

$$N_\ell = q^\ell + 1 - \text{Tr } \text{Fr}^\ell = q^\ell + 1 - \sum_{i=1}^{2g} \alpha_i^\ell, \quad Z_C(u) = \frac{\det(1 - u\text{Fr})}{(1 - u)(1 - qu)}.$$

	graph	curve
count	closed non-backtracking walks	points over $\mathbb{F}_{q^\ell}$
leading term	$q^\ell + 1$	$q^\ell + 1$
error term	$+\sum \mu^\ell$	$-\sum \alpha_i^\ell$
pairing	$\mu \leftrightarrow q/\mu$	$\alpha_i \leftrightarrow q/\alpha_i$
interesting eigenvalues are	poles of ζ_X	zeros of Z_C
RH	$ \mu = \sqrt{q}$	$ \alpha_i = \sqrt{q}$

Everything matches except the sign of the error term, hence whether the interesting eigenvalues are poles or zeros. For the graph ζ_X is the reciprocal of a polynomial and every eigenvalue is a pole; for the curve the trivial eigenvalues 1 and q are poles and the interesting ones are zeros. This is the sign fixed in Section 2, $\mathcal{S}_n = -\sum_i \alpha_i^n$, and the sign of the explicit formula. For a variety of dimension n the pattern repeats as an alternating sum over $n + 1$ blocks, block i of modulus $q^{i/2}$; the one that matters is the middle, $i = n$.

16.3 What is cheap and what is not

ingredient	graph	variety
side B exists	T plus Ihara–Bass	Grothendieck’s trace formula
the pairing	$\mu\mu' = q$ from Ihara–Bass	Poincaré duality
trivial bound	Perron: $ \mu \leq q$	$ \alpha_i \leq q$
the bound $\sqrt{q}$	<i>false in general</i>	Deligne’s theorem

The first three rows are cheap. Rationality of the zeta function is the first row and nothing more; the trace formula is heavy technology but the same move, that the count is the trace of a power of a finite matrix. The pairing turns the one-sided bound into equality, so the whole problem is one-sided; and the trivial bound is just convergence of the Euler product for $|u| < 1/q$. The gap between q and $\sqrt{q}$ is the entire content.

The last row is where graph and variety part. There is no theorem that a $(q + 1)$ -regular graph is Ramanujan; most are not, the prism is a counterexample, and each Ramanujan construction carries its own proof: arithmetic (Lubotzky–Phillips–Sarnak [56], through RH for a modular curve), probabilistic (Friedman), or interlacing (Marcus–Spielman–Srivastava). So the right question is not how one proves the bound but what varieties have that graphs lack, which makes every one of them Ramanujan. Three things: varieties can be *multiplied*, *sliced* over a curve, and their interesting eigenvalues enter the count with a *minus sign*. A graph has none of the three.

16.4 The engine: power up, dominate, take roots

The trick the proof is built on, in its purest graph form: why the Ramanujan bound is $2\sqrt{q}$. On the infinite $(q + 1)$ -regular tree let c_{2k} count closed ordinary walks of length $2k$ from a root, so the spectral radius of the tree’s adjacency operator is $\lim_k c_{2k}^{1/2k}$. Such a walk is a Dyck path (about 4^k , up to a polynomial factor) together with a fresh neighbour at each of the k outward steps (q choices, $q + 1$ at the root), so $c_{2k} = 4^k q^k$ times a bounded factor times a polynomial in k , and $c_{2k}^{1/2k} \rightarrow 2\sqrt{q}$. Three moves: a nonnegative count dominates a power of an eigenvalue, $c_{2k} = \langle \delta_{\text{root}}, A^{2k} \delta_{\text{root}} \rangle$; the count is known up to a loss that does not grow exponentially in k ; and the $2k$ -th root as $k \rightarrow \infty$ kills every fixed loss. This computation is the tree side of the Ramanujan comparison. Deligne makes the same three moves twice; the nontrivial content is entirely in the second.

16.5 The pointwise theorem, in twisted-graph language

Deligne’s core theorem is not about a variety but about a curve carrying matrix-valued coefficients. A *twist* on a base graph B assigns to each directed edge e an invertible $d \times d$ matrix $\varrho(e)$ with $\varrho(\bar{e}) = \varrho(e)^{-1}$; a closed walk w has holonomy $\varrho(w)$, the ordered product along it, and the twisted zeta is the Artin–Ihara L -function of Definition 3.10,

$$L(u, \varrho) = \prod_P \det(1 - u^{\ell(P)} \varrho(P))^{-1} = \exp\left(\sum_w \frac{1}{\ell(w)} \text{Tr } \varrho(w) u^{\ell(w)}\right) = \det(1 - uT_\varrho)^{-1}.$$

The quantum Ihara zeta of Section 17 is this with B a bouquet of D loops and $\varrho = \text{Ad}(U_i)$ on the i -th loop; a quantum expander is a Ramanujan twist of a bouquet. Two kinds of eigenvalue now live on side B: the *global* ones, the eigenvalues of T_ϱ and poles of L , analogues of the μ ; and the *pointwise* ones, the eigenvalues of each $\varrho(P)$. The twist is *pure with constant* c when every eigenvalue of every $\varrho(P)$ has modulus $c^{\ell(P)}$: a unitary twist is pure with $c = 1$, Deligne's coefficient systems with $c = q^{w/2}$ for an integer weight w (Definition 4.16).

Deligne's pointwise theorem (his Theorem 3.2, in this language): if (a) all traces $\text{Tr } \varrho(w)$ are real, in fact rational; (b) ϱ preserves an alternating pairing up to the scalar $c^{2\ell(w)}$, so eigenvalues of $\varrho(P)$ come in pairs α_i and $c^{2\ell(P)}/\alpha_i$; and (c) the holonomies generate a group as large as (b) permits; then ϱ is pure with constant c .

The note argues in four steps, for a fixed prime cycle P and eigenvalue α_1 of $\varrho(P)$, the target being $|\alpha_1| \leq c^{\ell(P)}$ with (b) supplying equality. *Power up*: replace ϱ by $\varrho^{\otimes 2k}$, so traces become $(\text{Tr } \varrho(w))^{2k}$ and α_1^{2k} is an eigenvalue at P . *Dominate*: by (a) those traces are nonnegative, so every Euler factor of $L_{2k} := L(u, \varrho^{\otimes 2k})$ has nonnegative coefficients and constant term 1, the product dominates any single factor coefficientwise, and the factor at P has a pole at $|u| = |\alpha_1|^{-2k/\ell(P)}$ which a dominated series cannot outrun. *The known pole*: suppose the only poles of L_{2k} were the trivial ones, at $|u| = 1/(q c^{2k})$; then

$$|\alpha_1|^{-2k/\ell(P)} \geq 1/(q c^{2k}) \iff |\alpha_1| \leq c^{\ell(P)} q^{\ell(P)/2k}.$$

Take roots: as $k \rightarrow \infty$ the loss $q^{\ell(P)/2k}$ vanishes. The fixed loss is the single factor q in the trivial pole, paid once by the base curve however large k is.

A graph breaks the third step, and must, since the theorem is false for graphs. Take B a bouquet of two loops with $\varrho(a) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $\varrho(b) = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$: traces are integers, so (a); determinants are 1, so (b) with $c = 1$; and the two generate a group as large as $\text{SL}_2(\mathbb{Z})$ permits, so (c). Yet $\varrho(a)$ has eigenvalue $(3+\sqrt{5})/2 > 1$, so the twist is not pure: $L_{2k} = \det(1 - uT_{\varrho^{\otimes 2k}})^{-1}$ has a pole at $|u| = ((3+\sqrt{5})/2)^{-2k}$, far inside $1/q$, from a nontrivial eigenvalue of the twisted edge operator. For a graph every side-B eigenvalue is a pole, so nothing forces the poles to the trivial place. What makes the third step true for a curve is one hypothesis and one structural fact.

16.6 Big monodromy, and the abelian failure

The trivial poles of L_{2k} come from invariant vectors: a vector in the d^{2k} -dimensional coefficient space fixed by every holonomy. On such a vector the twist acts like no twist, so L_{2k} acquires a copy of the untwisted zeta with its pole at $1/q$, rescaled by the scalar the twist puts on the vector; if that scalar is known, so is the pole. The pairing (b) supplies invariants for free, by contracting the $2k$ tensor slots in pairs, each contraction giving a pole at exactly $1/(q c^{2k})$. Hypothesis (c), *big monodromy*, says these are the only invariants: if the holonomies fill out the whole group preserving the pairing, invariant theory makes every invariant of $\varrho^{\otimes 2k}$ a combination of pairwise contractions, so no other trivial pole appears.

Observation 16.1 (The abelian obstruction is inside Deligne's proof). [sketched] Commuting holonomies produce uncontrolled invariants in tensor powers, exactly as commuting Kraus unitaries never give an expander: commuting $\varrho(w)$ have common eigenvectors, and a common eigenvector with eigenvalue $\chi(w)$ becomes invariant in $\varrho^{\otimes N}$ as soon as $\chi^N = 1$, carrying an unknown χ -dependent scalar, so the trivial poles sit where the argument cannot control them. Any Ramanujan lift of the prime dilations must break the commutativity by a quotient, as [56] does with the arithmetic lattice and as Deligne does with the shears of a generic slicing.

In Deligne's setting the commuting case is a coefficient system with abelian monodromy, which can have arbitrary pointwise weights; (c) excludes it. In this project it is the observation of the

founding session that commuting Kraus unitaries never give an expander, the adjoint action of an abelian group having as many invariants as it has joint eigenprojectors: the same obstruction on both sides of the dictionary. Geometrically, big monodromy is verified through shears. The holonomy around each bad slice is the identity plus a rank-one nilpotent along one vector, those vectors are carried to one another by the holonomy group, and a theorem of Kazhdan and Margulis says a group generated by one conjugacy class of shears and acting without invariant subspaces is automatically as large as the pairing allows. Big monodromy is the statement that the bad slices are all alike, which holds because they are generic.

16.7 The sign

For a graph, side B is the one matrix T_ϱ and every eigenvalue is a pole. For a curve with coefficients the trace formula has two blocks of opposite sign,

$$\sum_{w: \ell(w)=\ell} \text{Tr } \varrho(w) = \text{Tr}(\text{Fr}^\ell \mid \text{invariant}) - \text{Tr}(\text{Fr}^\ell \mid \text{interesting}), \quad L(u, \varrho) = \frac{\det(1 - u\text{Fr} \mid \text{interesting})}{\det(1 - u\text{Fr} \mid \text{invariant})}.$$

The invariant block is the one big monodromy computes, and it produces the poles; the interesting block produces zeros, and zeros do not obstruct convergence, so the radius of convergence of L_{2k} is set by the invariant block alone. That is the third step, and it is the minus sign of the table above: positivity of counts is available on both sides, but on the graph side it constrains the sum of everything, which is the Perron bound and no more, while on the curve side the interesting eigenvalues are subtracted, so positivity bounds them above by the trivial ones and even tensor powers squeeze the bound to equality. The two-loop bouquet is not a curve because its walk counts cannot be written with the interesting part subtracted.

16.8 From pointwise to global: slice, bound, multiply

Slice. Take a one-parameter family of hyperplane sections of a variety X of dimension n ; the parameter runs over a curve and all but finitely many slices are smooth of dimension $n - 1$. Up to pieces already understood, the middle block of X is the interesting block of that base curve with coefficients, the coefficient at a parameter value being the middle block of the slice there. In graph terms this is the Artin–Ihara factorisation (Definition 3.10) used in Section 17, with the fibre replaced by a variety of one dimension less. The coefficients satisfy (a), (b), (c): the pairing is Poincaré duality on the slice, big monodromy is the shear argument, and the traces are rational because the zeta of a slice is rational with rational coefficients, off which the blocks known by induction are peeled. So they are pure with $c = q^{(n-1)/2}$.

Bound. With pure coefficients the Euler product over the base converges absolutely for $|u| < 1/(qc)$, there being about q^ℓ/ℓ prime cycles of length ℓ each with holonomy of norm c^ℓ . So the interesting block has no eigenvalue larger than $qc = q^{(n+1)/2}$, where RH wants $q^{n/2}$: a loss of one factor $\sqrt{q}$, paid by the base being a curve, independent of n .

Multiply. The product X^k has dimension nk and its middle block contains the k -fold tensor products of the middle block of X , so the products $\alpha_1 \cdots \alpha_k$ are among its eigenvalues. The bound applied to X^k still loses one factor $\sqrt{q}$, because the base of a slicing is always a curve, whence

$$|\alpha_i|^k \leq q^{(nk+1)/2} \implies |\alpha_i| \leq q^{n/2} q^{1/2k} \xrightarrow{k \rightarrow \infty} q^{n/2},$$

with duality giving the reverse inequality: the engine a second time, count, fixed loss, roots. The pointwise theorem is what makes the last move legitimate, since the slices of X^k have dimension

$nk - 1$, far above anything the induction on dimension knows, so their purity must come from rationality and big monodromy alone. That is why the core theorem is stated for coefficients on a curve: it certifies coefficients in any dimension at the price of one curve. Even for a curve C the argument passes through C^k for all k ; the surface $C \times C$ is where Weil’s proof lived, and the products are the mechanism. The blocks $i \neq n$ follow by induction from lower-dimensional sections and from duality.

16.9 What the proof does not do

It never exhibits a Hermitian structure. Weil’s proof for curves does: by the Hodge index theorem the intersection form on $C \times C$ is definite on the relevant part, which makes $\text{Fr}/\sqrt{q}$ unitary for an explicit positive inner product on the middle block and produces a Hilbert–Pólya operator (Definition 4.7). In graph terms, on the nontrivial part T is $\sqrt{q}$ times a unitary in some inner product, which for the block $\begin{pmatrix} \lambda & -q \\ 1 & 0 \end{pmatrix}$ holds exactly when $|\lambda| < 2\sqrt{q}$. Grothendieck’s plan was to prove a definite form in all dimensions, the standard conjectures, and deduce RH as Weil had; that plan is still open. Deligne bypassed it: no form, no operator, only positivity and a limit. RH over finite fields is therefore a theorem, while in dimension above one the inner product that would make Frobenius normal is not known to exist.

The Artin–Schreier computation of Section 14 sits on the Weil side of this divide: $EE^\dagger = qI$ is manifest from Parseval for the additive characters (Proposition 14.3), an explicit inner product. Deligne’s proof is what one has when no such structure is in sight. Two honesty notes the note records: the Ramanujan property of the LPS graphs of Section 13 needs only the curve case, since Eichler–Shimura transports Hecke eigenvalues to Frobenius eigenvalues of a modular curve, so Weil’s theorem suffices there and Deligne’s method is what extends it to higher weight; and the technology behind “side B exists” is the largest single piece of the proof by page count, though not by idea.

16.10 What this says for the project

Observation 16.2 (The three ingredients, and no Hermitian form). [sketched] Deligne’s proof uses three ingredients that a Hilbert–Pólya approach would have to replace: products, under which eigenvalues multiply while the trivial loss stays fixed; slicing over a curve, which keeps that loss at one factor $\sqrt{q}$ in every dimension; and the sign, which makes the interesting eigenvalues zeros rather than poles, so that positivity of the counts bounds them. It exhibits no Hermitian form and no operator.

- **Which ingredients ζ has.** The zeros are subtracted from the prime count, so ζ is on the curve side of the table, not the graph side, and the positivity $\Lambda(n) \geq 0$ is the right kind. The sign and the positivity are present; the product is absent, since there is no $X \times X$ with ζ as its slice, so the multiply step has nothing to act on. This is the standard diagnosis, stated in the terms of this section.
- **The number-field shadow of the pointwise theorem is Rankin–Selberg.** For a modular form the Euler product of $L(f \times \bar{f})$ has nonnegative coefficients and a known pole, and dominating one local factor gives $|a_p| \leq p^{(k-1)/2} \cdot p^{1/2}$, the same fixed loss; each symmetric-power L -function whose poles are known shaves it (Kim–Sarnak’s 7/64 is the current state). Knowing the poles of all the tensor-power L -functions is the analogue of big monodromy; over number fields that is Langlands functoriality, unproved.
- **The abelian obstruction is inside the proof** (observation 16.1), on both sides of the dictionary.

- **Ramanujan without Hermiticity is possible.** Deligne’s route and the interlacing route both reach the spectral bound with no Hilbert–Pólya operator. For the steering question of where Hermiticity enters, this is the cleanest evidence that it need not enter at all; what must enter is a positivity plus an operation under which eigenvalues multiply at fixed loss.

17 The Ihara–Bass Formula for an Arbitrary Kraus Family

This is the load-bearing section of the book. The classical Ihara–Bass formula (Definition 3.8) compresses a determinant on the edge space of a regular graph to one on the vertex space, and Definition 3.18 lifts the edge operator to superoperators. The question the founding session left open, and which TJO steered towards, is whether the compression survives when the Kraus operators are not unitary. It does, at the price of two u -dependent operators in place of the adjacency matrix and the degree.

Status. Theorem 17.1 and its three corollaries are tagged **proved**. The author is **claudé:fable-5.1**; the reviewers are of a different model, **claudé:opus**; and their reports in **notes/reviews/** — a round-1 algebra report, a round-1 provenance report, and a round-2 re-verification — return **VALID** on every verdict. Only one model family was declared for this campaign, so **proved** here means exactly that: author $\neq$ reviewer, inside one family. The two **numerical** rows are independent evidence.

17.1 Setting and conventions

- $\mathcal{V}$ is a finite-dimensional complex vector space, $N = \dim \mathcal{V}$. In the application $\mathcal{V} = M_n$ with the Hilbert–Schmidt inner product, so $N = n^2$.
- $D = 2m$ is even, and ι is a fixed-point-free involution on $[D] = \{1, \dots, D\}$, the *reversal*; we write $\bar{i} = \iota(i)$, so that $\bar{\bar{i}} = i$.
- $\mathcal{E}_1, \dots, \mathcal{E}_D \in \text{End}(\mathcal{V})$ are arbitrary linear maps. No invertibility, positivity or unitarity is assumed.
- The edge space is $\mathcal{W} = \mathcal{V} \otimes \mathbb{C}^D$, with $\mathbb{C}^D$ spanned by $|1\rangle, \dots, |D\rangle$; an element is $x = \sum_i x_i \otimes |i\rangle$ with $x_i \in \mathcal{V}$ its *component* at i .
- $T \in \text{End}(\mathcal{W})$ is the quantum Hashimoto operator of Definition 3.18, $T(v \otimes |i\rangle) = \sum_{j \neq \bar{i}} \mathcal{E}_i v \otimes |j\rangle$, and $\zeta_T(u) = \det_{\mathcal{W}}(1 - uT)^{-1}$, a rational function of $u \in \mathbb{C}$.
- $\|\cdot\|$ is the operator norm; any submultiplicative norm on $\text{End}(\mathcal{V})$ would do. Determinants carry the space they are taken over, $\det_{\mathcal{V}}$ and $\det_{\mathcal{W}}$.
- Sylvester’s identity $\det(1 - XY) = \det(1 - YX)$, for $X: \mathcal{V} \rightarrow \mathcal{W}$ and $Y: \mathcal{W} \rightarrow \mathcal{V}$, is used without further comment; so is the block determinant with identity diagonal blocks, $\det \begin{pmatrix} 1 & B \\ C & 1 \end{pmatrix} = \det(1 - CB) = \det(1 - BC)$ (Schur complement, then Sylvester).

The Kraus case. $\mathcal{V} = M_n$; $A_1, \dots, A_m \in M_n$ are arbitrary, $A_{m+k} = A_k^\dagger$, $\iota(k) = m + k$, and $\mathcal{E}_i = \text{Ad}(A_i): X \mapsto A_i X A_i^\dagger$. With column stacking (Section 2), $\text{Ad}(A) = \bar{A} \otimes A$ as an $n^2 \times n^2$ matrix, so $\text{Tr}_{\mathcal{V}} \text{Ad}(A) = |\text{Tr } A|^2$ and $\text{Ad}(A) \text{Ad}(B) = \text{Ad}(AB)$.

Auxiliary maps. Three maps do all the work: **R** propagates along an edge, **S** starts along every edge, **J** is the backtracking term.

$$\mathbf{R}: \mathcal{W} \rightarrow \mathcal{V}, \quad \mathbf{R}(v \otimes |i\rangle) = \mathcal{E}_i v; \quad \mathbf{S}: \mathcal{V} \rightarrow \mathcal{W}, \quad \mathbf{S}v = \sum_{j=1}^D v \otimes |j\rangle; \quad \mathbf{J}(v \otimes |i\rangle) = \mathcal{E}_i v \otimes |\bar{i}\rangle.$$

The two u -dependent operators on $\mathcal{V}$ are the *deformed adjacency* and the *deformed degree*,

$$\mathcal{A}(u) = u \sum_{i=1}^D \mathcal{E}_i (1 - u^2 \mathcal{E}_i \mathcal{E}_i)^{-1}, \quad \mathcal{D}(u) = u^2 \sum_{i=1}^D \mathcal{E}_i \mathcal{E}_i (1 - u^2 \mathcal{E}_i \mathcal{E}_i)^{-1}.$$

The zeta variable is u throughout (Section 2); the channel normalisation $\Sigma = \sum_i \mathcal{E}_i = D\Phi$ is the one fixed there, and nothing below depends on it.

17.2 Statement

Theorem 17.1 (Ihara–Bass for an arbitrary family of superoperators). *[proved] Let $u \in \mathbb{C}$ be such that $1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i$ is invertible for every $i \in [D]$. Then*

$$\det_{\mathcal{W}}(1 - uT) = \prod_{\{i, \bar{i}\}} \det_{\mathcal{V}}(1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i) \cdot \det_{\mathcal{V}}(1 + \mathcal{D}(u) - \mathcal{A}(u)),$$

the product being over the m unordered pairs $\{i, \bar{i}\}$, which is well defined because $\det(1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i) = \det(1 - u^2 \mathcal{E}_i \mathcal{E}_{\bar{i}})$. The hypothesis holds for all $|u| < (\max_i \|\mathcal{E}_{\bar{i}} \mathcal{E}_i\|)^{-1/2}$, and for all u if every $\mathcal{E}_{\bar{i}} \mathcal{E}_i = 0$; the identity holds pointwise at every such u , and since the left side is a polynomial of degree $\leq ND$ and the right side is rational, the two sides agree as rational functions.

Corollary 17.2 (Kraus families and positivity of side A). *[proved] For arbitrary $A_1, \dots, A_m \in M_n$ with the adjoint pairing above, $\det_{\mathcal{W}}(1 - uT) = \prod_{k=1}^m \det(1 - u^2 \text{Ad}(A_k^\dagger A_k)) \cdot \det(1 + \mathcal{D}(u) - \mathcal{A}(u))$, valid for $|u| < \min_k \|A_k\|^{-2}$; here $\mathcal{E}_{\bar{i}} \mathcal{E}_i$ is $\text{Ad}(A_k^\dagger A_k)$ for $i = k \leq m$ and $\text{Ad}(A_k A_k^\dagger)$ for $i = m + k$, both positive semidefinite. Moreover, for every $\ell \geq 1$,*

$$\text{Tr}_{\mathcal{W}} T^\ell = \sum_{\substack{(i_1, \dots, i_\ell) \in [D]^\ell \\ i_{k+1} \neq i_k \text{ (} k \in \mathbb{Z}/\ell \text{)}}} |\text{Tr}(A_{i_\ell} \cdots A_{i_1})|^2 \geq 0,$$

so $\zeta_T(u) = \exp(\sum_\ell u^\ell \text{Tr} T^\ell / \ell)$ has nonnegative Taylor coefficients, and $\zeta_T(u) = \prod_P \det_{\mathcal{V}}(1 - u^{\ell(P)} \text{Ad}(A_P))^{-1}$ over primitive cyclically non-backtracking classes P of length $\ell(P)$, with A_P the ordered product along P .

Corollary 17.3 (Inverse pairing: the unitary case). *[proved] If $\mathcal{E}_{\bar{i}} = \mathcal{E}_i^{-1}$ for all i — in particular for unitary Kraus operators with $U_{\bar{i}} = U_i^\dagger = U_i^{-1}$ — then, with $\Sigma = \sum_i \mathcal{E}_i$,*

$$\det_{\mathcal{W}}(1 - uT) = (1 - u^2)^{N(D-2)/2} \det_{\mathcal{V}}(1 - u\Sigma + (D-1)u^2).$$

For $\mathcal{V} = M_n$ and $\Sigma = D\Phi$ this is the formula recorded as item 1 of `HANDOFF.md` and, on the bouquet, equation (2.11) of Matsuura–Ohta [60].

Corollary 17.4 (Which factor carries the spectrum). *[proved] The poles of ζ_T lie among the roots of $\det_{\mathcal{V}}(1 + \mathcal{D}(u) - \mathcal{A}(u)) = 0$ together with the points $u^2 = 1/\lambda$ for $\lambda \in \text{spec}(\mathcal{E}_{\bar{i}} \mathcal{E}_i) \setminus \{0\}$; the latter may cancel against poles of the second factor, as they do in Corollary 17.3 (exponent $ND/2$ against $-N$, a total cancellation when $D = 2$). For $D = 2$ in general, $T = \mathcal{E}_1 \oplus \mathcal{E}_2$ on $\mathcal{W} = \mathcal{V} \oplus \mathcal{V}$, since the only non-backtracking successor of i is i itself, so $\det_{\mathcal{W}}(1 - uT) = \det_{\mathcal{V}}(1 - u\mathcal{E}_1) \det_{\mathcal{V}}(1 - u\mathcal{E}_2)$. Conversely a pole of the second factor, which can only sit where some $1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i$ is singular, is not a pole of ζ_T unless it survives multiplication by the first.*

17.3 Proof of the theorem

The proof is written in Lamport structure: each step is a claim followed by its own proof, and the last step assembles them.

Proof of Theorem 17.1. ASSUME: $u \in \mathbb{C}$ with $1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i$ invertible for all $i \in [D]$. PROVE: the displayed identity.

- (1)1. **The decomposition $T = \text{SR} - \text{J}$.** *Proof.* On $v \otimes |i\rangle$, $\text{SR}(v \otimes |i\rangle) = \sum_j \mathcal{E}_i v \otimes |j\rangle$ and $\text{J}(v \otimes |i\rangle) = \mathcal{E}_i v \otimes |\bar{i}\rangle$, so $(\text{SR} - \text{J})(v \otimes |i\rangle) = \sum_{j \neq \bar{i}} \mathcal{E}_i v \otimes |j\rangle = T(v \otimes |i\rangle)$. Both sides are linear and these vectors span $\mathcal{W}$. $\square$
- (1)2. **Block structure of J .** For each pair $p = \{i, \bar{i}\}$ put $\mathcal{W}_p = \mathcal{V} \otimes \text{span}\{|i\rangle, |\bar{i}\rangle\}$. Then $\mathcal{W} = \bigoplus_p \mathcal{W}_p$, $\text{J}\mathcal{W}_p \subseteq \mathcal{W}_p$, and in the ordered decomposition $\mathcal{W}_p = \mathcal{V} \oplus \mathcal{V}$ (component at i , component at $\bar{i}$), $\text{J}|_{\mathcal{W}_p} = \begin{pmatrix} 0 & \mathcal{E}_{\bar{i}} \\ \mathcal{E}_i & 0 \end{pmatrix}$. *Proof.* J sends the component at i to the component at $\bar{i}$ through $\mathcal{E}_i$, and the component at $\bar{i}$ to the component at i through $\mathcal{E}_{\bar{i}}$, using $\bar{\bar{i}} = i$. The direct sum is the partition of $[D]$ into ι -orbits, which have size two because ι is fixed-point-free. $\square$
- (1)3. **$1+u\text{J}$ is invertible, with $\det_{\mathcal{W}}(1+u\text{J}) = \prod_p \det_{\mathcal{V}}(1-u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)$.** *Proof.* By (1)2, $\det_{\mathcal{W}}(1+u\text{J}) = \prod_p \det \begin{pmatrix} 1 & u\mathcal{E}_{\bar{i}} \\ u\mathcal{E}_i & 1 \end{pmatrix} = \prod_p \det_{\mathcal{V}}(1-u^2\mathcal{E}_i\mathcal{E}_{\bar{i}}) = \prod_p \det_{\mathcal{V}}(1-u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)$, by the block determinant and then Sylvester. Each factor is nonzero by ASSUME. $\square$
- (1)4. **Explicit inverse on a pair.** Write $B = u\mathcal{E}_{\bar{i}}$, $C = u\mathcal{E}_i$, $P = (1 - BC)^{-1} = (1 - u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)^{-1}$ and $Q = (1 - CB)^{-1} = (1 - u^2\mathcal{E}_i\mathcal{E}_{\bar{i}})^{-1}$, both existing by ASSUME (the second is the hypothesis at the index $\bar{i}$). Then $(1+u\text{J})^{-1}|_{\mathcal{W}_p} = \begin{pmatrix} P & -BQ \\ -CP & Q \end{pmatrix}$. *Proof.* Multiply out: $\begin{pmatrix} 1 & B \\ C & 1 \end{pmatrix} \begin{pmatrix} P & -BQ \\ -CP & Q \end{pmatrix} = \begin{pmatrix} (1-BC)P & -BQ+BPQ \\ CP-CP & (1-CB)Q \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. A right inverse of a square operator on a finite-dimensional space is the inverse. $\square$
- (1)5. **Sylvester step.** $\det_{\mathcal{W}}(1-uT) = \det_{\mathcal{W}}(1+u\text{J}) \cdot \det_{\mathcal{V}}(1-u\text{R}(1+u\text{J})^{-1}\text{S})$. *Proof.* By (1)1, $1-uT = 1+u\text{J} - u\text{SR} = (1+u\text{J})(1-u(1+u\text{J})^{-1}\text{SR})$, the inverse existing by (1)3. Take determinants and apply Sylvester with $X = u(1+u\text{J})^{-1}\text{S}: \mathcal{V} \rightarrow \mathcal{W}$ and $Y = \text{R}: \mathcal{W} \rightarrow \mathcal{V}$. $\square$
- (1)6. **Computation of $M(u) := \text{R}(1+u\text{J})^{-1}\text{S}$.** It equals $\sum_{i=1}^D (\mathcal{E}_i - u\mathcal{E}_{\bar{i}}\mathcal{E}_i)(1 - u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)^{-1}$. *Proof.* For $v \in \mathcal{V}$, $\text{S}v$ has component v at every index. By (1)4, on the pair $p = \{i, \bar{i}\}$ the vector $(1+u\text{J})^{-1}\text{S}v$ has component $(P - BQ)v$ at i and $(-CP + Q)v$ at $\bar{i}$. Applying R , which applies $\mathcal{E}_i$ to the component at i , $\mathcal{E}_{\bar{i}}$ to the component at $\bar{i}$, and sums over all indices, and substituting B, C, P, Q , the pair p contributes

$$\mathcal{E}_i P - u\mathcal{E}_i \mathcal{E}_{\bar{i}} Q + \mathcal{E}_{\bar{i}} Q - u\mathcal{E}_{\bar{i}} \mathcal{E}_i P = (\mathcal{E}_i - u\mathcal{E}_{\bar{i}}\mathcal{E}_i)P + (\mathcal{E}_{\bar{i}} - u\mathcal{E}_i\mathcal{E}_{\bar{i}})Q.$$

The first summand is the i -th term of the claimed sum and the second is the $\bar{i}$ -th, since $Q = (1 - u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)^{-1}$. Summing over the m pairs gives the sum over all D indices. $\square$

- (1)7. **$1-uM(u) = 1 + \mathcal{D}(u) - \mathcal{A}(u)$.** *Proof.* Expand (1)6 linearly: $uM(u) = u \sum_i \mathcal{E}_i (1 - u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)^{-1} - u^2 \sum_i \mathcal{E}_{\bar{i}}\mathcal{E}_i (1 - u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)^{-1} = \mathcal{A}(u) - \mathcal{D}(u)$. $\square$
- (1)8. **QED.** *Proof.* Combine (1)5, (1)3 and (1)7. Steps (1)1–(1)7 hold pointwise at every u satisfying ASSUME, with no analyticity used. The bound $|u| < (\max_i \|\mathcal{E}_{\bar{i}}\mathcal{E}_i\|)^{-1/2}$ makes $\|u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i\| < 1$, so the Neumann series supplies the inverses and ASSUME holds there. The set of admissible u is cofinite: its complement is the zero set of the polynomial $\prod_i \det_{\mathcal{V}}(1 - u^2\mathcal{E}_{\bar{i}}\mathcal{E}_i)$, which equals 1 at $u = 0$ and so is not identically zero. Two rational functions agreeing on a cofinite set are equal. $\square$

$\square$

17.4 Proofs of the corollaries

Proof of Corollary 17.2. $\mathcal{E}_i \mathcal{E}_{\bar{i}} = \text{Ad}(A_{\bar{i}}) \text{Ad}(A_i) = \text{Ad}(A_{\bar{i}} A_i)$, which is $\text{Ad}(A_k^\dagger A_k)$ or $\text{Ad}(A_k A_k^\dagger)$; for $P \geq 0$ with eigenvalues p_a , $\text{Ad}(P) = \bar{P} \otimes P$ has spectrum $\{\bar{p}_a p_b\} \subseteq [0, \infty)$, and $\|\text{Ad}(A_k^\dagger A_k)\| = \|A_k\|^4$, so the hypothesis of Theorem 17.1 holds for $|u| < \min_k \|A_k\|^{-2}$; the two orderings of a pair have equal determinant by Sylvester, leaving one factor per k . For the trace formula, induction on ℓ gives $T^\ell(v \otimes |i_1\rangle) = \sum \mathcal{E}_{i_\ell} \cdots \mathcal{E}_{i_1} v \otimes |i_{\ell+1}\rangle$, summed over $(i_2, \dots, i_{\ell+1})$ with $i_{k+1} \neq \bar{i}_k$ for $k = 1, \dots, \ell$. Tracing over $\mathcal{W} = \mathcal{V} \otimes \mathbb{C}^D$ sets $i_{\ell+1} = i_1$, which adds the cyclic condition $i_1 \neq \bar{i}_\ell$, and leaves $\text{Tr}_{\mathcal{V}} \text{Ad}(A_{i_\ell} \cdots A_{i_1}) = |\text{Tr}(A_{i_\ell} \cdots A_{i_1})|^2$. Nonnegativity of the coefficients of $\log \zeta_T = \sum_\ell u^\ell \text{Tr } T^\ell / \ell$ implies nonnegativity for $\zeta_T = \exp(\cdot)$. The Euler product is the standard regrouping of cyclic words into powers of primitive classes, using $\sum_{k \geq 1} \frac{u^{k\ell(P)}}{k} \text{Tr } \text{Ad}(A_P)^k = -\log \det(1 - u^{\ell(P)} \text{Ad}(A_P))$, valid for $|u|$ small. $\square$

Proof of Corollary 17.3. With $\mathcal{E}_{\bar{i}} \mathcal{E}_i = 1$ each pair factor is $\det_{\mathcal{V}}(1 - u^2) = (1 - u^2)^N$, so the product is $(1 - u^2)^{ND/2}$; and $\mathcal{A}(u) = \frac{u}{1-u^2} \Sigma$, $\mathcal{D}(u) = \frac{Du^2}{1-u^2}$. Hence $1 + \mathcal{D}(u) - \mathcal{A}(u) = \frac{1}{1-u^2} (1 - u\Sigma + (D-1)u^2)$, whose determinant is $(1 - u^2)^{-N} \det_{\mathcal{V}}(1 - u\Sigma + (D-1)u^2)$. The total exponent is $ND/2 - N = N(D-2)/2$. The computation divides by $1 - u^2$ and so is carried out for $u^2 \neq 1$; both sides are polynomials in u , so the identity extends to $u^2 = 1$. $\square$

Proof of Corollary 17.4. Immediate from the factorisation: a zero of the polynomial $\det_{\mathcal{W}}(1 - uT)$ is a zero of the product, each factor is analytic away from the points named, and a polynomial has no poles, so every pole of the second factor is cancelled by a zero of the first. $\square$

17.5 Provenance and relation to the literature

Watanabe and Fukumizu [77] prove an Ihara–Bass formula for graph zeta functions with arbitrary matrix weights, in the shape $\det(I + \hat{\mathcal{D}} - \hat{\mathcal{A}}) \prod_{[e]} \det(I - u_e u_{\bar{e}})$, with $\hat{\mathcal{D}}$ and $\hat{\mathcal{A}}$ built from $(I - u_e u_{\bar{e}})^{-1}$ exactly as $\mathcal{D}(u)$ and $\mathcal{A}(u)$ are; the byte-verified quotes are in Section 23. On a one-vertex graph whose $2m$ directed edges are $[D]$ with reversal ι , and with $u_e = u_{\mathcal{E}_e}$, their right-hand side becomes the right-hand side of Theorem 17.1 after the push-through identity $\mathcal{E}_i(1 - u^2 \mathcal{E}_{\bar{i}} \mathcal{E}_i)^{-1} = (1 - u^2 \mathcal{E}_i \mathcal{E}_{\bar{i}})^{-1} \mathcal{E}_i$ and the relabelling $i \leftrightarrow \bar{i}$ in the degree term. Two further conventions differ and wash out of $\det(1 - u \cdot)$: they compose a prime cycle in the order opposite to our $\mathcal{E}_{i_\ell} \cdots \mathcal{E}_{i_1}$, and their edge operator weights the target edge where T weights the source. Reversal of words is a bijection of cyclically non-backtracking words because ι is an involution, and $\det(1 - XY) = \det(1 - YX)$.

The framing matters, and was corrected by the round-1 provenance review: their corollary is stated for loopless graphs (their graphs are hypergraphs with two-element hyperedges), so a bouquet is not one of their graphs and their corollary does not apply to it. Theorem 17.1 is therefore the loop-admitting bouquet analogue of their arbitrary-weight corollary, not an instance of it; their proof goes through a hypergraph theorem, the proof above is direct. The shape-matching itself is checked numerically only (`notes/reviews/scratch_wf_mo_conventions.py`).

Matsuura and Ohta [60] treat invertible edge weights on a general graph, which is Corollary 17.3 in that generality; on a bouquet with $X_e = U_e \otimes U_e^\dagger$ their equation (2.11) is the formula of `HANDOFF.md` item 1. No local source covers arbitrary weights on a bouquet.

17.6 Numerical checks

Numerical result 17.5 (Ihara–Bass for random unitary Kraus sets). `[numerical]` For Haar-random unitary Kraus sets closed under adjoint, Corollary 17.3 holds at $u \in \{0.13, 0.29 + 0.11i, -0.4\}$ with relative error at most $2.1 \cdot 10^{-15}$ for $(n, D) = (3, 4)$ and at most $3.9 \cdot 10^{-15}$ for $(n, D) = (4, 6)$. The four

largest eigenvalues of $\Sigma = D\Phi$ are 1.465, 1.850, 3.622, 4.000 for (3, 4) against the Ramanujan bound $2\sqrt{q} = 3.464$, and 2.782, 3.128, 3.594, 6.000 for (4, 6) against 4.472: the first instance violates the bound of Definition 3.17 and the second satisfies it. (`scripts/qihara.py`, `outputs/qihara.txt`.)

Numerical result 17.6 (The general formula in five configurations).

[numerical]

Theorem 17.1, in both the $1 - uM(u)$ and the displayed $1 + \mathcal{D}(u) - \mathcal{A}(u)$ form, at three values of u (one complex), with maximal relative error per configuration as in the table; the trace formula of Corollary 17.2 for $\ell \leq 4$ to $5.5 \cdot 10^{-12}$; and exact verification in rational arithmetic (sympy) for $(N, D) = (1, 4), (2, 4), (2, 6)$. (`scripts/qihara_general.py`, `outputs/qihara_general.txt`.)

configuration	parameters	max. relative error
Gaussian Kraus operators, adjoint pairing	$n = 3, D = 4$	$2.1 \cdot 10^{-15}$
canonical-form MPS tensors and adjoints	$n = 3, D = 4$	$2.0 \cdot 10^{-15}$
generic real superoperators, involution (1 2)(3 4)	$n = 3, D = 4$	$6.7 \cdot 10^{-16}$
single Kraus pair, with the direct-sum check	$n = 3, D = 2$	$1.4 \cdot 10^{-15}$
unitary Kraus, against Corollary 17.3	$n = 3, D = 4$	$1.3 \cdot 10^{-15}$

17.7 What this does and does not give

It gives the compression for *every* matrix product state transfer matrix $\Sigma = \sum_i \text{Ad}(A_i)$, with the physical index doubled by the adjoints. The price of dropping unitarity is that the degree term $(D - 1)u^2$ becomes the operator $\mathcal{D}(u)$ and the adjacency term $u\Sigma$ becomes the deformed channel $\mathcal{A}(u)$, both depending on u through $(1 - u^2 \text{Ad}(A_k^\dagger A_k))^{-1}$.

It does not give a Ramanujan reading by itself. Without $\mathcal{E}_i \mathcal{E}_i = 1$ there is no quadratic relation $\mu\mu' = q$ pairing the eigenvalues of T with those of Σ . What replaces it, and whether canonical form (Definition 3.14) buys anything, is conjecture 26.1.

Side A is unconditionally positive: $\text{Tr } T^\ell = \sum_P |\text{Tr } A_P|^2 \geq 0$ for any Kraus family, so the primes of this zeta are the primitive cyclically non-backtracking words in the A_k and $A_k^\dagger$, with real nonnegative multiplicities.

Observation 17.7 (RH for the quantum Ihara zeta is Hastings' bound).

[sketched]

For unitary Kraus operators closed under adjoint, Corollary 17.3 attaches to each eigenvalue λ of $\Sigma = D\Phi$ the quadratic $1 - \lambda u + qu^2 = (1 - \mu u)(1 - \mu' u)$ with $\mu\mu' = q = D - 1$ and $\mu + \mu' = \lambda$; the eigenvalues λ are real because Σ is Hilbert–Schmidt self-adjoint for an adjoint-closed set. Both roots then have $|\mu| = \sqrt{q}$ exactly when $|\lambda| \leq 2\sqrt{q}$. Hence every nontrivial pole of ζ_Φ lies on the circle $|u| = q^{-1/2}$ if and only if $D\lambda_2 \leq 2\sqrt{D - 1}$, which is Hastings' bound λ_H [33] in the adjacency units of Section 2: the Riemann hypothesis for this zeta is exactly the statement that Φ is a Ramanujan quantum expander (Definition 3.17). The note records this as an observation, not found in the literature in this form; it is tagged **sketched** because it has had no reviewer.

18 Weil Positivity for an Arbitrary Transfer Operator

This section answers the question TJO put on 2026-09-12: the reflection $\rho \mapsto 1 - \bar{\rho}$ pairs the mode of the Riemann channel at a zero with the mode at its mirror image, RH says every mode is its own partner, and Weil's criterion is the positivity of that pairing written on the primes. What does this mean for a completely arbitrary Kraus family, finite-dimensional or continuous? The answer is in three layers. Without any duality, Weil positivity of the rescaled ring norms is exactly a one-sided bound, the spectral-gap half of RH. A duality of the spectrum, the functional equation, turns the bound into a circle and the Weil form into the pairing of each mode with its reflected partner; this identity is the finite form of the inflow identity. For Kraus families the two pairings of the Kraus index buy the two halves separately: inverse pairing gives the functional equation, adjoint pairing gives reality of the transfer spectrum, and only unitarity gives both.

Status. The theorems are the prover's, `codex:gpt-6-astra` (`notes/weil-positivity/astra-proofs.md`, Lament structure, correction ledger in its T7), reviewed adversarially by `claude:opus` (`notes/reviews/weil-positivity`), two model families; all verdicts **VALID**, two wording items fixed. The orchestrator's two observations in Section 19 were reviewed in the same file. Independent numerics by a third agent (numerical result 19.17) found the same three drafting errors the prover corrected.

18.1 Setting

Definitions 4.24 and 4.25 fix the objects: an operator X on a finite-dimensional space, a trivial sub-multiset S_{triv} of its spectrum, the retained multiset $\mathcal{A}_{\text{ret}}$, a critical radius r , the rescaled trace sequence $\mathbf{n}_\ell = \sum_{\mu \in \mathcal{A}_{\text{ret}}} (\mu/r)^\ell$ with $\mathbf{n}_{-\ell} = \overline{\mathbf{n}_\ell}$, the Weil form $W(c) = \sum c_\ell \bar{c}_k \mathbf{n}_{\ell-k}$, the reflection $J(\mu) = r^2/\bar{\mu}$ and the mode-pairing form M . In the Kraus case $X = T$ and $\text{Tr } T^\ell = \sum_w |\text{Tr } A_w|^2$ is the ring-norm sum of Corollary 17.2, so $\mathbf{n}_\ell$ is side A rescaled and with the trivial modes subtracted, and W is a form on rings. Nothing below assumes diagonalisability until Proposition 19.4.

18.2 Positivity is the one-sided bound

Theorem 18.1 (Finite Weil positivity). *[proved] For any X , any S_{triv} and any $r > 0$, the sequence $\mathbf{n}_\ell$ is positive definite if and only if $|\mu| \leq r$ for every $\mu \in \mathcal{A}_{\text{ret}}$, algebraic multiplicities included and zero eigenvalues allowed.*

The proof is the Poisson kernel and a growth lemma. A retained $z = \mu/r$ with $|z| < 1$ contributes to $W(c)$ the integral of $|f_c(e^{i\vartheta})|^2$ against the Poisson kernel $(1 - |z|^2)/|1 - ze^{-i\vartheta}|^2 > 0$, a point $|z| = 1$ contributes $|f_c(z)|^2$, and $z = 0$ contributes $\sum |c_\ell|^2$; so the bound gives positivity. Conversely a positive definite sequence has $|\mathbf{n}_\ell| \leq \mathbf{n}_0$ from its 2×2 minors, while a retained mode outside the disc makes the sequence unbounded, because the modes of maximal modulus $R > r$ contribute R^ℓ times a trigonometric sum whose mean square over $\ell < L$ tends to the sum of squared multiplicities. The point of the theorem is what it does not say: an eigenvalue inside the disc passes. Weil positivity alone is a spectral-gap statement, and the circle needs a second input.

18.3 A duality makes it two-sided: the inflow identity

Theorem 18.2 (Reflection duality and the inflow identity). *[proved] If $\mathcal{A}_{\text{ret}}$ is invariant under J (multiplicities preserved; in particular $0 \notin \mathcal{A}_{\text{ret}}$), then $W(c) = M(c)$ for every finitely supported c , and the following are equivalent: $W \geq 0$; $|\mu| = r$ for every $\mu \in \mathcal{A}_{\text{ret}}$; $J(\mu) = \mu$ for every $\mu \in \mathcal{A}_{\text{ret}}$.*

The identity $W = M$ is the finite inflow identity: the left side is built from traces, i.e. from rings, the right side pairs each mode with its partner under the reflection. It holds whether or

not the modes are on the circle; what the circle adds is that every partner is the mode itself, so $M(c) = \sum |f_c(\mu/r)|^2$. The equivalence follows from Theorem 18.1: $|\mu| \leq r$ and $|J(\mu)| \leq r$ force $|\mu| = r$. This is exactly how RH follows from the one-sided bound $\operatorname{Re} \rho \leq \frac{1}{2}$ together with the functional equation.

Proposition 18.3 (Orbit contributions; an off-circle pair makes the form negative). [proved]
Under the hypotheses of Theorem 18.2, a J-fixed mode z of multiplicity m contributes $m|f_c(z)|^2$ to $W(c)$ and a two-element orbit $\{z, Jz\}$ contributes $2m \operatorname{Re}(f_c(z)\overline{f_c(Jz)})$; if any two-element orbit exists there is a c with $W(c) < 0$.

This is Weil’s off-line pair argument in finite form; the prover’s proof interpolates f_c to vanish at every other retained value, which is what makes the single negative orbit dominate the whole form.

Proposition 18.4 (Reciprocal symmetry without conjugation). [proved]
If $\mathcal{A}_{\text{ret}}$ contains no zero and is invariant under the reciprocal map $\mu \mapsto r^2/\mu$ alone, positivity is still equivalent to $|\mu| = r$ throughout $\mathcal{A}_{\text{ret}}$. If $\mathcal{A}_{\text{ret}}$ is also closed under complex conjugation it is J-invariant and $W = M$; without conjugation symmetry $W = M$ can fail. The reciprocal map is holomorphic on $\mathbb{C}^\times$, not linear.

18.4 Kraus families: which pairing buys which half

Proposition 18.5 (Conjugation symmetry needs no pairing). [proved] *For any family $\mathcal{E}_i = \operatorname{Ad}(B_i)$ with arbitrary $B_i \in M_n$ and any fixed-point-free reversal, the antiunitary involution $x \mapsto x^\dagger$ on $\mathcal{V} = M_n$, and its componentwise version on $\mathcal{W}$, commute with Σ and with T . Hence $\operatorname{spec}(\Sigma)$ and $\operatorname{spec}(T)$ are closed under complex conjugation with multiplicities. The nonnegative ring trace $\operatorname{Tr} T^\ell = \sum_w |\operatorname{Tr} B_w|^2$ of Corollary 17.2 likewise holds for every such family, with no pairing hypothesis: its proof uses only $\operatorname{Ad}(A)\operatorname{Ad}(B) = \operatorname{Ad}(AB)$ and $\operatorname{Tr} \operatorname{Ad}(A) = |\operatorname{Tr} A|^2$.*

The draft sent to the prover asked for an inverse-paired counterexample to conjugation closure; none exists inside the Kraus class, and the independent numerics found the same. Likewise the trace formula $\operatorname{Tr} T^\ell = \sum_w |\operatorname{Tr} B_w|^2$ of Corollary 17.2 never used the pairing: it is a property of the Ad form. So every Kraus-type family, however paired, has a positive side A and a real-structured side B.

Theorem 18.6 (Inverse pairing gives the functional equation). [proved] *Let $\mathcal{E}_i = \mathcal{E}_i^{-1}$ on an N -dimensional space, $D = 2m \geq 4$, $q = D - 1$, $k = N(D - 2)/2$. Then $\operatorname{spec}(T)$ is the multiset $\{+1^{[k]}, -1^{[k]}\}$ together with, for each eigenvalue $\varkappa$ of Σ of multiplicity $a_\varkappa$, the two roots of $\mu^2 - \varkappa\mu + q$ counted $a_\varkappa$ times. The retained multiset after removing $\{+1^{[k]}, -1^{[k]}\}$ has $2N$ elements, no zero, and is invariant under $\mu \mapsto q/\mu$; it is conjugation-invariant iff $\operatorname{spec}(\Sigma)$ is. For inverse-paired Kraus matrices both hold, so the retained multiset is J-invariant for $r = \sqrt{q}$.*

The bookkeeping is where the draft was loose: a double root at $\varkappa = \pm 2\sqrt{q}$ gives $2a_\varkappa$ copies of $\pm\sqrt{q}$, and $\varkappa = \pm D$ gives extra copies of ± 1 beyond the k from the $(1 - u^2)$ factor.

Proposition 18.7 (Both pairings at once is unitarity). [proved] *$\operatorname{Ad}(B^\dagger) = \operatorname{Ad}(B)^{-1}$ iff $B^\dagger B = 1$ iff B is unitary; a family is simultaneously adjoint-paired and inverse-paired iff every B_i is unitary. A scalar $B^\dagger B = \kappa 1$ gives only $\operatorname{Ad}(B^\dagger) = \kappa^2 \operatorname{Ad}(B)^{-1}$; $B = 2$ disproves “unitary up to a scalar”.*

Theorem 18.8 (The Weil criterion for Kraus families). [proved] *(i) For an adjoint-paired Kraus family, any S_{triv} and any r , Weil positivity of $\mathfrak{n}_\ell(T; S_{\text{triv}}, r)$ is exactly the one-sided bound $|\mu| \leq r$*

off S_{triv} ; nonnegativity of the ring traces alone does not give it. (ii) For unitary, adjoint-paired B_i with $D \geq 4$, $r = \sqrt{D-1}$, $\Phi = \Sigma/D$, and $S_{\text{triv}} = \{+1^{[k]}, -1^{[k]}\}$ together with $\{q, 1\}$ once for each eigenvalue $+1$ of Φ and $\{-q, -1\}$ once for each eigenvalue -1 : Weil positivity holds iff every other eigenvalue λ of Φ satisfies $|\lambda| \leq 2\sqrt{D-1}/D$ iff every retained mode is J-fixed. Here λ is real because Φ is Hilbert–Schmidt self-adjoint.

Part (ii) is Hastings’ bound (cited fact 19.11) recovered as Weil positivity, and it is the quantum-channel case of observation 17.7: the Ramanujan property is the statement that the rescaled ring norms form a positive definite sequence.

Proposition 18.9 (An adjoint-paired family with no duality). *[proved]* For $n = 2$, $D = 4$,
 $B_1 = B_3 = \text{diag}(1, 2)$, $B_2 = B_4 = 1$, the retained spectrum of T is invariant under $\mu \mapsto c/\mu$ for no nonzero c . The Weil form is nevertheless defined and its positivity is the one-sided bound.

So for a general channel “every mode is its own partner” has, literally, no partner to refer to: the reflection exists on the punctured plane but need not send a mode to a mode. What remains is Theorem 18.1: a spectral-gap statement, the analogue of $\text{Re } \rho \leq \frac{1}{2}$ without the functional equation.

19 Weil Positivity, Continued: Continuous Time, Hilbert–Pólya, the Zeta Entry, Prior Art

This section continues Section 18 with the same objects (Definition 4.27); the status paragraph there applies.

19.1 Continuous time: cMPS and Lindblad generators

Theorem 19.1 (Continuous Weil positivity is a half-plane bound). *[proved] For a linear operator $\mathcal{L}$ on a finite-dimensional space, a trivial set S_{triv} and $a \in \mathbb{R}$, the rescaled continuous trace F of Definition 4.27 is continuous on $\mathbb{R}$ and positive definite iff $\text{Re } \varpi \leq a$ for every retained ϖ . A retained mode with $\text{Re}(\varpi - a) = -b < 0$ and $\text{Im } \varpi = \omega$ contributes the Lorentzian $2b/(b^2 + (x - \omega)^2)$ to the spectral measure, a mode on the line a point mass, and positive definiteness forces $|F| \leq F(0)$, which a mode to the right of the line violates by continuous mean-square growth.*

Theorem 19.2 (Line duality and the exact continuous pairing). *[proved] If the retained multiset is invariant under $R_a(\varpi) = 2a - \bar{\varpi}$, then for $g = \sum_j c_j \delta_{t_j}$ and $\hat{g}_{\mathcal{L}}(\varpi) = \sum_j c_j e^{t_j(\varpi - a)}$,*

$$\sum_{j,k} c_j \bar{c}_k F(t_j - t_k) = \sum_{\varpi \in \mathcal{A}_{\text{ret}}} \hat{g}_{\mathcal{L}}(\varpi) \overline{\hat{g}_{\mathcal{L}}(2a - \bar{\varpi})},$$

and F is positive definite iff $\text{Re } \varpi = a$ for every retained ϖ .

This is the continuous inflow identity: the autocorrelation of the rescaled trace equals the pairing of each mode with its reflection in the line, Weil’s form in the form $\sum_{\varpi} \hat{g}(\varpi) \overline{\hat{g}(1 - \bar{\varpi})}$.

Theorem 19.3 (Dyson expansion as continuous ring norms). *[proved] For $\mathcal{L}(x) = Kx + xK^\dagger + \sum_j R_j x R_j^\dagger$ on M_n , with no trace preservation assumed,*

$$\text{Tr}_{M_n} e^{t\mathcal{L}} = \sum_{k \geq 0} \sum_{j_1, \dots, j_k} \int_{0 < t_1 < \dots < t_k < t} \left| \text{Tr} \left(e^{(t-t_k)K} R_{j_k} e^{(t_k-t_{k-1})K} \dots R_{j_1} e^{t_1 K} \right) \right|^2 dt_1 \dots dt_k \geq 0,$$

absolutely convergent, the $k = 0$ term being $|\text{Tr } e^{tK}|^2$. Moreover $\mathcal{L}$ commutes with $x \mapsto x^\dagger$, so $\text{spec}(\mathcal{L})$ is conjugation-closed.

The free propagators between jumps were missing from the draft; without them the product is wrong for noncommuting K and R_j . This is the continuous side A: a cMPS with matrices K, R_j has ring norms given by this sum, and the transfer generator $\mathcal{L}$ is its side B. For a Hamiltonian generator $\mathcal{L} = -i[H, \cdot]$ with $H = H^\dagger$ every eigenvalue is purely imaginary and the line statement with $a = 0$ is trivial; that is the Hilbert–Pólya situation, and the content of RH in this language is the existence of a presentation in which the transfer generator is a scalar plus i times a Hermitian operator, which the next two propositions make precise.

19.2 The Hilbert–Pólya inner product, and what positivity does not see

Proposition 19.4 (Unitarisability on the retained space). *[proved] Let S_{triv} be a union of full eigenvalue classes, V' the sum of the generalised eigenspaces of the retained classes, $X' = X|_{V'}$. Then X' is diagonalisable with all eigenvalues of modulus r iff there is an inner product on V' in which X'/r is unitary.*

Proposition 19.5 (Skew-adjoint realisation). *[proved] With the same convention, $\mathcal{L}' - a$ is skew-adjoint in some inner product on V' iff $\mathcal{L}'$ is diagonalisable with all eigenvalues on $\operatorname{Re} \varpi = a$; equivalently $\mathcal{L}' = a + iH$ with H Hermitian in that inner product.*

Proposition 19.6 (Weil forms do not see Jordan blocks). *[proved] For the Jordan block $X = r \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$ with $S_{\text{triv}} = \emptyset$, $n_\ell = 2$ for all ℓ and the sequence is positive definite, although X/r is not unitarisable; likewise $\mathcal{L} = a + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Weil positivity holds with multiplicities and is blind to Jordan structure; a Hilbert–Pólya inner product needs semisimplicity in addition.*

This separates two statements that are usually run together. “All modes on the critical circle” is a statement about the characteristic polynomial and is equivalent to Weil positivity. “There is a Hilbert space in which the transfer operator is r times a unitary” is stronger by exactly the absence of Jordan blocks. For the Riemann channel the distinction is the multiplicity of the zeros: a multiple zero of S gives a Jordan block of $Z(t)$ on K_S .

19.3 The zeta instance and the prior art

Observation 19.7 (The zeta dictionary entry). *[sketched] For the compressed semigroup $Z(t)$ with modes $\eta = -\bar{\rho}/2$, the critical real part is $a = -\frac{1}{4}$, the reflection $\rho \mapsto 1 - \bar{\rho}$ becomes $\eta \mapsto -\frac{1}{2} - \bar{\eta}$, the transform is $\hat{g}_\zeta(s) = \int g(t) e^{(s-1/2)t/2} dt$, and Weil’s form $\sum_\rho \hat{g}(\rho) \overline{\hat{g}(1 - \bar{\rho})}$ is the autocorrelation of the centred trace distribution, whose full-line extension is $T_Z(-u) = e^{u/2} \overline{T_Z(u)}$. The trivial modes at 0 and $-\frac{1}{2}$ and the Gamma-factor terms are not eigenmodes of $Z(t)$; they stay on the geometric side of the explicit formula. The passage from finitely many modes to ζ needs the completed-zeta data, a distributional trace realisation, the full explicit formula and the infinite Weil converse, with Bochner–Schwartz in place of Bochner.*

The prover’s T6 also derives that in the explicit-formula convention of Section 9 the prime part of the centred trace carries weight $-2\Lambda(n)n^{-1/2}$ at $\pm 2\log n$; this is observation 9.6.

Cited fact 19.8 (Weil’s criterion). *[cited] RH holds iff the Weil scalar product $\sum_\rho F(\rho) \overline{G(1 - \bar{\rho})}$ is positive on the admissible test class [12, 51].*

Quoted (prov:cc20-weilcrit, prov:lagarias05-weil-scalar-product): In fact, following \cite{yoshida}, it is enough to prove the negativity of the right-hand side of \eqref{explicit form} for functions g with compact support in the (locally compact) multiplicative group $\mathbb{R}_{>0}$. Furthermore, given any finite set of complex numbers $F \subset \{0, 1\}$, $F \cap \mathbb{Z} = \emptyset$, using the notation $g^\sharp(x) := x^{-1} g(x^{-1})$, $\bar{g}(x) = \overline{g(x)}$, one has (Appendix \ref{apppositivity}, Proposition \ref{mainprop}) \begin{equation} \text{RH} \iff \sum_v \sum_{W \in \mathcal{W}} \sum_{g \in C_c^\infty} (g * \bar{g}^\sharp)(v) \leq 0, \quad \text{for all } g \in C_c^\infty \text{ with } \sum_z \tilde{g}(z) = 0, \quad \text{for all } z \in F. \end{equation}

Given $F(s), G(s) \in \mathcal{S}$ we define the Weil scalar product associated to the automorphic representation π by $\langle F, G \rangle_\pi := \sum_{\rho \in Z(\pi)} F(\rho) \overline{G(1 - \bar{\rho})}$. The sum on the right counts zeros with

Cited fact 19.9 (Li’s criterion). *[cited] RH iff $\lambda_n \geq 0$ for all $n \geq 1$; each condition $\lambda_n \geq 0$ encodes Weil positivity for one test function [76, 51].*

Quoted (prov:voros06-li-criterion, prov:lagarias05-li-is-weil): \emph{Li’s criterion} for the Riemann Hypothesis (RH) states that the latter is true if and only if a specific real sequence $\{\lambda_n\}_{n=1,2,\dots}$ has \emph{all its terms positive} \cite{LI1,BL}.

\begin{theorem} \label{th31} Let π be an irreducible cuspidal unitary automorphic representation of $GL(N)$, with associated ξ -function $\xi(s, \pi)$ and with Weil scalar

that is, $\langle \phi, \phi \rangle_{G,a} > 0$ for all non-zero $\phi \in L^2(-a, a)$ for every $0 < a < \infty$. $\end{theorem}$

showed that the RH is true if and only if the distribution W is non-negative definite, that is,

Cited fact 19.14 (PT symmetry and real spectrum).

[cited] A PT-symmetric

non-Hermitian Hamiltonian has real spectrum when the symmetry is unbroken, i.e. its eigenstates are also PT eigenstates [5].

Quoted (prov:bbm17-pt-real): (iii) Although $\hat{H}$ is not Hermitian, $\hat{H}$ is $\mathcal{CPT}$ symmetric; that is, $\hat{H}$ is invariant under parity-time reflection (in the sense to be defined), which means that the eigenvalues of $\hat{H}$ are either real or else occur in complex-conjugate pairs. If $\hat{H}$ has *maximally broken* $\mathcal{CPT}$ symmetry; that is, if all of its eigenvalues are *pure-imaginary complex-conjugate pairs*, then the eigenvalues of $\hat{H}$ are real and the Riemann hypothesis follows. (iv) While $\hat{H}$ is not Hermitian (symmetric)

Positivity criteria for RH are old (Weil; Li; the kernel form of Suzuki), and Huang’s criterion is the graph case. What is not in the literature found is the statement for an arbitrary transfer operator, the separation of the one-sided bound from the duality, and the Kraus dichotomy. Huang’s criterion is termwise rather than Toeplitz; the next observation reconciles the two.

19.4 Two observations by the orchestrator

Observation 19.15 (Huang’s criterion is the boundedness form). [proved] In finite dimensions, for a conjugation-closed retained multiset, the following are equivalent: positive definiteness of n_ℓ ; boundedness; $|n_\ell| \leq n_0$ for all ℓ ; and the termwise criterion $\operatorname{Re} n_\ell \leq n_0$ for all $\ell \geq 1$. For a connected non-bipartite $(q+1)$ -regular graph with $r = \sqrt{q}$ and $S_{\text{triv}} = \{+1^{[m-n]}, -1^{[m-n]}, q, 1\}$ (m edges, n vertices), Huang’s sequence is exactly $h_k = n_0 - n_k$ with $n_0 = 2(n-1)$, so Huang’s criterion is the boundedness form of Weil positivity. The Toeplitz form is what survives when the trace is a distribution and boundedness has no meaning.

Observation 19.16 (The Kraus dichotomy). [proved] For any Ad-type family the rings are positive and $\operatorname{spec}(\Sigma)$ and $\operatorname{spec}(T)$ are conjugation-closed. (b) Inverse pairing buys the functional equation: with $S_{\text{triv}} = \{+1^{[k]}, -1^{[k]}\}$ and $r = \sqrt{D-1}$, “all retained modes on the circle” is equivalent to: every eigenvalue $\varkappa$ of Σ is real with $|\varkappa| \leq 2\sqrt{D-1}$. (c) Adjoint pairing buys Hilbert–Schmidt self-adjointness of Σ , so $\operatorname{spec}(\Sigma)$ is real for free; $\operatorname{spec}(T)$ need not be real, and in general there is no functional equation (Proposition 18.9). (d) A family with both pairings is unitary; there the eigenvalue $\varkappa = D$ of Σ (from $\Phi(1) = 1$) always violates the band, so (b) is applied with the enlarged trivial set of Theorem 18.8(ii), $S_{\text{triv}} = \{+1^{[k]}, -1^{[k]}\} \uplus \{q^{[d+]}, 1^{[d+]}\} \uplus \{(-q)^{[d-]}, (-1)^{[d-]}\}$, and reduces to Hastings’ bound alone. Reality of $\operatorname{spec}(\Sigma)$ is the Hilbert–Pólya half; the functional equation is the other half; a Kraus family gets the two from two different pairings, and gets both only by being unitary.

The dichotomy reopens conjecture 26.1 in a sharper form: inverse-paired non-unitary families (for instance $B_i = GU_i G^{-1}$ with a common similarity G , which keeps Σ similar to a self-adjoint operator) have a genuine two-sided Ramanujan property with a positive side A, and their non-Hermitian Σ with real spectrum is the transfer-matrix version of an unbroken PT symmetry (cited fact 19.14).

Numerical result 19.17 (Numerical checks).

[numerical]

scripts/weil_positivity.py (outputs/weil_positivity.txt): random adjoint-paired families ($n = 2, 3$; $D = 4, 6$; 300 cases) satisfy “one-sided bound $\Rightarrow$ Toeplitz positivity at $L = 40$ ” without

exception, the threshold radius approaches $\max |\mu|$ off S_{triv} from below like $L^{-3/2}$ (relative gap $3 \cdot 10^{-2}$, $5 \cdot 10^{-3}$, $4.7 \cdot 10^{-4}$ at $L = 40, 160, 640$), the retained spectra are conjugation-closed to 10^{-14} and not reciprocal-invariant; Haar-unitary families reproduce Corollary 17.3 to 10^{-15} , the J-invariance, and “Weil positivity iff Hastings’ bound” on four passing and two violating families, with $W = M$ to 10^{-14} ; an inverse-paired non-unitary family has reciprocal-invariant, conjugation-closed spectrum and nonnegative ring traces to 10^{-15} ; random Lindblad-type generators have $\text{Tr } e^{t\mathcal{L}} \geq 0$ matching the Dyson expansion to 10^{-7} and Gram positivity iff $\max \text{Re } \varpi \leq a$; the Jordan block passes with n_ℓ identical to its diagonal companion. Three of thirty checks fail: they are the three drafted statements the prover corrected.

Two remarks on the numerics. Positivity at a fixed finite L is weaker than the theorem: a mode slightly outside the circle is masked by the positive contribution of the interior modes until L is large. And the $D = 4$ Pauli family $\{X, Z, X^\dagger, Z^\dagger\}$ is an exact Ramanujan example with Φ -spectrum $\{1, 0, 0, -1\}$, exercising the -1 branch of the trivial set with retained modes $\pm i\sqrt{3}$.

20 An Ihara Zeta for Complexes I: the Literature and the Two Flows

TJO’s question of 2026-09-13 was whether the Hashimoto operator, the Bass identity and “RH $\Leftrightarrow$ Ramanujan” survive when a graph becomes a finite simplicial complex, and whether the answer grades by cell dimension into an even and an odd sector. The literature answer is negative in that generality and positive on a thin class: finite quotients of affine buildings.

Status. The theorems here and in Sections 21 and 22 are the prover’s, `codex:gpt-6-astra` (notes/complex-zeta/astra-proofs.md, correction ledger T8.2, hypothesis register T8.3), carried as `sketched` pending the adversarial review; the numerics are an independent third agent’s. Each `asm`: row is an external theorem used but not reproved, byte-cited in `db/claims.tsv`.

20.1 The prototype: PGL_3

Cited fact 20.1 (Kang–Li: the PGL_3 complex zeta and its Bass identity). [cited] For a torsion-free cocompact type-preserving $\Gamma \subset \mathrm{PGL}_3(F)$, the zeta of X_Γ counts tailless primitive closed geodesics made of same-type edges, by algebraic length (0804.2305:main.tex:215-220), and $Z(X_\Gamma, u) = \frac{\{(1-u^3)^{-\chi(X_\Gamma)}\}}{\det(I - A_1u + qA_2u^2 - q^3u^3I)\det(I + L_Bu)}$, (0804.2305:main.tex:236-239); equivalently $1/D_E$ with $D_E = \det(I - L_Eu)\det(I - L_E^t u^2)$ (:247-248). Kang–Li–Wang remove the regularity hypothesis and clear denominators (0809.1401v1:main.tex:1414-1416; operators at :342-368). Note the base $(1 - u^3)$, the *plus* sign in $\det(I + L_Bu)$ and the u^2 on the transposed edge factor [42, 43].

Cited fact 20.2 (Kang–Li–Wang: RH is four-way equivalent to Ramanujan, on three circles). [cited] X_Γ is a Ramanujan complex; \item [(2)] The nontrivial zeros of $\det(I - A_1u + qA_2u^2 - q^3u^3I)$ have absolute value q^{-1} ; \item [(3)] The nontrivial zeros of $\det(I + L_Bu)$ have absolute values 1 , $q^{-1/2}$ and $q^{-1/4}$; and \item [(4)] The nontrivial zeros of $\det(I - L_Eu)$ have absolute values q^{-1} and $q^{-1/2}$. (0804.2305:main.tex:285-295). Unlike the graph case RH is not a single circle: three for the chamber factor, two for the edge factor. In type $\tilde{C}_2$ it degrades to bands, $|u| = q^\alpha$ with $-2 \leq \alpha \leq -1$ (1109.3854:zetagsp4-final.tex:145-157) [25].

20.2 All ranks: an alternating product over facet dimension

Cited fact 20.3 (Kang–Yu: the uniform PGL_n identity is a torsion). [cited] $(1-u^n)^{-\chi(X_\Gamma)}$, $L(\Gamma, q^{(n-1)/2}u) = \prod_{k=1}^{n-1} Z_k^{-\epsilon_k(X_\Gamma, u)^{-1}}$, (2607.21262:main.tex:2019-2023), each factor a reciprocal determinant $\det(I - (-1)^{k+1}T_k)^{-1}$ of a successor operator (:2304-2312) with the parity sign $\epsilon = (-1)^{(k+1)l_G}$ (:1971-1981). The Euler factor is literally a torsion: the alternating product of $\det \Phi_i$ on cochains equals the same on cohomology, where Φ_i acts by $1 - u^n$ (:2463-2470), a strategy that It is inspired instead by Hoffman’s reformulation of Bass’s proof of the Ihara identity in terms of torsion of complexes~\cite{Hof}. (:354-356). The source states *no* RH [45].

Cited fact 20.4 (Lubetzky–Lubotzky–Parzanchevski: the branching geodesic flow). [cited] The higher-dimensional Hashimoto operator is multi-valued: The geodesic flow T on $\mathrm{UT}^j(\mathcal{B})$ is defined as follows: $T(\sigma)$ for $\sigma = (v_0, (v_0, \dots, v_j))$ in $\mathrm{UT}^j(\mathcal{B})$, with $(v_i) \equiv (v_0) + i$, consists of all the cells $\sigma' = (v_1, (v_1, \dots, v_j, w))$ in $\mathrm{UT}^j(\mathcal{B})$ such that (1702.05452:rw_ramanujan_complex.tex:1009-1019), the two

conditions being consecutive colours and geodcity (the union must not be a cell); in dimension one this is non-backtracking. A k -out-regular digraph is Ramanujan when every eigenvalue has $|\lambda| \leq \sqrt{k}$ or $|\lambda| = k$ (:333-336) [55].

Cited fact 20.5 (The higher-rank RH is one-sided, and the interior really is occupied). [cited]Then $Z_{\{T, fX\}}(u)$ satisfies the Riemann Hypothesis; that is, if $Z_{\{T, fX\}}(u)$ has a pole at k^{-s} then $|s|=1$ or $\operatorname{Re} s \leq \frac{1}{2}$. (1702.05452:rw_ramanujan_complex.tex:410-416), for every building. The inequality is not an equality: the source states that poles with $|\operatorname{Re} s| < \frac{1}{2}$ occur, already with $|\operatorname{Re} s| = 0$ in the graph case and with $0 < |\operatorname{Re} s| < \frac{1}{2}$ in higher dimension (:1420-1424).

Cited fact 20.6 (Kamber: the converse, as an L_p criterion). [cited]With one Bernstein–Lusztig operator h_{β_i} per simple coweight and $\zeta = \prod_i \det(1 - h_{\beta_i} u^{l(\beta_i)})^{-1}$ (1701.00154:L_p_Expander_Complexes.tex:352-358) one gets an if-and-only-if: The complex XX is an L_p -expander if and only if every pole λ of $\zeta_{\hat{B}_\phi}$ satisfies $|\lambda| \leq q^{-(p-1)/p}$ or $|\lambda| = q$. (:359-363), Ramanujan meaning L_2 -expander. “Any complex” there means any quotient of an affine building [41].

20.3 What happens off the buildings

Cited fact 20.7 (Storm: the hypergraph zeta is a bipartite graph zeta). [cited]Reading a complex as a hypergraph does give a zeta (math/0608761:Ihara-SelbergofHypergraph.tex:173-174), with a Bass identity (:463) and an if-and-only-if Ramanujan theorem (:627-630) — but all of it because $[\zeta_{\text{hyperH}}(u) = Z_{\{B_{\text{hyperH}}\}}(\sqrt{u})]$ (:430): it is the Ihara zeta of the incidence bipartite graph in $\sqrt{u}$, blind to the simplicial structure beyond incidence [70].

Cited fact 20.8 (Benard–Chaubet–Dang–Schick: a general triangulation has a zeta, but no Ramanujan). [cited]On any triangulation of a compact oriented n -manifold the geodcity rule is exactly LLP’s — consecutive $(n-1)$ -simplices must not bound a common n -simplex (2303.11226:zeta_triangulations9.tex:165-172) — and The combinatorial zeta function $\zeta_{\mathcal{T}}(z) = \prod_{\gamma \in \mathcal{P}} (1 - \varepsilon_\gamma z^{|\gamma|})$, $\zeta_{\mathcal{T}}$ is a polynomial of degree $|\mathcal{T}^{(n-1)}|$ vanishing to order $b_1(M)$ at $z = (n+2)^{-1}$ (:192-205). A Fried-type torsion theorem: that point is not $q^{-1/2}$, and there is no spectral-gap statement [4].

Cited fact 20.9 (No Ihara-type zeta of a general finite complex exists). [cited]Four sources say so in their own words: There is no Ihara-formula for higher rank up to date. (math/0407509:main.tex:174, 2004); Lubotzky’s 2018 ICM survey still calls attaching zetas to high-dimensional complexes a direction of research “with the hope that” Ramanujanness can be read off RH (1712.02526:main.tex:497); Hong–Kwon name “the lack of a unified zeta function” in higher rank (2411.15489:main.tex:354); and Kang–Yu claim only that theirs is To our knowledge, it is the first Ihara-type identity uniform in (n) for this family. (2607.21262:main.tex:249-250). This is a record of what was searched, not an impossibility theorem [15, 57, 34].

20.4 The graded reading, in the literature’s own words

Cited fact 20.10 (Deitmar: a Deligne-type alternating product, and a positive Euler exponent). [cited]For locally symmetric spaces the Ruelle zeta is an alternating product over exterior degree, $Z_{\{P, \phi\}}^R(s) = \prod_{l=0}^{\dim N}$

$Z_{\{P, \sigma_1, \phi\}}(s+1|\alpha) \sim (-1)^1$, (dg-ga/9511006:main.tex:1153-1168), and the divisor is a *degree-weighted* alternating sum $-\sum_p p(-1)^p \dim H^p$ (:1194-1212), the plain Euler sum vanishing identically. Deitmar and Kang take the Euler factors with a positive exponent, because that exponent *is* an Euler number (1303.6848:main.tex:803) [14, 16].

Cited fact 20.11 (Dyatlov–Zworski: odd upstairs, even downstairs, order $-\chi$). [cited] For a negatively curved oriented surface, $\zeta_R(s) = \frac{\zeta_1(s)}{\zeta_0(s)\zeta_2(s)}$, $\text{quad } s \in \mathbb{C}$. so that the order at 0 is $m_1(0) - m_0(0) - m_2(0)$ (1606.04560:zazi.tex:843-855), and $s^{\chi(\Sigma)} \zeta_R(s)$ is holomorphic and nonzero at $s = 0$ (:68-77). Even sectors contribute 1 each, the odd sector b_1 : the continuous prototype of the graded reading [22].

Cited fact 20.12 (Hashimoto’s special value: the trees sit in the vertex determinant). [cited] With $h_X(u) = \det(I - Au + (D - I)u^2)$ and $\kappa(X)$ the tree number, $h_X'(1) = -2 \chi(X) \kappa(X)$, (2310.15619:main.tex:950-955). The complexity is carried by the Laplacian factor, not by the Euler factor $(1 - u^2)^\chi$, whose role is to cancel its zero at $u = 1$ [46].

Cited fact 20.13 (Knill: graph analytic torsion is a super pseudodeterminant). [cited] Torsion $\mathcal{A}(G)$ agrees with the super pseudo determinant $\mathcal{S}(\text{rm SDet})(D) = \prod_k \text{rm Det}(D_k)^{(-1)^k}$ of the Dirac blocks $D_k = d_k * d_k$ of the Dirac operator $D = d + d^*$ and is the ratio of rooted spanning trees on even-dimensional simplices to those on odd simplices (2201.09412:reidemeister.tex:42-52). Knill never mentions the Ihara zeta, nor the Ihara literature SDet [47].

Cited fact 20.14 (Matsuura–Ohta: a fermionic proof of Bass’s identity on a graph). [cited] One Grassmann field on vertices and two per oriented edge, a Dirac operator off-diagonal between them, and two Schur evaluations of one determinant give $(1 - t^2)^{n_E - n_V} \det \Delta$ (2501.08803:main.tex:750-755) and $\zeta_{\Gamma}(q, u)^{-1} = (1 - t^2)^{n_E - n_V} \det \Delta_{\{q, u\}} = \det \left((I_{2n_E} - qB_u) \right)$, . (:811-826). The Euler exponent is the fermionic determinant of the edge-reversal involution, $\det(I - tJ) = (1 - t^2)^{n_E}$ [61].

20.5 Hypothesis register

Assumption 20.15 (H-POINT: building successor conventions). [assumed] Pointed facets, cyclically increasing ordered type, algebraic step length λ_0 , link opposition, and quotient incidences, as in Definition 6.2.

Assumption 20.16 (H-LLP: branching flows and the digraph bound). [assumed] UT^j with consecutive colours and the geodicty exclusion; and, for an equivariant collision-free b -regular branching operator on a Ramanujan complex, $|\lambda| \leq \sqrt{b}$ off the peripheral $|\lambda| = b$, with interior radii genuinely occurring (cited facts 20.4 and 20.5).

Assumption 20.17 (H-KL: the rank-three identity and its four-way RH). [assumed] With $P_3(u) = I - A_1u + qA_2u^2 - q^3u^3I$, $D_E = \det(I - uL_E) \det(I - u^2L_E^t)$, $D_B = \det(I + uL_B)$: $D_B/D_E = (1 - u^3)^{\chi} / \det P_3$, the four-way equivalence of cited facts 20.1 and 20.2, and the L -normalisation of Kang–Li–Wang [44].

Assumption 20.18 (H-KY: the all-rank pointed cochain identity). [assumed] The deformed pointed cochains carry $d(u)$, a degree -1 map $\delta(u)$ and $\Phi_i = (1 - u^n)I + d_{i-1}\delta_i + \delta_{i+1}d_i$, with $\det \Phi_0 = \det P_n = \det \sum_j (-1)^j q^{j(j-1)/2} u^j A_j$ and $\det \Phi_i = (1 - u^n)^{if_i} \det(I - s_i T_i(u))$, $i \geq 1$ (cited fact 20.3).

Assumption 20.19 (H-KY-LOCAL: the local factorisation behind equivariant use). [assumed] $\Phi_i = cW_i^{-1}J_i(I - s_i\Sigma_i)J_i^{-1}Q_i^{-1}$ with Q_i unipotent in the ordered-type filtration, J_i cyclic rotation, Σ_i carrying the same weighted closed paths as T_i , and $\Phi_0 = P_n$ as an *operator* (2607.21262:main.tex:2516-2576, 2586-2640, 2930-2968, 2973-3091).

Assumption 20.20 (H-KL-TABLE: the Iwahori-spherical constituents). [assumed] The unitary Iwahori-spherical types, their fixed-space dimensions and determinant roots, the Steinberg multiplicity $3\chi-3$, and the representation-wise conjugate-pair cancellations (0809.1401v1:main.tex:391-424, 1132-1154, 1171-1209, 1425-1461).

Assumption 20.21 (H-LSV: the joint Hecke spectrum and the explicit complexes). [assumed] The colour-shift convention, commuting normality with $A_k^* = A_{n-k}$, the tempered joint spectrum Aspec_n with its type-trivial exceptions, and the explicit Ramanujan Cayley clique complexes on $\text{PGL}_n(\mathbb{F}_{q^e})$ (Definitions 6.4 and 6.5).

Assumption 20.22 (H-STRONG: Iwahori-spherical temperedness). [assumed] LLP's Ramanujan condition asks every infinite-dimensional *Iwahori-spherical* constituent of $\ell^2(\Gamma \backslash G)$ to be tempered; this can be strictly stronger than the vertex condition, and the LSV examples satisfy it (1702.05452:rw_ramanujan_complex.tex:196-214, 674-701).

Assumption 20.23 (H-KAMBER: the normalised one-sided criterion). [assumed] p -temperedness is equivalent to $|\theta| \leq q^{l(\beta_i)(p-1)/p}$ for every eigenvalue θ of h_{β_i} (cited fact 20.6); the bound is on eigenvalues, so it inverts to a *lower* bound $|u| \geq q^{-(p-1)/p}$ on pole radii.

20.6 The two flows are not the same flow

Theorem 20.24 (Ordered flow versus building flow, exactly, in rank three). [proved-conditional] *Let X be a simplicial type-preserving torsion-free finite quotient of the PGL_3 building. Then (i) the colour-1 restriction of the edge successor relation is L_E and the colour-2 one is conjugate by edge reversal to L_E^t , so $T_1(u) = uL_E \oplus u^2L_E^t$ and $D_E(u) = \det(I - uL_E) \det(I - u^2L_E^t)$; (ii) $T_2(u) = uL_B$ on the $3f_2$ cyclically ordered pointed chambers; (iii) the unrestricted T_1 of Definition 6.1 is not this block sum even after replacing u^2 by u — its outdegree is $2q^2 + q$ against q^2 for L_E , and its directed-colour classes are not invariant; (iv) T_2 has two invariant cyclic orientations of dimension $3f_2$, transposes of one another, so $\det_{H_2}(I + uT_2) = D_B(u)^2$. The u^2 is an algebraic length, not a matrix square, and the $3!$ orderings do not realise L_B once.*

Theorem 20.25 (Which zeta reduces to which). [proved-conditional] *With the pointed data of Definition 6.2, $Z_{\text{pt}} = D_B/D_E = (1 - u^3)^{\chi L_{\text{vertex}}}$, while $Z_{\text{KL}} = 1/D_E$ and $L_{\text{vertex}} = 1/\det P_3$. The graded total is therefore the completed vertex L -function, not Kang–Li's edge-only zeta; with the unrestricted ordered operators it is neither. The orchestrator's draft asserted that the total graded zeta equals Kang–Li's $Z(X_\Gamma, u)$; the prover refuted that, and the above is the correction.*

21 An Ihara Zeta for Complexes II: the Graded Geodesic Determinant

What survives on an arbitrary finite complex is a rational compression, not a polynomial Bass identity; what survives on a building is the full Hecke and torsion apparatus. Notation and the hypothesis rows **asm:h-*** are those of Section 20; Section 22 carries the twisted and quantum layer, which sits on the building side, not the general one.

21.1 Torsion, compression, and the obstruction

Theorem 21.1 (Finite cochain determinant cancellation). *[proved]* Let $(C^\bullet, d)$ be a bounded finite-dimensional cochain complex over $K = \mathbb{C}(u)$, h any degree -1 map, $c \in K^\times$, and suppose each $F_i = cI + d_{i-1}h_i + h_{i+1}d_i$ is invertible. Then $\prod_i \det(F_i|C^i)^{(-1)^i} = \prod_i \det(F_i|H^i)^{(-1)^i} = c^{\sum_i (-1)^i \dim C^i}$. Neither $h^2 = 0$ nor $h = d^*$ is required. The exponent is the dimension count of the complex used: $\chi(X)$ for ordinary cochains, $\sum_i (-1)^i (i+1)! f_i$ for all ordered cells, $\sum_i (-1)^i (i+1) f_i$ for pointed cochains — already $\chi = 1$ against 0 on a single edge.

Theorem 21.2 (The ordinary Euler factor is a difference of two weighted sums). *[proved-conditional]* Under $H\text{-}KY$, $\prod_i \det \Phi_i^{(-1)^i} = (1 - u^n)^{\chi_{\text{pt}}}$ with $\chi_{\text{pt}} = \sum_i (-1)^i (i+1) f_i$, and subtracting the local $\sum_{i \geq 1} (-1)^i i f_i$ contributions leaves $\mathcal{Z}_{\text{pt}}(X, u) = (1 - u^n)^{\chi(X)} / \det P_n(u)$. The bare cochain superdeterminant carries the weighted exponent; only the final flow/vertex comparison carries the ordinary χ . The draft's identification of the cochain product's exponent with χ was wrong.

Theorem 21.3 (Universal ordered-cell Bass–Schur identity). *[proved]* For every finite abstract simplicial complex and every $1 \leq k \leq d$, with R_k the first-vertex deletion, S_k the append map, C_k cyclic rotation, $F_k = C_k + R_{k+1}S_{k+1}$ and $K_k(z) = I + zF_k$: $T_k = S_k R_k - F_k$ and

$$\det_{H_k}(I - zT_k) = \det_{H_k} K_k(z) \det_{H_{k-1}}(I - zR_k K_k(z)^{-1} S_k).$$

This is an explicit rational compression by one dimension, retaining vertices, ordered edges and the upper-cell cyclic factor; it is not a degree- $(d+1)$ polynomial on vertices.

Corollary 21.4 (Bass for every graph, with no regularity assumption). *[proved]* If $d = 1$ then $F_1 = J$ is edge reversal, $\det(I + uJ) = (1 - u^2)^{f_1}$, $R_1 S_1 = A$, $R_1 J S_1 = \text{Deg}$, and $\det(I - uT_1) = (1 - u^2)^{f_1 - f_0} \det_{H_0}(I - uA + u^2(\text{Deg} - I))$. Isolated vertices and trees are included by rational cancellation.

Proposition 21.5 (The tetrahedron obstruction: no universal cubic collapse). *[proved]* On $X = \partial\Delta^3$, $(f_0, f_1, f_2) = (4, 6, 4)$, $\chi = 2$, $T_1 = 0$, $\det(I + uT_2) = (1 - u^4)^6$ and $\mathcal{Z}_{\text{ord}} = (1 - u^4)^6$. There is no matrix polynomial $P(u)$ of any size or degree with $\mathcal{Z}_{\text{ord}} = (1 - u^3)^\chi / \det P(u)$: that would force $\det P = (1 - u^3)^2 / (1 - u^4)^6$, which has poles at $u = -1, \pm i$. The orchestrator's draft asked for a universal cubic vertex Bass identity on every 2-complex with local coefficients Q_1, Q_2 ; the prover refuted it here, and Theorem 21.3 is the surviving universal replacement.

21.2 The graded reading and what it does not say

Theorem 21.6 (The exact graded identity). *[proved]* With odd part $\bigoplus_{k \text{ odd}} H_k$, even part $\bigoplus_{k \text{ even}, k \geq 2} H_k$ and $\Theta = \bigoplus_k s_k T_k$, $\mathcal{Z}_{\text{ord}}(X, u) = \text{sdet}(I - u\Theta)^{-1}$ and $\log \mathcal{Z}_{\text{ord}} = \sum_m \frac{u^m}{m} \text{str}(\Theta^m) = \sum_m \frac{u^m}{m} \sum_k s_k^{m+1} \text{Tr}(T_k^m)$. The order at $u_0 \neq 0$ is the odd minus the

even algebraic multiplicity of u_0^{-1} ; zero eigenvalues give no finite zero or pole, and common even/odd factors cancel. The draft's "the zeros are exactly the odd eigenvalues" is false without the reciprocal, sign, multiplicity and cancellation qualifications.

Proposition 21.7 (Cancellation: the reduced vertex L -function has no finite zeros). *[proved-conditional]* Under H -KL, $L_{\text{vertex}}(u) = D_B / ((1-u^3)^x D_E) = 1 / \det P_3(u)$. In the factorised expression the chamber determinant is a numerator and the edge determinants denominators, but the reduced function is a reciprocal polynomial: every finite zero of D_B cancels, with multiplicity, against an edge or Euler factor. Calling the chamber zeros uncanceled zeros of L_{vertex} , as the draft did, is therefore wrong.

Proposition 21.8 (The cohomology dictionary is parity only). *[proved]* The reindexing $i = k-1$ matches the exponents of the Grothendieck–Lefschetz convention $\prod_i \det(I - u \text{Fr} | H^i)^{(-1)^{i+1}}$, but does not identify H_k or $\mathbb{C}^{P_k}$ with cohomology of X , nor assign Weil weight $k-1$: a 3-cycle has six ordered-edge states and one-dimensional H^0 , and the chamber radii are not those of a single pure weight.

Proposition 21.9 (Which constituents supply the three chamber circles). *[proved-conditional]* Under H -KL and H -KL-TABLE, on a Ramanujan quotient: tempered principal series give chamber roots $\pm q^{-1/2} \alpha_i^{-1/2}$ and edge roots $q^{-1} \alpha_i^{-1}$; the tempered nonspherical type gives one chamber root at $q^{-1/2}$ and two at $q^{-1/4}$; Steinberg twists give the unit circle; one-dimensional constituents give trivial roots at q^{-1} . So the circle $q^{-1/2}$ is not "the odd sector", and a single constituent can contribute to two circles at once.

21.3 An incidence matrix, and why the total is a Berezinian

Theorem 21.10 (One explicit incidence matrix with two Schur evaluations). *[proved]* On $H_{k-1} \oplus H_k \oplus H_{k+1}$ let $\mathbb{D}_k(z)$ have diagonal $(I, I + zC_k, I)$ and off-diagonal couplings $-R_k, -zS_k, zR_{k+1}, -S_{k+1}$ between adjacent dimensions. Then $\det \mathbb{D}_k(z) = \det_{H_k}(I - zT_k) = \det K_k(z) \det_{H_{k-1}} M_k(z)$. Eliminating the edge mass first instead gives an honest coupled vertex/triangle block, which disappears when there are no triangles.

Theorem 21.11 (The total object needs a ratio, not one fermion determinant). *[proved]* Giving the whole auxiliary space of $\mathbb{D}_k(s_k u)$ parity $k+1$ and setting $\mathbb{D}_{\text{flow}} = \bigoplus_k \mathbb{D}_k(s_k u)$ gives $\text{sdet } \mathbb{D}_{\text{flow}}(u) = \mathbb{Z}_{\text{ord}}(X, u)^{-1}$. An ordinary Berezin integral gives $\det D$, not a reciprocal or a superdeterminant merely because D has chiral blocks; the construction uses separate auxiliary copies of some cell spaces, so cell chirality and flow superparity are distinct gradings. The draft's universal ordinary fermion determinant with flow parity equal to cell chirality does not exist.

21.4 Arbitrary weights and Kraus rings

Theorem 21.12 (Arbitrary weights preserve the compression). *[proved]* With the connection of Definition 6.6 and S_k, C_k replaced by their weighted versions, $T_k^E = S_k^E R_k - F_k^E$ and the identity of Theorem 21.3 holds verbatim over $V \otimes H_k$, with no positivity, pairing, invertibility or flatness hypothesis and no inverse of any E_e . Any other successor rule with the same overlap admits the same construction, by letting F be the sum of its forbidden weighted transitions.

Theorem 21.13 (Ring traces, primitive product, and positivity without pairing). *[proved]* $\text{Tr}(T_k^E)^m = \sum \text{Tr}_V(E_m \cdots E_1)$ over based closed walks in the state digraph; for $V = M_n$ and $E_e = \text{Ad}(B_e)$ this is $\sum |\text{Tr}(B_m \cdots B_1)|^2 \geq 0$. Adjoint pairing is not needed: complete positivity of the step maps suffices, as in Corollary 17.2. The unsigned factor is $\prod_{[\gamma]} \det_V(I - u^{|\gamma|} E_\gamma)^{-1}$ over primitive classes; for the signed factor replace E_γ by $s_k^{|\gamma|} E_\gamma$.

Proposition 21.14 (The scope of the positive-ring no-go).

[proved] A single finite transfer matrix contributes a reciprocal polynomial, so it has no nonconstant reduced numerator; in this finite model a numerator comes from net odd multiplicity between sectors. Positivity of a numerical ring sequence alone does not force this: $(1-u)/(1-2u)$ has nonnegative coefficients and positive logarithmic coefficients $(2^m - 1)/m$, yet a genuine numerator.

Observation 21.15 (The natural object, and what “natural” means here). [proved] Definition 6.3 is functorial under isomorphisms preserving states, lengths, successor incidences and transports; it is not proved unique among possible choices of higher-dimensional geodesic. For a graph the default is Hashimoto’s operator and there is only an even flow sector; for a building quotient it is the completed vertex product of Theorem 21.2; the Kang–Li zeta is obtained by retaining only the edge factor.

21.5 The three smallest exact tests

Numerical result 21.16 (Three smallest exact tests, on the ordered default). [numerical]
 C_3 (unfilled): $\det(I - uT_1) = (1 - u^3)^2$, no chamber factor, total $(1 - u^3)^{-2}$. Filled Δ^2 : both flows vanish, total 1 — there is no forced $(1 - u^3)$ factor. $\partial\Delta^3$: $T_1 = 0$, $\det(I + uT_2) = (1 - u^4)^6$, total $(1 - u^4)^6$. The optional clique-only fourth test, the octahedron $K_{2,2,2}$, gives $(1 - u^6)^8/(1 - u^4)^6$, again with a pole at $u = -1$ in the would-be vertex expression. Checked exactly with `sympy`.

22 An Ihara Zeta for Complexes III: Twists, Quantum Expanders, Numerics

The twisted layer of Section 21. Arbitrary weights already keep the compression (Theorem 21.12) and already give nonnegative ring traces (Theorem 21.13); what needs care is *which* twisted object is the Artin one, and what a Ramanujan complex buys for a channel.

22.1 Gauge triviality and the voltage quotient

Proposition 22.1 (A representation connection on the full complex is pure gauge). *[proved] On a Cayley complex the forward weights $E_{(g,gs)} = \varrho(s)$ are generally not flat (flatness asks $\varrho(t)\varrho(s) = \varrho(st)$); the covariant convention is $\varrho(s^{-1})$, and those weights are pure gauge, $F_g = \varrho(g^{-1})$, so $\det(I - uT_k^E) = \det(I - uT_k)^{\dim V}$: every closed straight walk of the full complex has identity holonomy. The four-cycle with $G = \mathbb{Z}/4$ and one-dimensional π separates the full twist $(1-u^4)^2$ from the trivial-block determinant $(1-u)^2$, refuting the draft's claim that the full twist is a representation block.*

Theorem 22.2 (The Artin decomposition of a voltage quotient). *[proved] With the data of Definition 6.7 and a free action on states, $\det(I - s_k T_k(u)) = \prod_{\varrho} \det(I - s_k T_{k,\varrho}(u))^{d_{\varrho}}$. For $R = \bigoplus_{\varrho} a_{\varrho} \varrho$ the Artin determinant has exponents a_{ϱ} , not d_{ϱ} : $R = \pi \otimes \bar{\pi}$ need not embed in a single regular representation (for $\pi = 1 \oplus 1$, $R = 1^{\oplus 4}$). What always transfers is spectral support. An equivariant matrix identity restricts to every block; a scalar determinant equality need not.*

Theorem 22.3 (Exact Euler-factor bookkeeping on finite-group blocks). *[proved-conditional] Under H-KY and H-KY-LOCAL, with $w_{i,\varrho} = \det(W_i|C_{i,\varrho})$, $\mathcal{Z}_{\varrho}(u) = F_{\varrho}(u)/\det(P_n|C_{0,\varrho})$ where $F_{\varrho} = c^{m_{0,\varrho}} \prod_{i \geq 1} w_{i,\varrho}^{(-1)^i}$ and $\prod_{\varrho} F_{\varrho}^{d_{\varrho}} = c^{\chi(X)}$. Only if the action is free on unpointed simplices does this collapse to $F_{\varrho} = c^{d_{\varrho}\chi/|G|}$; with simplex stabilisers the fractional power is not a rational determinant and must not be asserted.*

22.2 Quantum expanders of type $\tilde{A}_{n-1}$

Theorem 22.4 (Harrow-type transfer, with the correct exceptional space). *[proved-conditional] Under H-LSV, for every finite-dimensional unitary π : (i) the Φ_k are commuting normal unitary trace-preserving channels with $\Phi_k^* = \Phi_{n-k}$, whose common fixed space is the commutant $\pi(G)'$, of dimension one exactly when π is irreducible; (ii) they satisfy Definition 6.8 relative to $\mathcal{E}_R$; (iii) $P_{n,R}(u)$ restricted to $\mathcal{E}_R^{\perp}$ has all zeros on $|u| = q^{-(n-1)/2}$; (iv) under H-KL the rank-three quotient Artin flow factors inherit the radii of cited fact 20.2; (v) under H-KY, H-STRONG and H-LLP, general rank gives only the one-sided bound of Proposition 22.6. Removing only the channel fixed points, as the draft did, leaves type characters that violate the tempered bound.*

Proposition 22.5 (The exact converse criterion; one faithful π is not enough). *[proved] Let $\mathcal{B}_X$ be the irreducibles with a nonexceptional joint eigenvalue outside Aspec_n . Then R is Ramanujan iff $\mathcal{B}_X \cap \text{supp}(R) = \emptyset$, and X is vertex-Ramanujan iff $\mathcal{B}_X = \emptyset$; coverage of every nonexceptional irreducible by $\pi \otimes \bar{\pi}$ is sufficient for the converse. Faithfulness is not: for $G = \mathbb{Z}/28$, $S = \{\pm 1, \pm 3\}$, the Cayley graph is not Ramanujan yet a faithful character has $\pi \otimes \bar{\pi} = 1$ and a vacuous quantum condition.*

Proposition 22.6 (One-sided RH, in the correct variable). *[proved-conditional] Under H-LLP, for a b -regular flow with zeta $\det(I - uT)^{-1}$, a pole at $u = b^{-s}$ has $\text{Re } s = 1$ when $|\lambda| = b$*

and $\operatorname{Re} s \leq \frac{1}{2}$ when $|\lambda| \leq \sqrt{b}$, the imaginary part being defined only modulo $2\pi/\log b$; the source's printed $|s| = 1$ cannot literally replace $\operatorname{Re} s = 1$. Under *H-KAMBER* the actual nontrivial pole radii satisfy $|u| \geq q^{-(p-1)/p}$: the quoted eigenvalue bound inverts.

Proposition 22.7 (Which rank-three factors have a duality). [proved-conditional] With $A_2 = A_1^*$ and $p_a(u) = 1 - au + q\bar{a}u^2 - q^3u^3$, the zero multiset is invariant under $u \mapsto q^{-2}/\bar{u}$, before imposing Ramanujan; the principal-series parts of the edge and chamber factors inherit dualities at q^{-2} and q^{-1} . The full edge and chamber factors do not: a tempered nonspherical block has one root at radius $q^{-1/2}$ and two at $q^{-1/4}$, so no $u \mapsto c/\bar{u}$ or $u \mapsto c/u$ can fix or exchange them. Higher-rank *RH* has no universal functional equation.

Proposition 22.8 (Weil positivity on a retained block is a bound until a duality is supplied). [proved] For a retained finite block M with algebraic eigenvalue multiset $\{\mu_j\}$ and $r > 0$, the sequence $\nu_m = \sum_j (\mu_j/r)^m$ is positive definite exactly when $|\mu_j| \leq r$ throughout; invariance of the multiset under $\mu \mapsto r^2/\bar{\mu}$ then forces $|\mu| = r$. This is the finite statement of Theorem 18.1, applied to each retained ordinary block, and it does not transfer automatically to an alternating product with negative spectral multiplicities.

22.3 Numerics on the smallest $\tilde{A}_2$ Ramanujan complex

Numerical result 22.9 (The smallest $\tilde{A}_2$ Ramanujan complex). [numerical] `scripts/a2_complex_zeta.py` (`outputs/a2_complex_zeta.txt`), LSV algorithm at $q = 3$, $d = 3$, $e = 1$: $|S_1| = 13$, $\langle S_1 \rangle = \operatorname{PGL}_3(\mathbb{F}_q[3])$ of order 5616, $S_2 = S_1^{-1}$, $f = (5616, 73008, 97344)$, $\chi = 29952 = |G| \cdot 16/3$, every vertex link the incidence graph of $\operatorname{PG}(2, 3)$ (26 vertices, 4-regular, bipartite, girth 6). Two facts not anticipated in the brief: the complex is *not* 3-colourable ($\operatorname{PGL}_3(\mathbb{F}_q[3])$ is simple), so Kang–Li's type-preservation hypothesis (I) fails on it; and all 13 generators have order 3, so the 24336 triangles carry $\mathbb{Z}/3$ stabilisers and G is not free on unpointed 2-cells. A type-preserving 3-fold cover with $f_0 = 16848$, $\chi = 89856$ is built alongside.

Numerical result 22.10 (The vertex spectrum is tempered). [numerical] $A_2 = A_1^t$ exactly, A_1 normal, $[A_1, A_2] = 0$ (exact sparse check). Of the 5616 eigenvalues of A_1 exactly one is trivial ($\lambda = 13$); all 5615 others are tempered by Cohn's self-inversive criterion, maximum defect $2.0 \cdot 10^{-14}$, and $\max |\lambda| = 7.126860684 < 3q = 9$. So the complex is Ramanujan and cited fact 20.2(2) holds outright.

Numerical result 22.11 (The corrected identity, and the drafted one failing). [numerical] Traces computed exactly as integers by the free G -action: $\operatorname{Tr} L_E^m = 0$ for $m \leq 5$, then 657072 at $m = 6$ up to $1.5 \cdot 10^{17}$ at $m = 18$; $\operatorname{Tr} L_B^m = 0$ for $m \leq 14$, then 16426800 at $m = 15$. Verified exactly to u^{18} , on the 5616-vertex base *and* on the 16848-vertex type-preserving cover: $\det P_3(u) \det(I + uL_B) = (1 - u^3)^\chi \det(I - uL_E) \det(I - u^2L_E^t)$, with degree 308880 on both sides and the residual fitting χ exactly (29952, resp. 89856) at every order. The placement written in the orchestrator's brief fails at u^3 (residual -59904) and fails the degree count (398736 against 219024); $\det(I - uL_B)$ in place of $\det(I + uL_B)$ first fails at u^{15} . The identity holds on the base even though type preservation fails there.

Numerical result 22.12 (Flow operators: the prover's cross-checks). [numerical] L_E has outdegree exactly $q^2 = 9$ and L_B outdegree $q = 3$, whereas the unrestricted ordered edge rule has outdegree $2q^2 + q = 21$: it is not Kang–Li's operator. On all six orderings of a chamber, $\operatorname{Tr} T_2^m = 2 \operatorname{Tr} L_B^m$ for $m = 1, \dots, 18$, i.e. $\det(I + uT_2) = \det(I + uL_B)^2$. On $\partial\Delta^3$, $T_1 = 0$ and T_2 is a permutation of cycle type 4^6 . All three confirm Theorem 20.24 and Proposition 21.5.

Numerical result 22.13 (The three circles, exactly, on two blocks). [numerical] Trivial block: $|\lambda(L_E)| \in \{9 (\times 1), \sqrt{3} (\times 12)\}$ and $|\lambda(L_B)| \in \{3, \sqrt{3} (\times 12), 3^{1/4} (\times 24), 1 (\times 15)\}$. Twelve-dimensional block: $|\lambda(L_E)| \in \{3 (\times 36), \sqrt{3} (\times 120)\}$ and $|\lambda(L_B)| \in \{\sqrt{3} (\times 192), 3^{1/4} (\times 240), 1 (\times 192)\}$. Exactly the radii $\{1, 3^{1/4}, 3^{1/2}\}$ for L_B and $\{3, \sqrt{3}\}$ for L_E predicted by cited fact 20.2, and nothing else; on the full space ARPACK’s forty largest moduli for L_E are 9 once and 3 forty times.

Numerical result 22.14 (The twelve-dimensional quantum twist). [numerical] π the 12-dimensional irrep (the permutation action on the 13 points of $\mathbb{P}^2(\mathbb{F}_q[3])$ minus the constant). The channels $\Phi_k(X) = \frac{1}{13} \sum_{s \in S_k} \pi(s) X \pi(s)^\dagger$ on $M_n[12]$ are unital, trace-preserving and commuting, with exactly one eigenvalue 1; the largest nontrivial $|13\mu| = 6.481723$, below the classical deltoid radius $3q = 9$ and below Hastings’ bound $2\sqrt{D-1} = 6.928203$ at $D = 13$. All 143 nontrivial joint eigenvalues lie inside the tempered deltoid (Cohn defect $2.2 \cdot 10^{-15}$). This is a Ramanujan quantum expander of type $\tilde{A}_2$, and it cannot fail: $\Phi_1 = A_1^{(\pi \otimes \bar{\pi})}/13$, so its spectrum is a sub-multiset of $\text{spec}(A_1)$.

Numerical result 22.15 (Gauge triviality, voltage blocks, and the Euler bookkeeping). [numerical] The forward weights $\pi(s)$ are not flat (39 of the 52 chambers fail $\pi(t)\pi(s) = \pi(st)$); the covariant weights $\pi(s^{-1})$ on the full complex are pure gauge, $\det(I - uT_E^\pi) = \det(I - uL_E)^{12}$. The genuine Artin object is the voltage quotient — 13 states, 9 voltage successors each — whose block traces match the character sums and differ from the gauge-trivial twist, confirming Proposition 22.1 and Theorem 22.2. Writing the residual as $(1-u)^a(1-u^3)^c$: trivial block (13, 1), $\nu = 1 \oplus \pi$ block (13, 65), π block (0, 64) — so for π the naive $c = \chi \dim \pi / |G| = 64$ is exactly right, while the whole anomaly sits in the trivial block, where $\chi/|G| = 16/3$ is replaced by $(1-u)^{13}(1-u^3)^1$. For the trivial block this is an exact polynomial identity of degree 55 on both sides. This is Theorem 22.3’s stabiliser caveat, seen.

22.4 What this does and does not give

The natural object is Definition 6.3: a graded geodesic determinant of a complex *with specified geodesic data* — states, successor incidences, algebraic lengths, transports — and not an unqualified “Thara zeta of every complex”. On an arbitrary complex what is universal is Theorem 21.3, a rational compression that still mentions the forbidden transitions; the polynomial vertex collapse is a *building* phenomenon, and Proposition 21.5 shows it fails on a four-vertex sphere. Opposition, not non-incidence, is the building’s successor rule (Theorem 20.24), and the u^2 is a length.

The odd sector is real as a graded *presentation* and unreal as a divisor: Theorem 21.6 is exact, but on the building vertex side the chamber factor cancels completely (Proposition 21.7), so “fermionic zeros” names a net odd multiplicity that must be checked after cancellation. The parallel with the continuous Ruelle picture (cited fact 20.11) and with Knill’s supertrace (cited fact 20.13) is parity and alternating structure, not an identification of spaces or weights (Propositions 21.8 and 21.9).

The quantum version is the voltage-quotient Artin block, not a connection on the full cover, which is pure gauge (Proposition 22.1). Harrow transfer then gives Ramanujan quantum expanders of type $\tilde{A}_2$ for free from a Ramanujan complex (Theorem 22.4, numerical result 22.14) — the channel adds no spectral values. But higher-rank RH is one-sided (Proposition 22.6), the interior of the band is genuinely occupied (cited fact 20.5), and there is no universal duality to close it (Proposition 22.7): what one has is Weil-positivity-as-bound, Theorem 18.1 applied block by block (Proposition 22.8). None of this touches the Riemann side directly. It is a supply of exactly-solvable graded transfer operators with a Ramanujan property, not a channel whose zeta is $\zeta(s)$.

Conjecture 22.16 (Characterising vertex collapse for a general complex). [conjectured]

There is a combinatorial characterisation of the finite complexes X for which $B_X(u) = (1 - u^{d+1})^{\chi(X)} / \mathcal{Z}_{\text{ord}}(X, u)$ is the determinant of a matrix polynomial $I - uA + \sum_{j \geq 2} u^j Q_{j-1}$ with Q_{j-1}

built from local link data. Building quotients are sufficient (H-KY); Proposition 21.5 gives a necessary polynomiality condition; matching generalised-polygon link parameters is not known to suffice, and no recognition theorem for Bruhat–Tits universal covers is assumed here.

23 Prior Art for the Quantum Ihara Zeta

The search of 2026-09-12 used three literature agents (graph zetas and L -functions; quantum expanders and free probability; direct phrase search), after which every load-bearing sentence was rechecked against the arXiv T_EX source under `refs/src/` (fetched by `refs/fetch_sources.sh`, hashes in `refs/manifest.sha256`). Each quote below is reproduced verbatim from `db/provenance.tsv` and carries its locus; sources with no local T_EX are unverified and are used for attribution only.

23.1 The object under examination

The object is the one fixed in Definition 3.18: for a unital channel $\Phi(X) = \frac{1}{D} \sum_i U_i X U_i^\dagger$ with unitary Kraus operators closed under adjoint, the non-backtracking edge superoperator T on $M_n \otimes \mathbb{C}^D$ and the zeta $\zeta_\Phi(u) = \det(1 - uT)^{-1}$. The founding session made four claims about it. First, an Ihara–Bass identity,

$$\det(1 - uT) = (1 - u^2)^{n^2(D-2)/2} \det(1 - u D\Phi + (D-1)u^2). \quad (4)$$

Second, that all nontrivial poles of ζ_Φ lie on $|u| = (D-1)^{-1/2}$ if and only if Φ is a Ramanujan quantum expander (Definition 3.17). Third, that ζ_Φ is the Artin–Ihara L -function (Definition 3.10) of the bouquet with $D/2$ loops twisted by $\varrho: g_i \mapsto \text{Ad}(U_i)$, with $\text{Tr } T^\ell = \sum_w |\text{Tr } U_w|^\ell$ over cyclically reduced words w of length ℓ . Fourth, that the Weil representation of $\text{SL}_2(\mathbb{F}_p)$ with LPS generators gives an instance where the analogue of the Riemann hypothesis holds exactly. Alongside these sits the matrix-product-state reading (channel as transfer matrix, ζ_Φ as a generating function of ring norms, one correlation length as the Riemann hypothesis, AKLT giving K_4).

23.2 Verdict

claim	status	where
1, Ihara–Bass with $\text{Ad}(U_i)$ weights	known	Matsuura–Ohta 2022, eqs. (2.8) to (2.11) with $X_e = U_e \otimes U_e^\dagger$, loops allowed; the arbitrary-weight formula of Watanabe–Fukumizu 2011 for loopless graphs; twisted zetas back to Sunada 1986 and Hashimoto 1989 (unverified, secondary)
3, trace formula, free-group L -function	known	Matsuura–Ohta 2022 eq. (2.10); Matsuura–Ohta 2026, where the object is named an Artin–Ihara L -function for a representation extended to $U(N_c)$
2, poles on the critical circle iff Hastings–Ramanujan	not found	zero hits for any zeta or Ihara combined with channel, Kraus, CP map or expander; Matsuura–Ohta contain none of the words Ramanujan, expander, Riemann hypothesis; Bordenave–Collins have the operator but no determinant and no zeta
4, exactly Ramanujan channels from LPS	mechanism known, instance not	Harrow 2008 plus LPS; Iyer, Jain, Jordan and Somma (Feb 2026) state it with $SU(2)$ irreps and call explicit Ramanujan quantum expanders a longstanding open problem; no Weil-representation instance and no zeta
MPS and ring-norm reading, $\text{AKLT} = K_4$	not found	no hits for Ihara or zeta combined with matrix product state, transfer matrix or AKLT

23.3 Matsuura and Ohta

Cited fact 23.1 (Extended Ihara zeta with invertible matrix weights). [cited] Suppose $\{X_e\}_{e \in E}$ are invertible matrices of size K living on each edge. (2204.06424:main.tex:476), with $\zeta_G(q; X) \equiv \det \Bigl(\sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \Bigr) - q W_X \Bigl(\sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \Bigr) - q A_X + q^2 (D - \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X)) \Bigl(\sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \Bigr)^{-1}$ (2204.06424:main.tex:479-482) for the edge-matrix determinant, $\zeta_G(q; X) = \exp \left(\sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \right) - q W_X \left(\sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \right) - q A_X + q^2 (D - \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X)) \left(\sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \right)^{-1}$ (2204.06424:main.tex:503-505) for the trace formula over reduced cycles, and the three-term identity $\zeta_G(q; X) = (1 - q^2)^{-(n_V - n_E)K} \det \Bigl(\sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \Bigr) - q A_X + q^2 (D - \sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X)) \Bigl(\sum_{k=1}^{\infty} \frac{1}{k} \sum_{C \in R_k} \text{Tr } P_C(X) \Bigr)^{-1}$ (2204.06424:main.tex:516-519), proved by the classical route: The following proof is parallel to the one given in [bass1992ihara] for the original Ihara zeta function (2204.06424:main.tex:533-534). This is claim 1 with arbitrary invertible matrix weights in place of $\text{Ad}(U_i)$, and claim 3 in the shape of a sum over cyclically reduced classes.

Cited fact 23.2 (Adjoint weights and the name Artin–Ihara L -function). [cited] The weights they apply the formula to are the adjoint ones, with $X_e = U_e \otimes U_e^\dagger$ (2204.06424:main.tex:696); the 2026 paper names the resulting object an Artin–Ihara L -function for a representation R of a finite group extended to $U(N_c)$, taking R to be the adjoint representation with $\varrho_{\text{adj}}(U_e) = U_e \otimes U_e^\dagger$ (2607.27935:main.tex:508-510, with the L -function at 2607.27935:main.tex:381 and its Bartholdi-deformed three-term formula at 2607.27935:main.tex:401-404). So the twisting representation of claim 3 is theirs, up to the column-stacking convention of Section 2.

Their construction admits loops and multiple edges: a loop contributes $X_e + X_e^{-1}$ to the weighted adjacency matrix and 2 to the degree. Specialising to the bouquet with $n_V = 1$, $n_E = D/2$, $K = n^2$ and graph degree D , the exponent $(n_V - n_E)K$ of their eq. (2.11) becomes $-n^2(D - 2)/2$, the bracket becomes $1 - uA_X + (D - 1)u^2$ with $A_X = \sum_i \bar{U}_i \otimes U_i = D\Phi = \Sigma$, and inverting eq. (2.8) turns ζ_X^{-1} into $\det(1 - uT)$. That is Equation (4), so the note records the verdict SAME for claims 1 and 3. The companion paper [59] carries the same adjoint weights into a matrix-weighted Bartholdi zeta, of which the Ihara case is one value of the Bartholdi parameter, and is SAME by the same route; [60] and [62] are the loci above.

23.4 Watanabe and Fukumizu

Cited fact 23.3 (Ihara–Bass for arbitrary matrix weights, loopless). [cited] The edge zeta is built from matrix weights $\{u_e\}_{e \in E}$ with $u_e \in \text{Mat}_{r_t(e) \times r_o(e)}$, with the cycle weight $\pi(\frac{p}{q}) := u_{e_1} \cdots u_{e_k}$ (1103.0605:section3.tex:307-310 and 1103.0605:section3.tex:315), and the corollary `cor:IBfornonhyper` reads $Z_G(\{u\})^{-1} = \det(I + \hat{\mathcal{M}}_{\{D\}}(\{u\}) - \hat{\mathcal{M}}_{\{A\}}(\{u\})) \prod_{e \in E} \det(I - u_e u_{\bar{e}})$ (1103.0605:section3.tex:333-335), of which the authors say Corollary `\ref{cor:IBfornonhyper}` gives the extension of the result to graphs with arbitrary weights. (1103.0605:section3.tex:373). No inverse relation is imposed between u_e and $u_{\bar{e}}$, which is the feature Theorem 17.1 needs.

Their graphs are hypergraphs, A *hypergraph* $H = (V, F)$ consists of a set of vertices V and a set of *hyperedges* F . (1103.0605:section2.tex:17-18), whose hyperedges are subsets of V of size two: no loops and no multiple edges, so a bouquet is not one of their graphs. Theorem 17.1 is the loop-admitting analogue of the corollary, of the same shape after $u_e = u\mathcal{E}_e$, but the corollary does not cover the bouquet and its scalar reduction does

not evaluate correctly there; the note also records that the hypergraph-free proof it points to (the NeurIPS 2009 supplement) was not fetched. Verdict RELATED: the closest prior art for arbitrary weights, not a special case, and with no representation-theoretic or quantum reading [77].

23.5 Bordenave and Collins

Cited fact 23.4 (Matrix-valued non-backtracking operator, no zeta). [cited]The operator is $B = \sum_{j \neq i} a_j \otimes S_i \otimes E_{\{ij\}}$, (1801.00876:tenseur9.tex:784), introduced because Our proof relies on the development of a matrix version of the non-backtracking operator theory and a refined trace method. (1801.00876:tenseur9.tex:158), and aimed at this application: This extension is especially relevant in the context of quantum expanders. (1801.00876:tenseur9.tex:165). The only place the tradition names the identity is in the later work, This formula is closely related to various spectral identities known as Ihara-Bass identities (2304.05714:PT-RC_reloaded.tex:2568). The same operator as T , then, and the spectral correspondence, but as a resolvent identity rather than a determinant: no zeta and no Riemann hypothesis.

The verdict is RELATED. The follow-ups are [8] for general unitaries and [7] for the operator-valued identity; [6] also records, at 1801.00876:tenseur9.tex:608-610, that a family of tensor-squared permutations is a nearly optimal quantum expander in Hastings' sense.

23.6 Hastings and Harrow

Cited fact 23.5 (Transfer of the spectral gap from a Cayley graph). [cited] $\lambda_2(\mathcal{E}) \leq \lambda_2(W_\Gamma)$. (0709.1142:main.tex:213), under the dictionary degree becomes the number of Kraus operators, the spectral gap becomes the gap of the quantum operation (0709.1142:main.tex:72-73). Here λ_2 of the channel is its second largest singular value as a map on density matrices and λ_2 of W_Γ that of the Cayley graph transition matrix, so taking Γ to be LPS generators in $\text{PSL}_2(\mathbb{F}_p)$ gives exactly Ramanujan channels from any irreducible representation.

The bound itself, $\lambda_H = 2\sqrt{D-1}/D$, is Hastings' [33], quoted in Definition 3.17; that paper has no zeta and no non-backtracking operator, and the word Ramanujan does not occur in [31]. Verdict SAME for the mechanism behind claim 4, NOT for anything zeta-shaped.

23.7 Iyer, Jain, Jordan and Somma

Cited fact 23.6 (Exact Ramanujan quantum expanders from LPS). [cited]it is possible to construct exact Ramanujan quantum expanders from $SU(2)$ unitaries derived from the LPS construction~\cite{lubotzky1988ramanujan,harrow2007quantum}. (2602.15180:main.tex:1508), and, of the explicit construction, Our quantum circuits can be used to construct explicit Ramanujan quantum expanders, a longstanding open problem. (2602.15180:main.tex:241). The mechanism is the one of cited fact 23.5, carried by $SU(2)$ irreducible representations rather than by the Weil representation, and no zeta is attached.

Verdict RELATED for claim 4 [38]. The note records that their Appendix C example, $p = 5$ and $D = 6$, uses the same norm-5 quaternions as the script behind Section 13. Separately, [39] amplifies arbitrary expanders to almost Ramanujan ones, short of the bound and with no zeta: verdict NOT.

23.8 Quantum-walk zetas and the secondary sources

Cited fact 23.7 (Attribution of the twisted zeta and of the three-term formula). [cited] A zeta function of a regular graph G associated with a unitary representation of the fundamental group of G was developed by Sunada [Sunada1, Sunada2]. (2105.02677:main.tex:87–89); for graph coverings, Stark and Terras further introduced notions of *Galois coverings* and *Artin–Ihara L -functions* for graphs (2309.15873:AIM_Final.tex:131); and for the determinant, the three-term determinant formula from [Theorem 18.15]{Terras:2011} applied to the Artin–Ihara L -function (2405.04361:v3.tex:427). The twisted zeta is therefore Sunada’s [71], sharpened by Hashimoto [32] and Bass [2]; the L -function of a covering is Stark and Terras’ [69], with the three-term formula at Theorem 18.15 of [73]. None of these primary sources is local.

In Stark and Terras the representation is one of the Galois group of a covering of finite graphs, hence of a finite group, whereas $\text{Ad}(U_i)$ for generic unitaries generates an infinite group: the object of Definition 3.18 is not literally a Stark–Terras L -function, although the determinant formula is the same. The secondary sources [23] and [28] serve this attribution only; verdict RELATED.

The quantum-walk zeta line [50], [49], [48] builds zetas from the unitary evolution matrix of a quantum walk rather than from Kraus operators, [49] extending to open quantum random walks on the torus by Fourier analysis; verdict NOT for claims 1 to 4, the value here being the Sunada pointer. Ben-Aroya, Schwartz and Ta-Shma [3] is the one unverified source near claim 2: an agent read from the publisher PDF that they believe their first construction has the potential of being improved to a quantum Ramanujan expander.

23.9 What was not found, and the limits of the search

Observation 23.8 (Verdict of the prior-art search). [sketched]

No source found attaches a zeta function, or an analogue of the Riemann hypothesis, to a quantum channel; none states that the poles of ζ_Φ lie on the critical circle if and only if Φ meets the Hastings bound λ_H ; none gives the matrix-product-state ring-norm reading or the identification of AKLT with K_4 . The Ad-weighted Ihara–Bass formula and the free-group L -function reading are known (cited facts 23.1 and 23.2), arbitrary matrix weights are known for loopless graphs (cited fact 23.3), and the matrix-valued non-backtracking operator is known (cited fact 23.4).

The consequence for this book: [60], [77] and the Sunada–Hashimoto line are cited for the object and the determinant, while the Ramanujan reading of Section 17, the dictionary of Section 14 and the Weil–LPS instance of Section 13 are this notebook’s own. Theorem 17.1 is the bouquet analogue of cited fact 23.3, proved there from scratch; no local source covers arbitrary non-invertible weights on a bouquet.

These arXiv metadata queries returned zero hits: "Ihara zeta" AND "quantum channel"; "Ihara zeta" AND "completely positive"; "zeta" AND "quantum expander"; "zeta" AND "Kraus operators"; abs:"zeta function" AND abs:"superoperator"; "Ruelle zeta" AND "quantum channel"; "Ihara" AND "matrix product state"; "AKLT" AND "zeta"; "non-backtracking" AND "superoperator"; "Ihara" AND ("quantum Ramanujan" OR "Ramanujan quantum"). Web and Scholar searches for the phrase quantum Ihara zeta, for Ihara zeta with noncommutative or quantum graphs (Weaver, Duan–Severini–Winter), for Ihara zeta with matrix product states or tensor networks, and a MathOverflow search for Ihara zeta and quantum expanders, returned nothing on point. The standing limit: arXiv search is metadata-only, so a formula inside a paper body would not surface, and the absence claims of observation 23.8 are weaker than its presence claims.

23.10 Corrections

One agent attributed arXiv:2309.15873 to Kudo and Li. The $\text{\TeX}$ header of the local source, at 2309.15873:AiM_Final.tex:86,90, gives the authors as Mason Eyler and Jaiung Jun; [23] and every citation of it were corrected on the evening of 2026-09-12.

24 The Continuous Dictionary

TJO’s question: replace channels by Lindblad generators. The simplest continuous instance is the generator of rotations on a circle of circumference $\log p$, whose trace is a Poisson comb at multiples of $\log p$; the Selberg analogue should be a Lindbladian built from the vector fields on $\mathrm{PSL}_2(\mathbb{R})$, with the Laplacian as the small operator, and a formal analogue of the comb. This section records what survives when the statements are written down and proved. The proofs are in `notes/selberg/astra-proofs.md`, authored by the second model family (codex, `gpt-6-astra`, reasoning effort xhigh) as a constructive prover from the brief `notes/selberg/astra-brief.md`, and reviewed adversarially by a Claude Opus refuter; the symbolic and numerical checks are in `scripts/selberg_lindblad.py`. Conventions: Definitions 4.18 to 4.23; $\Delta = -y^2(\partial_x^2 + \partial_y^2)$ is positive; Γ is torsion-free and cocompact except in the modular computation.

24.1 Standard inputs

Six facts are used as assumptions, printed as such because no local source quotes them in full; every claim below that depends on one prints `-conditional`.

Assumption 24.1 (Fuchsian geometry).

[assumed]

A torsion-free cocompact Fuchsian group acts freely and properly discontinuously on $\mathbb{H}$; primitive hyperbolic conjugacy classes correspond to primitive oriented closed geodesics with period equal to translation length; hyperbolic balls have area $2\pi(\cosh R - 1)$.

Assumption 24.2 (Laplace spectrum of a compact surface). [assumed] Δ on a connected compact hyperbolic surface has discrete spectrum $0 = \lambda_0 < \lambda_1 \leq \dots \rightarrow \infty$, written $\lambda_j = \frac{1}{4} + r_j^2$ with $r_j \geq 0$ real for $\lambda_j \geq \frac{1}{4}$, $r_j = i\nu_j$ with $0 < \nu_j < \frac{1}{2}$ for $0 < \lambda_j < \frac{1}{4}$, and $r_0 = i/2$.

Assumption 24.3 (Selberg zeta is entire with known divisor).

[assumed] Z_{Sel} extends

to an entire function with zero divisor $\sum_{j \geq 0} ([\frac{1}{2} + ir_j] + [\frac{1}{2} - ir_j]) + (2g - 2) \sum_{n \geq 0} (2n + 1)[-n]$ and no other zeros or poles.

Assumption 24.4 (Closed realisations).

[assumed] Complete volume-preserving flows

have skew-adjoint unitary generators, and Δ on a compact manifold has its self-adjoint realisation and heat semigroup.

Assumption 24.5 (Standard facts about ζ).

[assumed] ζ continues meromorphically

with $\zeta(0) = -\frac{1}{2}$, $\zeta(1 + w) = 1/w + \gamma_E + O(w)$, the functional equation $\zeta(z) = \chi(z)\zeta(1 - z)$ with $\chi(z) = \pi^{z-1/2}\Gamma((1 - z)/2)/\Gamma(z/2)$, no zeros on $\mathrm{Re} z = 1$, and $|\zeta'/\zeta(1 + \epsilon + iv)| \leq C \log^2(3 + |v|)$ for $|v| \geq 1$, $0 \leq \epsilon \leq \epsilon_0$.

Assumption 24.6 (Standard facts about Γ).

[assumed] Γ is meromorphic

and zero-free with simple poles at the nonpositive integers, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, and (Γ'/Γ) has the Gauss integral representation and $O(\log)$ vertical growth in right half-planes.

24.2 Cited facts

Cited fact 24.7 (Guillemin flat trace formula).

[cited] Dyatlov and Zworski

[21] state, for an Anosov flow, $\mathrm{tr} \, e^{-itP} = \sum_{\gamma} \frac{T_{\gamma} \, \delta(t - T_{\gamma})}{|\det(I - \mathcal{P}_{\gamma})|}$, $\forall t > 0$ (1306.4203:zeta.tex:260–264, sum over closed orbits including repetitions, $T_{\gamma}^{\#}$ the primitive period). In our notation this is Definition 4.20.

The correction to the worklog's wording: the prime comb is an orbital (locally finite) trace, and the phrase “Poisson trace of the direct sum” does not name a trace-class operator. The missing ingredient is transverse expansion: a circle has no Poincaré map, so no Jacobian, so no resonances.

24.4 The Lindbladian of the vector fields is the Casimir, and the Laplacian on the K -invariant sector

Proposition 24.17 (Multiplication observables see a sum of squares). *[proved]* For real vector fields X_j acting as derivations of a smooth function algebra and jump operators $L_j = -iX_j$, on a common invariant domain, $\sum_j (L_j M_f L_j - \frac{1}{2}\{L_j^2, M_f\}) = -\frac{1}{2} \sum_j [L_j, [L_j, M_f]] = M_{\frac{1}{2} \sum_j X_j^2} f$; with the plus sign, since the X_j are formally anti-self-adjoint.

Proposition 24.18 (Casimir on the K -invariant sector). *[proved]* For right- K -invariant $f(g) = F(g \cdot i)$: $\mathbf{W}f = 0$, $\Omega f = y^2(F_{xx} + F_{yy})$ (the constant is $c = 1$), and $\frac{1}{2}(\mathbf{H}^2 + \mathbf{E}^2)f = 2\Omega f = -2\Delta F$. On all functions $\frac{1}{2}(\mathbf{H}^2 + \mathbf{E}^2) = 2\Omega + \frac{1}{2}\mathbf{W}^2$: the Casimir carries the compact direction with a minus sign and is not a diffusion Lindbladian on all of $L^2(\Gamma \backslash G)$; the two-jump Lindbladian with $L_1 = -i\mathbf{H}$, $L_2 = -i\mathbf{E}$ preserves the K -invariant sector and equals -2Δ there.

The second-order computation is the whole content: along $g_{x,y} \exp(u\mathbf{H}) \cdot i = x + iye^{2u}$ and $g_{x,y} \exp(u\mathbf{E}) \cdot i = x + y \tanh 2u + iy \operatorname{sech} 2u$, one finds $\mathbf{H}^2 f = 4y^2 F_{yy} + 4y F_y$ and $\mathbf{E}^2 f = 4y^2 F_{xx} - 4y F_y$, and the first-order drifts cancel. Squaring the projected tangent vectors alone would miss the drift and give the wrong operator.

Proposition 24.19 (The quantum Lindbladian of a representation). *[proved]* For a unitary representation π with anti-self-adjoint generators $A_j = d\pi(X_j)$ and jumps $L_j = -iA_j$, on finite-rank smooth operators, $\sum_j (L_j X L_j - \frac{1}{2}\{L_j^2, X\}) = \frac{1}{2} \sum_j B_j^2 X$ with $B_j X = [A_j, X]$ the generators of $\pi \otimes \bar{\pi}$ on Hilbert–Schmidt operators; for $\mathfrak{sl}_2$ and the two non-compact jumps it is $2\Omega_{\pi \otimes \bar{\pi}} + \frac{1}{2}B_W^2$, equal to $2\Omega_{\pi \otimes \bar{\pi}}$ on diagonal- K -invariant operators.

This is the continuous Harrow construction: a channel from a representation becomes a Lindbladian from a representation, and its gap is a question about $\pi \otimes \bar{\pi}$.

Conjecture 24.20 (Gap of the representation Lindbladian). *[open]* For an irreducible unitary π of $\mathrm{SL}_2(\mathbb{R})$, or of $\mathrm{SL}_2(\mathbb{F}_p)$, the two-jump Lindbladian $2\Omega_{\pi \otimes \bar{\pi}} + \frac{1}{2}B_W^2$ has a spectral gap governed by the decomposition of $\pi \otimes \bar{\pi}$; for the Weil representation this is the continuous shadow of the Weil–LPS channels of Section 13.

25 The Continuous Dictionary II: Selberg Tower and the Cusp Comb

Continuation of Section 24, same conventions and the same proof file `notes/selberg/astra-proofs.md`.

25.1 The geodesic flow: continuous Ihara–Bass

Proposition 25.1 (Poincaré Jacobian).

[proved]

For a closed geodesic of length ℓ the linearised Poincaré map on $\mathfrak{sl}_2/\mathbb{R}H$ is $\text{Ad}(a_\ell^{-1})$ with multipliers $e^\ell, e^{-\ell}$, so $|\det(1 - P_\gamma^k)| = (e^{k\ell} - 1)(1 - e^{-k\ell}) = 4 \sinh^2(k\ell/2) = e^{k\ell}(1 - e^{-k\ell})^2$; the circle has an empty transverse space and factor 1.

Proposition 25.2 (Flat trace, Selberg tower, sign).

[proved-conditional] For

$\text{Re } \varsigma > 0$, $\Lambda_\text{fl}(\varsigma) = \sum_\gamma \sum_{k \geq 1} \ell_\gamma e^{-\varsigma k \ell_\gamma} / (4 \sinh^2(k \ell_\gamma / 2))$ converges absolutely, the tower $D_{\text{tow}}(\varsigma) = \prod_{j \geq 1} Z_{\text{Sel}}(\varsigma + j) = \prod_\gamma \prod_{m \geq 0} (1 - e^{-(\varsigma+m+1)\ell_\gamma})^{m+1}$ converges absolutely without zeros, and $\Lambda_\text{fl} = +D'_{\text{tow}}/D_{\text{tow}}$, equivalently $D_{\text{tow}}(\varsigma) = \exp \left(- \int_0^\infty t^{-1} e^{-\varsigma t} \text{Tr}^b e^{-t\mathcal{X}} dt \right)$.

The tower comes from $1/(4 \sinh^2(x/2)) = \sum_{m \geq 0} (m+1) e^{-(m+1)x}$: the Jacobian of the two transverse directions is what a graph does not have. The sign was drafted as a minus and is a plus; the graph counterpart of D_{tow} is $\det(1 - uT)$, not the zeta.

Proposition 25.3 (Divisor of the tower and the first band). [proved-conditional] D_{tow} continues to an entire function whose zeros are $\varsigma = -\frac{1}{2} - k \pm ir_j$ ($k, j \geq 0$) and $\varsigma = -N$ ($N \geq 1$) with order $(2g-2)N^2 + 2$; Λ_fl has simple poles there with residue the order. The nonconstant first band $\mathcal{B}_0$ is exactly the pole multiset in $-1 < \text{Re } \varsigma < 0$, and $\mathcal{B}_0 \subset \{\text{Re } \varsigma = -\frac{1}{2}\}$ iff every r_j , $j \geq 1$, is real iff Δ has no eigenvalue in $(0, \frac{1}{4})$.

The constant mode has $r_0 = i/2$ and contributes $\varsigma = 0$ and -1 ; it must be removed, exactly as the constant adjacency mode is removed in the Ramanujan condition of a graph. With it removed, the continuous Ramanujan property is Selberg's $\frac{1}{4}$ property, and the band structure is the one Dyatlov, Faure and Guillarmou prove for the operator-theoretic resonances (cited fact 24.8); Proposition 25.3 makes no claim about operator realisations, only about the poles of the continued Λ_fl .

Proposition 25.4 (Graph comparison and the Ruelle quotient).

[proved-conditional] For a

finite graph with Hashimoto matrix T , $\text{Tr } T^m = \sum_\gamma \sum_{k \ell_\gamma = m} \ell_\gamma$ and $\det(1 - uT) = \prod_\gamma (1 - u^{\ell_\gamma})$, with no Jacobian and no tower, and $\partial_\varsigma \log \det(1 - e^{-\varsigma} T) = \sum_m \text{Tr } T^m e^{-\varsigma m}$ has the sign of $\partial_\varsigma \log D_{\text{tow}}$. For the surface, $\zeta_R(s) = Z_{\text{Sel}}(s)/Z_{\text{Sel}}(s+1)$ (telescoping the product) and $D_{\text{tow}}(\varsigma)/D_{\text{tow}}(\varsigma+1) = Z_{\text{Sel}}(\varsigma+1)$.

25.2 The cusp of the modular surface is the prime comb

Proposition 25.5 (Exact Fourier identity for the scattering multiplier). [proved-conditional] m_{sc} is real, even, smooth and of polynomial growth, and

$$C_{\text{cusp}} = -P + A_{\text{arch}}, \quad P(t) = \sum_{n \geq 2} \frac{\Lambda(n)}{n} [\delta(t - 2 \log n) + \delta(t + 2 \log n)],$$

where $\langle A_{\text{arch}}, h \rangle = \frac{1}{2} \int h - (\gamma_E + \log \pi) h(0) + \int_0^\infty \frac{e^{-u} h(0) - \frac{1}{2} e^{-u/2} (h(u) + h(-u))}{1 - e^{-u}} du$, so $A_{\text{arch}}(t) = \frac{1}{2} - 1/(4 \sinh(|t|/2))$ away from 0. The prime atoms are dips; the constant $\frac{1}{2}$ is the pole of ζ at 1 (the ordinary boundary value of $\zeta'/\zeta(1+2ir) + \zeta'/\zeta(1-2ir)$ differs from its Abel boundary distribution by $\pi \delta_0$), not a digamma term.

The route: $\varphi'/\varphi(s) = (\Gamma'/\Gamma)(s - \frac{1}{2}) - (\Gamma'/\Gamma)(s) + 2\zeta'/\zeta(2s - 1) - 2\zeta'/\zeta(2s)$; the functional equation moves $\zeta'/\zeta(2ir)$ to $-\zeta'/\zeta(1 - 2ir)$ plus Gamma terms, and the $(\Gamma'/\Gamma)(ir)$ terms cancel exactly, leaving $m_{\text{sc}} = a + b$ with $b(r) = \zeta'/\zeta(1 + 2ir) + \zeta'/\zeta(1 - 2ir)$; the Dirichlet series of b at $1 + \epsilon$ transforms termwise to the comb, and the $\epsilon \downarrow 0$ limit produces $-\pi\delta_0$ on the frequency side.

Numerical result 25.6 (Symbolic and numerical checks).

[numerical]

`scripts/selberg_lindblad.py` (output `outputs/selberg_lindblad.txt`): exact sympy, Lindbladian $= \frac{1}{2}X^2$; the left-invariant fields derived from the Iwasawa coordinates satisfy the $\mathfrak{sl}_2$ brackets, and on K -invariant F : $Hf = 2yF_y$, $Ef = 2yF_x$, $Wf = 0$, $(H^2 + E^2)f = 4y^2(F_{xx} + F_{yy})$, constant $c = +1$, Lindblad constant 2; quantum tensor identity to $7 \cdot 10^{-15}$, and for the three-dimensional $\mathfrak{sl}_2$ irrep the Lindbladian spectrum is $\{0, -4^{\times 3}, -12^{\times 5}\} = -2j(j + 1)$; Poincaré determinant $2 - 2\cosh(k\ell)$, absolute value $4\sinh^2(k\ell/2)$; cusp transform with a Gaussian window ($G_w = 40$, $R = 6G_w$): dips at $2\log n$ with amplitude ratio to $\Lambda(n)/n$ of mean -31.91545 , spread 10^{-3} , against the predicted $-2G_w/\sqrt{2\pi} = -31.91538$; residual after the exact archimedean background and the additive constant is $3 \cdot 10^{-8}$ over $t \in [0.5, 7]$. The constant came out as 1 because the script's multiplier is $-\text{Re}(\varphi'/\varphi)$, twice m_{sc} .

Proposition 25.7 (What the rate- $\frac{1}{4}$ damping identifies).

[proved-conditional] On

$(0, \infty)$, $e^{-t/4}P_{\text{ch}} = P_+ = -(C_{\text{cusp}} - A_{\text{arch}})|_{(0, \infty)}$. This is an equality of prime atomic measures after subtracting A_{arch} and choosing positive weights; it is not an equality of the full cusp distribution with a semigroup trace, nor of operators or spectra, and it implies nothing about the Riemann hypothesis.

The sentence in the previous session, “the cusp term of the modular trace formula is the trace of the Riemann channel damped at rate $\frac{1}{4}$ ”, is therefore corrected: what is proved is that the prime measure written in the Riemann-channel note, damped at rate $\frac{1}{4}$, is the prime part of the cusp distribution. The full trace of $Z(t)$ is not established as a pure prime comb (its own explicit formula has pole, archimedean and sign bookkeeping), and the rate $\frac{1}{4}$ is the uniform rate only under the Riemann hypothesis.

25.3 The dictionary

Observation 25.8 (Continuous dictionary).

[sketched-conditional] Row by row: edge

space $\leftrightarrow \Gamma \backslash G$; Hashimoto operator $T \leftrightarrow$ geodesic generator $\mathcal{X}$; adjacency $A \leftrightarrow$ Casimir Ω , equal to $-\Delta$ on the K -invariant sector; Ihara–Bass $\det(1 - uT) = (1 - u^2)^{|E| - |V|} \det(1 - uA + qu^2) \leftrightarrow$ the tower $D_{\text{tow}} = \prod_{j \geq 1} Z_{\text{Sel}}(\zeta + j)$ with zeros in the bands $-\frac{1}{2} - k \pm ir_j$; Ramanujan $\leftrightarrow$ no eigenvalue in $(0, \frac{1}{4})$; the primes of ζ enter only through the cusp of the modular surface, as the scattering term, not as closed geodesics.

discrete	continuous	where
edge space $\mathcal{W}$	$\Gamma \backslash \text{PSL}_2(\mathbb{R})$	Definition 4.18
T	$\mathcal{X} = H/2$	Definition 4.18
$\text{Tr } T^m$ (no Jacobian)	$\text{Tr}^b e^{-t\mathcal{X}}$, weights $1/4 \sinh^2$	Propositions 25.1 and 25.4
A , $\Sigma = D\Phi$	Ω , -2Δ on K -invariants	Proposition 24.18
$\det(1 - uT)$	$D_{\text{tow}}(\zeta)$, sign $+D'_{\text{tow}}/D_{\text{tow}}$	Proposition 25.2
$\mu\mu' = q$	bands $-\frac{1}{2} - k \pm ir_j$	Proposition 25.3
Ramanujan	$\text{spec } \Delta \cap (0, \frac{1}{4}) = \emptyset$	Proposition 25.3
mixed-unitary channel	two-jump Lindbladian of $\pi \otimes \bar{\pi}$	Proposition 24.19
primes as rings	cusp term, dips $\Lambda(n)/n$ at $2\log n$	Proposition 25.5

25.4 Not established

A literal operator-theoretic compression from $\Gamma \backslash G$ to the K -invariant sector that turns D_{tow} into a Laplacian determinant (the continuous Ihara–Bass as an identity of operators rather than of divisors); an anisotropic-space realisation of the trace resonances; a trace formula for the full compressed semigroup $Z(t)$; any gap for conjecture [24.20](#); anything about the Riemann hypothesis or the $\frac{1}{4}$ property of a given surface.

26 Open Problems and Dead Routes

Dead routes are results. Both lists mirror `HANDOFF.md`, which is the live version.

26.1 Open problems, in order

Conjecture 26.1 (Ramanujan without unitarity). *[open]* *There is a Ramanujan-type reading of the poles of ζ_T for non-unitary Kraus families, replacing the relation $\mu\mu' = q$ of the unitary case, and canonical form $\sum_k A_k^\dagger A_k = 1$ constrains the poles.*

What is known: Theorem 17.1 replaces qu^2 by the operator $\mathcal{D}(u)$ and $u\Sigma$ by the deformed channel $\mathcal{A}(u)$, both built from $(1 - u^2 \text{Ad}(A_k^\dagger A_k))^{-1}$, so no quadratic relation pairs the eigenvalues of T with those of Σ . Cheap next step: a random-MPS pole-radius hunt (bond dimension 2 to 4, physical dimension 2) to see whether the poles cluster on one circle at all.

Stinespring form of the compressed semigroup. Conjecture 9.10 (stated in Section 9): a Stinespring form of $Z(t)$ on K_S whose Kraus (jump) operators are the prime dilations.

The single-Kraus lift exists trivially. A Holevo-form covariant Lindbladian with the von Mangoldt jump measure lives on the full regular representation, has symbol ζ^t , and is not the compressed object. A negative answer with a reason is a result. Suggested route (HANDOFF step 2): write $Z(t)$ explicitly through Cauchy kernels and test a generator of Holevo jump form with jumps at $k \log p$ on K_S , blind, in a second model family.

Hecke identification of the Weil–LPS spectra. Conjecture 13.4 (stated in Section 13): the joint spectra of the commuting channels $\Phi_q^\pm$ are Hecke eigenvalues of weight-2 forms for the quaternion algebra ramified at $\{2, \infty\}$, via Jacquet–Langlands.

Needs LMFDB and the Jacquet–Langlands bookkeeping; the visible structure (joint eigenvalue pairs, small-degree algebraic integers, multiplicities equal to those of irreducibles of $\text{PSL}_2(\mathbb{F}_p)$ in $W_\pm \otimes \overline{W}_\pm$) is consistent with it.

Conjecture 26.2 (Assembling over p). *[open]* *The Weil–LPS channels for different p assemble into one object whose rings are the primes and whose transfer spectrum is the zeros of ζ .*

This is the parent campaign’s DG-GLOBAL question in this guise. Nothing is known beyond the obstruction recorded below.

Conjecture 26.3 (The Phantasm has both halves). *[open]* *There is a $\mathbb{Z}_2$ -graded Lindbladian (or a type III generalisation at $\beta = 1$) whose stationary state is the Bost–Connes KMS state, with entanglement Hamiltonian $\log N$, and whose odd relaxation modes are the nontrivial zeros: the object having both halves of observation 11.17. The rebound-state equation of the SHW renewal picture with a mixed rebound state is its finite-dimensional shadow; the vacuum rebound state is excluded by Proposition 11.8.*

This replaces item 0b of the reframe programme. The fixed point constrains the Lindbladian only partially, and at $\beta = 1$ the fixed point is not normal (Proposition 11.10), so K_S with its pure vacuum cannot be the whole bond; the identification of K_S with the Bost–Connes bond that Section 9 records as missing is the first obstacle. A second open item from the same campaign is the unrestricted infinite parity question, observation 10.13.

Conjecture 26.4 (Galois-graded bond at finite level).

[open] The model space of a congruence subgroup $\Gamma_0(N)$, whose scattering determinant involves the Dirichlet L -functions mod N , carries the zeros of $L(s, \chi)$ organised by the characters χ of $(\mathbb{Z}/N)^\times$, with the Shor map U_a as the level- N Galois symmetry acting on the bond; the graded bond of observation 11.18 is realised at finite level.

This is a computation, not yet done; it is the Riemann-side test of the trivial-versus-nontrivial-character reading of the grading that the Artin–Schreier super-transfer matrix exhibits.

26.2 Dead routes

Observation 26.5 (Prime-by-prime ansätze are blind to zeros).

[sketched] The direct sum over primes of rotations on circles of circumference $\log p$ has trace $\sum_n \Lambda(n) \delta(s - \log n)$ by Poisson summation and dynamical zeta equal to the Euler product; for any finite set S of primes its spectrum is the set of poles of the partial Euler product. The same holds for product states and commuting dilations. Zeros exist only in the infinite product, so no prime-by-prime ansatz can see them.

Recorded 2026-09-12 from the circle-flow discussion. Two remarks kept: in the Witt convention $L_1 - L_{-1}$ is the dilation flow (source and sink) and L_0 the rotation; the only relation between primes is additive.

Observation 26.6 (Symmetry protection fixes signs, not moduli).

[sketched] The content of an L -function that has the three properties of a symmetry-protected topological invariant (a phase, quantised, robust under symmetric perturbation) is its sign data: the root number as a product of local ϵ -factors with the product formula as anomaly cancellation, the sign of a quadratic Gauss sum (the Arf invariant over $\mathbb{F}_2$, the Gauss–Milgram central-charge phase over $\mathbb{Z}/p$), and Hilbert-symbol reciprocity as a local-to-global cocycle condition. The Riemann hypothesis is modulus data. Moduli are not protected: a Ramanujan graph ceases to be Ramanujan under a small symmetric perturbation, and the cubic Cayley graphs of $\mathrm{PSL}_2(\mathbb{F}_p)$ on the generators of $\mathrm{PSL}_2(2, \mathbb{Z})$ fail the Ramanujan bound for $p \geq 11$ while keeping every symmetry. Cohomological protection therefore cannot deliver RH.

Recorded 2026-09-12 from the SPT side quest. What survives of the bulk–boundary lesson is the positivity form of Weil’s criterion: the reflection $\rho \mapsto 1 - \bar{\rho}$ pairs the mode at a zero with the mode at its mirror image, RH says every mode is its own partner, and Weil’s form is the pairing of each mode with its partner, written through the explicit formula as a form on the primes. Its finite-dimensional, arbitrary-Kraus version is the subject of Section 18. Two relabellings, not new content: the Weil–LPS channels live on the edge Hilbert space of the $\mathbb{Z}/p \times \mathbb{Z}/p$ cluster-state phase (the Heisenberg group is that edge’s projective representation and $\mathrm{SL}_2(\mathbb{F}_p)$ its automorphism group, acting by the Weil representation); and Bost–Connes at $\beta = 1$ is a critical point of a spontaneous breaking of the Galois symmetry, not a symmetry-protected phase.

Further routes closed or flagged in the founding session: the functional integral framing of Selberg as a continuum limit (TJO: too far); calling $-\log E$ the entanglement Hamiltonian (an error, corrected in observation 8.7); an algebraic-integer recognition of the Weil–LPS joint eigenvalues from five-digit output (tautological, since the adjacency matrix is integral); the Dwork transfer operator as a route to the Riemann hypothesis for curves (observation 14.4).

27 Reproducibility Map

Every numerical claim in this book points to a script under `scripts/` whose standard output is captured under `outputs/`. All scripts run from the repository root with `python3` and need only `numpy`, `scipy`, `mpmath` and `sympy`. Random data are seeded. The local CI re-runs the fast scripts and diffs their output against the captured files.

script	what it establishes	output, claims
<code>qihara.py</code>	Quantum Ihara–Bass identity for Haar-random unitary Kraus sets, $(n, D) = (3, 4), (4, 6)$, at complex u ; Ramanujan check of Σ	<code>outputs/qihara.txt</code> ; <code>num:qihara-unitary</code>
<code>bcmpo.py</code>	Bost–Connes phase operators $e(a/b)$ as MPOs over the prime chain against Hurwitz-zeta sums; character-sector transfer eigenvalues against $L(\beta, \bar{\chi})/\zeta(\beta)$	<code>outputs/bcmpo.txt</code> ; <code>num:bc-mpo</code>
<code>scat.py</code>	Unimodularity of $S(\tau)$, its zeros at $\gamma_n/2 - i/4$, contractivity in the lower half plane, prime-phase identity	<code>outputs/scat.txt</code> ; <code>num:scattering-symbol</code>
<code>ringnorm.py</code>	Windowed sum over the first 3000 zeros against the full explicit formula; dips at $\log n$ of depth $\Lambda(n)/\sqrt{n}$	<code>outputs/ringnorm.txt</code> ; <code>num:ringnorm-3000</code>
<code>weil_lps.py</code>	Weil–LPS channels for $(p; q)$ in seven cases: intertwining check, Ramanujan bound, Harrow containment, Hecke relations, commutation, joint spectra	<code>outputs/weil_lps_p*.txt</code> ; <code>num:weil-lps-ramanujan</code>
<code>weil_lps_hashimoto.py</code>	Direct edge superoperator for $(13, 17)$, odd block: Ihara–Bass to 10^{-14} , moduli $\{17, \sqrt{17}, 1\}$	<code>outputs/weil_lps_hashimoto_p13_q17_odd.t</code> ; <code>num:weil-lps-rh</code>
<code>artin_schreier_mps.py</code>	Brute-force exponential sums against $\text{Tr } E^n$ for quadratic Artin–Schreier data; $EE^\dagger = qI$	<code>outputs/artin_schreier_mps.txt</code> ; <code>num:artin-schreier-mps</code>
<code>qihara_general.py</code>	Theorem 17.1 in both forms, five configurations, float to 10^{-15} and exact rational arithmetic	<code>outputs/qihara_general.txt</code> ; <code>num:qihara-general</code>
<code>artin_schreier_super.py</code>	Sign law, endpoint criteria, corrected Frobenius blocks, super-transfer super-traces against point counts, symplectic matrices and the Weil character formula, Prony on the counts; 1969 checks	<code>outputs/artin_schreier_super.txt</code> ; <code>num:artin-schreier-super</code>
<code>riemann_graded_trace.py</code>	Windowed supertrace of the Riemann graded generator against $2P_+$ with 3000 zeros; ladder alternatives; flipped-pair negativity; finite vacuum-decay spectra; prime-chain $M/M/1$ bands	<code>outputs/riemann_graded_trace.txt</code> ; <code>num:riemann-graded-trace</code>
<code>bc_entropy.py</code>	Entropy of the Bost–Connes Gibbs density and its prime and bond cutoffs; Hagedorn level count	<code>outputs/bc_entropy.txt</code> ; <code>num:bc-entropy</code>

Data

`data/zeros3000.npy`: the first 3000 imaginary parts of the zeta zeros, from `mpmath.zetazero`.
`data/ringnorm.npz`: the arrays of the ring-norm test.

Reference sources

`refs/fetch_sources.sh` downloads the arXiv TeX source of every cited paper into `refs/src/<id>/` (not redistributed) and writes `refs/manifest.sha256`. Every quote in `db/provenance.tsv` is byte-checked by the gate at its stated `<id>:<file>:<line>`.

Reviews

`notes/reviews/` holds the adversarial review reports; a `proved` claim names its review file in `db/claims.tsv` and the gate checks that the file contains `VERDICT <id>: VALID`.

28 Lab Log

The dated worklogs under `docs/worklog/` are the primary chronology; `HANDOFF.md` is the live state. This section is their mirror at the level of decisions.

2026-09-10/11: founding session

One conversation in the parent repository `arithmetic-quantum-mechanics`. Trajectory: Bost–Connes basics; why not Wick-rotate to Hilbert–Pólya; an elliptic curve over $\mathbb{F}_2$ as a finite instance; permutations and the one determinant identity; automata and rationality; the actual Riemann hypothesis as a flow with orbit periods $\log p$; the Ihara zeta of regular graphs as the vestigial Selberg zeta; Shor’s unitary as the level- N Galois symmetry of Bost–Connes; quantum expanders and the quantum Ihara zeta; matrix-product states and the ring-norm reading; the Bost–Connes system as a matrix-product operator over the prime chain; the Holevo–Werner lift; and the computation of the scattering symbol, its zeros, and the 3000-zero ring-norm test.

Corrections TJO made along the way, kept as working rules: stay elementary, no jargon dumps, one step at a time; no curves or field extensions when a permutation will do; “QFT” did not mean functional integrals; length quantisation should come from a flatness-type condition.

Afternoon: the Weil–LPS channels computed for seven primes. Bugs found and fixed: quaternion enumeration starting at an odd coordinate (caught by the divisor-sum count); parity as a second fixed point on the full space (resolved by splitting into the two Weil blocks); the Hecke relation failing on SL_2 by signs (resolved on PSL_2).

Evening: the matrix-product formulation of the Weil conjectures for quadratic Artin–Schreier curves. Bugs: a wrong irreducibility test for composite n ; the even- n sign, which was the L -function normalisation, not a bug.

2026-09-12

Morning: circle flows discussed (no code); the expository note on Deligne’s proof written; repository made public under AGPL-3.0.

Afternoon: prior-art search on the quantum Ihara zeta, three Opus literature agents, then byte-verification against arXiv TeX sources (`refs/`). Finding: the object and its determinant formula are known (Matsuura–Ohta; Watanabe–Fukumizu for loopless graphs; Sunada); the RH-for-channels reading, the MPS dictionary and the Weil-representation instance were not found. Then the Ihara–Bass formula for arbitrary Kraus families proved in Lamport style and taken through two rounds of adversarial review by Opus reviewers; promoted to **proved**. Standing preference recorded: TeX source, not PDF, for every quotation.

Evening: this lab book set up following the parent repository’s sharded pattern, with the notation, definitions, claims and provenance databases under `db/`, the generated macro and status files, the gate `scripts/labbook_check.py`, the local CI `scripts/ci_local.sh` and its pre-commit hook. Notation and definitions across the seven notes de-duplicated into the databases; the clashes and their resolutions are listed in Section 2.

Night: the Selberg dictionary campaign (codex prover, Opus refute review, shards 09b/09c), and the Weil-positivity campaign of the second afternoon session (shards 08b/08c). Evening: TJO set the Bost–Connes reframe as the central priority and named the chased object the Phantasm.

2026-09-12, night: the Phantasm campaign

Orchestration: one codex prover session (gpt-6-astra, xhigh; brief `notes/riemann-cmps/astra-brief.md`, proofs `notes/riemann-cmps/astra-proofs.md`, 1356 lines, labels T1–T8), three Opus lanes in parallel (Artin–Schreier numerics, Riemann-side numerics, TeX sources with 57 byte-verified quotes and 17 new arXiv sources), then an Opus REFUTE review. Results: Section 10, Section 11, Section 15, the definitions shard Section 5, and the restatements of Proposition 9.5 (HANDOFF item 1c) and Proposition 14.1. Corrections along the way: the founding session’s $\alpha_i = -\lambda_i(E)$ was found false by the numerics lane; the orchestrator’s replacement (a conjugated law) was found false by the same lane in two further cases; the prover produced the exact law (radical sign, Stickelberger on the complement) and the numerics lane confirmed it in 624 further checks. TJO’s steering during the campaign: the fixed point of the bond Lindbladian is literally the entanglement spectrum, so the Gibbs entropy and its cutoffs are entanglement data (recorded; the Hagedorn level density rules out a CFT reading); and the Phantasm must carry the Bost–Connes symmetry $\hat{\mathbb{Z}}^\times$, which led to the trivial-versus-nontrivial-character reading of the grading and to conjecture 26.4.

2026-09-13: finite BC symmetry constraints

TJO clarified that the BC state is the prescribed stationary state of an unknown Lindbladian, constrained by Galois symmetry and adelic symplectic structure. The finite inverse problem was formulated as a linear section of the conditional Choi cone in Section 12. At criticality its dimensions were computed for Galois, linear Weil, and additional Weyl covariance; the last must be distinguished from merely carrying the symplectic representation. The finite BC phase marginal does not specify a matrix-state extension, and an exact finite representation of the full BC algebra cannot retain nontrivial phase observables.

An explicit $p = 7$ covariant GKLS generator has unequal odd decay rates. Compatible two-prime models retain a free coupling even with fixed local semigroups. The numerical script passes 427 checks and is included in local CI. Full written derivations were supplied by Codex but no independent review was performed, so the eight new analytic claim rows are **sketched**; the ninth is **numerical**. The arithmetic map to the scattering bond and the adelic limit remain open.

References

- [1] Daniel Alpay, Fabrizio Colombo, David P. Kimsey, and Irene Sabadini. An extension of Herglotz’s theorem to the quaternions. *Journal of Mathematical Analysis and Applications*, 421:754–778, 2015. arXiv:1403.0079.
- [2] Hyman Bass. The ihara-selberg zeta function of a tree lattice. *International Journal of Mathematics*, 3:717–797, 1992. no local source.
- [3] Avraham Ben-Aroya, Oded Schwartz, and Amnon Ta-Shma. Quantum expanders: motivation and constructions. *Theory of Computing*, 6:47–79, 2010. no local source.
- [4] Leo Benard, Yann Chaubet, Nguyen Viet Dang, and Thomas Schick. Combinatorial zeta functions counting triangles, 2023.
- [5] Carl M. Bender, Dorje C. Brody, and Markus P. Müller. Hamiltonian for the zeros of the Riemann zeta function. *Physical Review Letters*, 118:130201, 2017. arXiv:1608.03679.
- [6] Charles Bordenave and Benoît Collins. Eigenvalues of random lifts and polynomials of random permutation matrices. *Annals of Mathematics*, 190, 2019.
- [7] Charles Bordenave and Benoît Collins. Norm of matrix-valued polynomials in random unitaries and permutations, 2023.
- [8] Charles Bordenave and Benoît Collins. Strong asymptotic freeness for independent uniform variables on compact groups associated to non-trivial representations. *Inventiones Mathematicae*, 237, 2024.
- [9] J.-B. Bost and A. Connes. Hecke algebras, type iii factors and phase transitions with spontaneous symmetry breaking in number theory. *Selecta Mathematica*, 1:411–457, 1995. no local source.
- [10] Nick Bultinck, Dominic J. Williamson, Jutho Haegeman, and Frank Verstraete. Fermionic matrix product states and one-dimensional topological phases. *Phys. Rev. B*, 95:075108, 2017.
- [11] Alain Connes. Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Mathematica*, 5:29–106, 1999. arXiv:math/9811068.
- [12] Alain Connes and Caterina Consani. Weil positivity and trace formula, the archimedean place. *Selecta Mathematica*, 27:77, 2021. arXiv:2006.13771.
- [13] Alain Connes and Matilde Marcolli. From physics to number theory via noncommutative geometry. Part I: Quantum statistical mechanics of $\mathbb{Q}$ -lattices, 2004.
- [14] Anton Deitmar. Geometric zeta-functions of locally symmetric spaces, 1995.
- [15] Anton Deitmar and J. William Hoffman. The ihara-selberg zeta function for $p\mathrm{gl}_3$ and hecke operators, 2004.
- [16] Anton Deitmar and Ming-Hsuan Kang. Geometric zeta functions for higher rank p -adic groups, 2013.
- [17] Pierre Deligne. La conjecture de weil. i. *Publications Mathématiques de l’IHÉS*, 43:273–307, 1974. no local source.

- [18] Christopher Deninger. Arithmetic geometry and analysis on foliated spaces, 2005.
- [19] Bernard Dwork. On the rationality of the zeta function of an algebraic variety. *American Journal of Mathematics*, 82:631–648, 1960. no local source.
- [20] Semyon Dyatlov, Frédéric Faure, and Colin Guillarmou. Power spectrum of the geodesic flow on hyperbolic manifolds. *Analysis & PDE*, 8, 2015.
- [21] Semyon Dyatlov and Maciej Zworski. Dynamical zeta functions for anosov flows via microlocal analysis. *Annales scientifiques de l'École normale supérieure*, 49, 2016.
- [22] Semyon Dyatlov and Maciej Zworski. Ruelle zeta function at zero for surfaces. *Inventiones Mathematicae*, 210, 2017.
- [23] Mason Eyler and Jaiung Jun. Artin-ihara l-functions for hypergraphs, 2023.
- [24] L. D. Faddeev and B. S. Pavlov. Scattering theory and automorphic functions. *Journal of Soviet Mathematics*, 3, 1975. no local source; Russian original 1972.
- [25] Yang Fang, Wen-Ching Winnie Li, and Chian-Jen Wang. The zeta functions of complexes from $sp(4)$. *International Mathematics Research Notices*, 2013.
- [26] Philipp Fleig, Henrik P. A. Gustafsson, Axel Kleinschmidt, and Daniel Persson. *Eisenstein series and automorphic representations*. Cambridge University Press, 2018.
- [27] Joshua S. Friedman, Jay Jorgenson, and Lejla Smajlović. An evaluation of the central value of the automorphic scattering determinant, 2016.
- [28] Sohan Ghosh and Anwesh Ray. On the iwasawa theory of cayley graphs, 2024.
- [29] Shamgar Gurevich and Ronny Hadani. The geometric Weil representation, 2007.
- [30] Jutho Haegeman, J. Ignacio Cirac, Tobias J. Osborne, and Frank Verstraete. Calculus of continuous matrix product states. *Phys. Rev. B*, 88:085118, 2013.
- [31] Aram W. Harrow. Quantum expanders from any classical cayley graph expander. *Quantum Information and Computation*, 8, 2008.
- [32] Ki-ichiro Hashimoto. Zeta functions of finite graphs and representations of p-adic groups. In *Automorphic forms and geometry of arithmetic varieties*, volume 15 of *Advanced Studies in Pure Mathematics*, pages 211–280. 1989. no local source.
- [33] M. B. Hastings. Random unitaries give quantum expanders. *Physical Review A*, 76:032315, 2007.
- [34] Soonki Hong and Sanghoon Kwon. Edge zeta function and closed cycles in the standard non-uniform complex from pgl_3 , 2024.
- [35] Roger Howe. On the character of Weil's representation. *Trans. Amer. Math. Soc.*, 177:287–298, 1973. no local source.
- [36] Hau-Wen Huang. Ihara zeta function, coefficients of Maclaurin series, and Ramanujan graphs. *International Journal of Number Theory*, 16:1–12, 2020. arXiv:1905.13485.

- [37] Yasutaka Ihara. On discrete subgroups of the two by two projective linear group over p-adic fields. *Journal of the Mathematical Society of Japan*, 18:219–235, 1966. no local source.
- [38] Vishnu Iyer, Siddhartha Jain, Stephen Jordan, and Rolando Somma. Efficient quantum circuits for high-dimensional representations of $\mathrm{su}(n)$ and ramanujan quantum expanders, 2026.
- [39] Fernando Granha Jeronimo, Tushant Mittal, Sourya Roy, and Avi Wigderson. Almost ramanujan expanders from arbitrary expanders via operator amplification, 2022.
- [40] Bernard L. Julia. Statistical theory of numbers. *Number Theory and Physics (Les Houches 1989)*, *Springer Proc. Phys.*, 47:276–293, 1990. no local source; cited from memory for the Hagedorn reading of the Bost–Connes transition.
- [41] Amitay Kamber. l_p -expander complexes, 2017.
- [42] Ming-Hsuan Kang and Wen-Ching Winnie Li. Zeta functions of complexes arising from $\mathrm{pgl}(3)$. *Advances in Mathematics*, 256:46–103, 2014.
- [43] Ming-Hsuan Kang, Wen-Ching Winnie Li, and Chian-Jen Wang. The zeta functions of complexes from $\mathrm{pgl}(3)$: a representation-theoretic approach. *Israel Journal of Mathematics*, 177:335–348, 2010.
- [44] Ming-Hsuan Kang, Wen-Ching Winnie Li, and Chian-Jen Wang. Zeta and l -functions of finite quotients of apartments and buildings. *Israel Journal of Mathematics*, 228:79–117, 2018.
- [45] Ming-Hsuan Kang and Jiu-Kang Yu. Zeta functions of pgl_n over non-archimedean local fields, 2026.
- [46] Sohren Kleine and Daniel Vallieres. Spanning trees in $\mathbb{Z}$ -covers of a finite graph and mahler measures, 2023.
- [47] Oliver Knill. Analytic torsion for graphs, 2022.
- [48] Takashi Komatsu, Norio Konno, and Iwao Sato. The scattering matrix with respect to an hermitian matrix of a graph, 2021.
- [49] Takashi Komatsu, Norio Konno, and Iwao Sato. Walk/zeta correspondence, 2021.
- [50] Norio Konno and Iwao Sato. On the relation between quantum walks and zeta functions. *Quantum Information Processing*, 11, 2012.
- [51] Jeffrey C. Lagarias. Li coefficients for automorphic L-functions. *Annales de l’Institut Fourier*, 57:1689–1740, 2007. arXiv:math/0404394.
- [52] Peter D. Lax and Ralph S. Phillips. *Scattering Theory for Automorphic Functions*. Princeton University Press, 1976. no local source.
- [53] Abraham Lempel and Marcelo J. Weinberger. Self-complementary normal bases in finite fields. *SIAM J. Discrete Math.*, 1:193–198, 1988. no local source.
- [54] Rudolf Lidl and Harald Niederreiter. *Finite Fields*. Cambridge University Press, 2nd edition, 1997. no local source.
- [55] Eyal Lubetzky, Alexander Lubotzky, and Ori Parzanchevski. Random walks on ramanujan complexes and digraphs. *Journal of the European Mathematical Society*, 22, 2020.

- [56] A. Lubotzky, R. Phillips, and P. Sarnak. Ramanujan graphs. *Combinatorica*, 8:261–277, 1988. no local source.
- [57] Alexander Lubotzky. High dimensional expanders, 2017.
- [58] Jens Marklof. Selberg’s trace formula: an introduction. In *Hyperbolic Geometry and Applications in Quantum Chaos and Cosmology*. Cambridge University Press, 2012.
- [59] So Matsuura and Kazutoshi Ohta. Graph zeta functions and wilson loops in kazakov-migdal model, 2022.
- [60] So Matsuura and Kazutoshi Ohta. Kazakov-migdal model on the graph and ihara zeta function. *JHEP*, 09:178, 2022.
- [61] So Matsuura and Kazutoshi Ohta. Fermions and zeta function on the graph, 2025.
- [62] So Matsuura and Kazutoshi Ohta. Generalized kazakov-migdal models on graphs via artin-ihara l-function and random partitions, 2026.
- [63] Brian Maurizi. Extending Landau’s theorem on Dirichlet series with non-negative coefficients, 2010.
- [64] Arash Momeni and Alexei Venkov. Zeta functions and regularized determinants related to the selberg trace formula, 2011.
- [65] Saulius Norvidas. On positive definite functions and Bochner’s theorem. *arXiv preprint*, 2020. arXiv:2009.02802.
- [66] Oleg Reinov and Qaisar Latif. Grothendieck–Lidskii theorem for subspaces and factor spaces of L_p -spaces, 2011.
- [67] Inken Siemon, Alexander S. Holevo, and Reinhard F. Werner. Unbounded generators of dynamical semigroups, 2017.
- [68] Barry Simon. *Trace Ideals and Their Applications*. American Mathematical Society, 2nd edition, 2005. no local source.
- [69] H. M. Stark and A. A. Terras. Zeta functions of finite graphs and coverings. *Advances in Mathematics*, 121:124–165, 1996. no local source.
- [70] Christopher K. Storm. The zeta function of a hypergraph. *Electronic Journal of Combinatorics*, 13, 2006.
- [71] Toshikazu Sunada. L-functions in geometry and some applications. In *Curvature and topology of Riemannian manifolds*, volume 1201 of *Lecture Notes in Mathematics*, pages 266–284. Springer, 1986. no local source.
- [72] Masatoshi Suzuki. Aspects of the screw function corresponding to the Riemann zeta function. *Journal of the London Mathematical Society*, 108:1448–1487, 2023. arXiv:2206.03682.
- [73] Audrey Terras. *Zeta Functions of Graphs: A Stroll through the Garden*. Cambridge University Press, 2010. no local source.
- [74] Teruji Thomas. The character of the Weil representation. *J. London Math. Soc.*, 77:221–239, 2008.

- [75] Frank Verstraete and J. Ignacio Cirac. Continuous matrix product states for quantum fields. *Phys. Rev. Lett.*, 104:190405, 2010.
- [76] André Voros. Sharpenings of Li’s criterion for the Riemann hypothesis. *Mathematical Physics, Analysis and Geometry*, 9:53–63, 2006. arXiv:math/0506326.
- [77] Yusuke Watanabe and Kenji Fukumizu. Loopy belief propagation, bethe free energy and graph zeta function. *Journal of Machine Learning Research*, 2011.